

The Energy Balance Project

What an Economy Must Keep Alive

**The physical theory and design of an Energy
Balance Unit**

A continuous explanatory edition

Expanded theoretical integration

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AUTHOR-REVIEW MANUSCRIPT

October 2026

Preface

Every generation inherits a world that makes its activity possible. Money helps people exchange, coordinate and plan within that world. Yet a completed sale cannot, by itself, tell us whether the water line will still work, whether a field will still produce food or whether a community can absorb its next disturbance. Those are physical questions. This book asks what an economy would look like if its account of action had to answer to them as well.

We begin with the ordinary reasons money arose, and with the incentives it now creates. From there the argument moves through light, energy, thermodynamics and the living body's continuous work of homeostasis. The point of these comparisons is not to pretend that an economy is a photon or an organism. It is to look for a disciplined way to connect local actions to the conditions that keep a larger system functioning.

The proposed Energy Balance Unit enters only after that problem is clear. We declare what physical state is represented, how an action changes it, and what potential expresses a chosen functional comparison. The exact finite EBU is the decrease of that potential across the complete accepted change. An integral along the change shows why an initial slope is not enough and why the finite result can be calculated without asking one actor to know the state of the whole planet.

The later chapters ask what happens when actions share sources, when physical effects cross several steps, and when the field itself changes. They also ask how a future EBU system could keep its physical permissions, measurements, attributions and actor decisions distinct. An equation may be exact while the world chosen for it remains a hypothesis. We therefore explain the mathematics fully, then show what an implementation and an independent test would have to establish before anyone could claim more.

The physical consequence, its mathematical value, a personal attribution and an institution's rule for using that attribution will remain distinguishable throughout. Keeping them connected without making them identical is the discipline on which EBU's promise depends. We can begin, then, where people already make their choices: with money, the work it does for us, and the physical conditions it cannot describe by itself.

How to read the extended argument

The opening chapters build the physical vocabulary without assuming advanced mathematics. The central sequence then moves from paths and finite changes to coalition tables, recursive interaction and exact discrete Taylor structure. It asks how a nonlinear response or a generator supplies the endpoints before adding the narrower probability and entropy interpretations. Each major result is followed by what it could make possible for an EBU system and what remains a matter of evidence or institutional choice.

The final chapters return to routes, attribution, measurement, implementation and access. The appendices contain the longer calculus and generator proofs, statistical counterexamples and a worked claim review. They can be read after the main argument without breaking its continuity.

A definition declares an object. A theorem or corollary derives a consequence under stated assumptions. Constructed examples teach the mathematics. Physical interpretations require their named models, while hypotheses and future studies remain open. This edition integrates independently cleared source mathematics; it remains an author-review book requiring a separate independent book audit.

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What does an economy keep alive?

1.1 Why people made money

No household can make everything it needs, and no community can rely on everyone needing the same thing at the same moment. People have therefore had to exchange, remember obligations, compare unlike contributions and settle claims across time. The solutions did not appear in one place on one day. Historical records include accounts kept on clay tablets, measured commodities, weighed metal and, later, coins. Money has taken several forms; its history is not a simple story in which coins suddenly replaced a world of direct barter.[1]

Whatever its form, money can make a remarkable kind of cooperation possible. It gives people a common unit in which to state an obligation, a means of paying it and a way to carry a claim into the future. A carpenter can be paid without having to take precisely the food that the customer happens to have that day. A hospital can buy supplies from people it has never met. These functions depend on trust and institutions, but they free cooperation from the need for every exchange to be settled by an immediate swap.[2] We should begin with that achievement, not with the claim that money was a mistake.

The question is what a monetary account tells us *after* a payment succeeds. Suppose a town's clinic has paid its electricity bill and ordered medicine, but no water reaches its treatment room because a pipe has failed. The payment establishes that a claim was discharged under an agreement. The service requires a working chain of reservoirs, pipes, power, maintenance and human attention. If one link fails, a financially completed transaction can coexist with incomplete physical provision. If someone repairs the pipe without charging anyone, the reverse can occur: a vital service returns without a sale.

There is no paradox. Money records and coordinates claims; water, care and functioning equipment are conditions in the world. An economy should make it possible for people to live, recover, care for one another and hand a functioning world to those who follow them. How can it tell whether it is doing so? The word *tell* does not require a central mind. It asks what information an arrangement records and what responses that information encourages. A clinic may monitor its water, a town may protect a catchment, and a govern-

ment may regulate a pollutant. The ordinary monetary result of an exchange does not, by itself, report whether these life-supporting physical conditions improved or deteriorated.

1.2 A measure can be excellent at the wrong question

The versatility that makes money useful also limits what one amount can report. A sum can travel across goods, people and years precisely because it does not describe every physical property of the underlying transaction. Two products sold for the same price can require different material inputs, leave different wastes and contribute very differently to health or resilience. Equal revenue is not equal physical consequence.

An enterprise deciding whether to repair a pipe faces costs and expected returns. If it cannot recover the repair cost, the financially rational choice for that enterprise may be delay, even when the delay imposes a larger physical cost on households. The opposite possibility matters too: maintenance may be profitable and beneficial at the same time. There is no universal rule that financial gain destroys function. There is simply no identity that makes the sign of a financial return equal the sign of a physical change.

The mismatch can be corrected in particular cases. Law, public services, contracts, standards, taxes and community obligations can bring neglected effects into a decision. Their existence is important: the present economy is not a lawless machine whose only instruction is profit. But each correction must know which physical consequence it is trying to address. Without that knowledge, one can reward an activity because it is lucrative and discover its hidden costs only after water, soil, equipment or health has been damaged.

1.3 Want, need and the power to pay

In a market, a person's expressed demand depends partly on purchasing power. A buyer who can pay for a second luxury delivery may exert more influence on a seller than someone who urgently needs a first delivery of safe water but cannot pay. This is not an accusation against the seller. It is a distinction between a transaction's signal and a person's need. Money answers how much can be offered under a set of rights and prices; need asks what is required for a function or a life.

The distinction becomes sharper when the future has no present purchaser. A child not yet born cannot bid for a stable climate or an intact aquifer. A forest's contribution to rainfall or habitat may support many people without appearing on the receipt for a nearby harvest. These effects can be repre-

sented through institutions, but a price tag on one immediate sale does not automatically carry them.

It is tempting to answer by saying that every need should be assigned a price. Pricing can help in some circumstances, but it does not settle how the need is measured, who has authority to set the price, whether those affected can pay, or whether a loss is reversible. A financially visible problem is not necessarily a physically solved problem. We need to know what the world actually contains and what an action actually changes.

The question of this book

Before asking whether an action has money, ask what it takes from the physical world, what it leaves there, and which continuing function depends on the change. A useful economy needs both a way to coordinate people and a way to remain answerable to those physical facts.

1.4 A world inherited rather than purchased

Each generation receives a planet it did not manufacture. Sunlight arrives, water circulates, organisms reproduce, and human beings maintain the networks through which these processes become shelter, food, care and knowledge. None of this means that nature is a perfectly balanced machine or that every existing arrangement should be held still. Living systems change. A sustainable inheritance is not a frozen photograph; it is the continued possibility of life-supporting function through change.

That possibility can be lost while accounts still rise. It can also be preserved by work whose immediate monetary return is small. A repaired wetland, a maintained water line or a nurse's unpaid care may not fit neatly into a calculation centred on sales. They nonetheless matter to what people can do tomorrow. The question is not whether money is useful; it is whether financial success alone is an adequate compass for a civilisation that must keep its physical home viable.

We will eventually build a mathematical account of particular physical actions. First we must understand why the familiar account can miss them. The next chapter follows the incentive faced by an ordinary decision-maker: what does a price reward, what does it leave outside the deal, and why can maximising a return become different from maintaining the conditions on which the return depends?

What money rewards, and what it cannot see

2.1 The decision at the workshop

A workshop can spend a day repairing a town's water line or use that day to finish an order for a customer who will pay immediately. Its workers need wages; its tools and electricity cost money; it cannot simply ignore its accounts. The paid order is therefore a serious claim on its time. The pipe is also a serious claim, because households and a clinic depend on it. The difficulty is not that one claim is real and the other imaginary. It is that the two arrive in different languages.

In a familiar business calculation, an activity is attractive when expected revenue exceeds the private costs borne by the decision-maker. Written schematically,

$$\text{expected private return} = \text{expected receipts} - \text{private expenditure.} \quad (2.1)$$

This is not a complete description of every person's motivation. People act from duty, affection, law, fear and curiosity as well as financial return. But when an institution must remain solvent, the sign of this difference has practical force. A repair that protects many homes can be difficult to undertake if the workshop cannot pay for it; an activity with a buyer can proceed even if some physical costs fall elsewhere.

Prices convey important information about willingness and ability to pay, costs, scarcity under existing rights and expectations. They are not a general sensor of the future condition of a catchment, a patient's health or the integrity of a bridge. The distinction is well known in economics as an *externality*: an action affects others in ways its market price does not include. The IMF's account of externalities uses pollution as a standard example, and its energy-subsidy work notes that retail fuel prices often do not cover environmental costs.[4, 5] Such an omitted effect does not become unreal because it is absent from a receipt.

2.2 A dollar is a claim, not a physical forecast

A bank deposit, a wage and a sale are parts of a monetary system that allocates claims and makes exchange possible. In modern banking, deposits can also be created through lending; the Bank of England describes the process and its constraints in detail.[3] None of these facts implies that money is unreal or that every loan requires growth forever. The relevant observation is simpler: a unit of currency is not stamped with the amount of clean water, usable soil, climatic stability or care that will exist when it is spent.

The workshop may receive the same amount for two orders. One consumes scarce spare parts and shortens the life of a pump; another repairs the pump. The money entering its account is equal, while the future physical options differ. A good set of prices, contracts and regulations can narrow that gap. Yet the physical consequence has to be known or estimated before it can be brought into a rule. One cannot repair a missing measurement by assuming that the monetary number already contains it.

The problem reaches across generations. Future people cannot sign today's contracts. Some long-lived damage will be carried by them even though no invoice names them. Present institutions can and do act on their behalf, but the incentive is an explicit political and ethical achievement, not a property of a banknote.

2.3 Demand, desire and indispensable service

A demand curve records offers under a distribution of purchasing power. It does not rank claims by biological urgency. A household can want an additional convenience while another lacks its first safe supply. If the first can pay and the second cannot, the former may have greater market demand. This does not mean producers are indifferent to suffering. It means that an economy organised around sale needs additional arrangements if indispensable functions are to remain available independently of the buyer's wealth.

Public water systems, health provision, subsidies, insurance and regulation are examples of those additional arrangements. Their existence makes a crude attack on money misleading. They also reveal the underlying issue: societies repeatedly have to connect financial decisions to outcomes that financial transactions alone do not guarantee. How well that connection works depends on what is measured, who can act and whose interests count.

One response is to adjust prices until every physical effect is priced. That may be useful for some effects, especially where they can be measured, attributed and governed. It is harder where a loss is irreversible, where a service has no buyer, or where a single price hides unequal exposure. We should therefore

treat money as a powerful coordination instrument with a defined job, not as a universal account of all the conditions that make life possible.

2.4 Where an incentive points

Imagine that a supplier can earn ten monetary units by selling water to a distant buyer and six by supplying a local clinic. If the accounts omit the clinic's health consequences, the larger sale looks better in the supplier's private calculation. If a rule pays for protecting access, the comparison changes. If a drought makes export physically dangerous, even a high offered price cannot create water. The three statements refer to different objects: private return, institutional incentive and physical possibility.

In this book we will eventually ask whether a physical account can make one of those objects more visible. It is not a promise that a new number should replace every price. It is a demand that an action's represented physical consequence be available for scrutiny when incentives are designed. A number that can be audited can influence a rule; a rule can influence a choice; repeated choices can influence a future. Each link matters, and none should be silently supplied by the previous one.

Two ledgers for one decision

Suppose a repair costs the workshop eight units of money and earns only five today. Its private return is -3 . If the repair prevents a measured loss of water, that physical fact does not turn -3 into $+3$. It gives a community a reason to design a payment, obligation or public service that makes the repair possible. The two accounts inform one decision without being equated.

2.5 The next difficulty: who can reach what exists?

Even perfect information about a product's price would not tell us whether everyone who needs it can obtain it. Goods can be present and still be inaccessible because of location, quality, rights, wealth or conflict. The next chapter examines this gap. It is where scarcity stops being a single total and becomes a question about people, routes and power.

Scarcity is more than an empty store

3.1 Water behind a locked gate

A reservoir may contain enough water to supply a town today, yet its clinic can still have none. The connecting pipe may be broken, the water may be unsafe, the owner may deny access, or the clinic may be unable to pay the charge. Calling all these situations simply *scarcity* hides the difference between them. A dry reservoir calls for a different response from a blocked pipe, and both differ from exclusion by price.

We can separate three questions. How much of a physical resource exists within the declared boundary? Can it be moved and used for the required function in time? Who is permitted and able to obtain it? A stock total answers the first question only. Infrastructure and quality enter the second; institutions and purchasing power enter the third. The FAO likewise treats water scarcity as involving physical supply, demand and access rather than one universal stock number.[7]

Some limits cannot be negotiated away. A town cannot distribute more drinkable water than it can obtain, treat or safely reuse over the relevant interval. Other limits are made by rules and may be changed. A full warehouse can stand beside hunger when the people who need food have no effective claim on it. This does not prove that all scarcity is artificial; it proves that physical abundance and social access are different measurements.

3.2 Competition is a choice of allocation rule

When several buyers want a constrained good, offers of money can decide who receives it. Competitive bidding is a clear way of rationing under a particular allocation rule. It can also make exclusion appear inevitable to those who cannot bid. Scarcity of the resource and scarcity of purchasing power then reinforce one another, although they are not the same thing.

The urge to gain an advantage is not a law of physics. Nor is cooperation an automatic consequence of shared need. People and organisations respond to the rules under which they operate. If an extra unit sold today increases private revenue while depletion tomorrow is borne by others, extracting and

selling can be rewarded. If maintenance, restraint and reliable service are rewarded, the feasible choices look different. The design of incentives matters because it changes which physically possible actions are attractive.

This is why it would be mistaken to call maximisation itself the enemy. Every decision uses some criterion, even if it is not written down. The important question is *what* is being maximised and *subject to which conditions*. Maximising a short-term monetary return without a condition on essential services is not equivalent to maximising the continuing ability to meet needs. The first can grow while the second falls.

3.3 Inequality changes exposure as well as income

Unequal access is not merely a philosophical worry. The World Bank's *Poverty, Prosperity, and Planet Report 2024* estimated that about 700 million people lived in extreme poverty under its then-used \$2.15-a-day line, and that progress in poverty reduction had slowed. The number is a dated estimate under a specified poverty definition, not a count of people who lack every physical resource. The same report describes the joint challenge of poverty, unequal opportunity and climate risk.[8]

The WHO's 2025 report on social determinants of health equity documents how conditions of birth, work, housing and access to power, money and resources shape health. Its comparison of life expectancy across places and social groups shows why a national average cannot stand in for everyone's lived condition.[11] A resource model that reports only a total can miss the distribution that determines whether a person can use the resource at all.

A stock can be adequate for a district as a whole and inadequate for one clinic. A road can be passable to people with vehicles and effectively closed to people who cannot travel. A treatment can exist and remain unavailable to a patient. The relevant physical account must have enough structure to keep these distinctions visible; a total alone is often too coarse.

3.4 Resources, grievance and conflict

Control of land, fuel, water and revenue can become part of a conflict. Exclusion and inequality can deepen grievances; damaged services can make an already fragile situation worse. The World Bank identifies economic exclusion among factors that can feed fragility, alongside governance and other causes.[9] The IPCC finds that climate variability and extremes can affect conflict risk in some contexts, but its assessment warns against attributing armed conflict to climate change alone; the link is conditional and weaker than major social and political drivers.[10]

The same care applies to the claim that monetary gain causes war. Armed conflicts have many histories and motives. Material advantage can be one incentive and control of resources one means of sustaining violence; neither statement is a universal explanation. A book about physical accounting should make the connection discussable without turning a grave human event into a slogan.

For our small district, imagine two towns drawing from one river during a drought. An account showing each withdrawal and the river's changing condition will not settle their rights or eliminate hostility. It may, however, prevent a negotiation from proceeding as though each town possessed the entire flow. Shared facts do not choose a just rule, but a just rule is harder to exercise when the facts are invisible.

Three shortages

Not enough usable resource, no practical route to it, and no effective right to receive it are different failures. A responsible response begins by finding which failure is present.

3.5 From a district to a planet

Two towns can sometimes import what they lack. Humanity as a whole cannot import a second biosphere on the schedule of a procurement contract. That does not make every natural process a fixed stock; sunlight, water cycles and regeneration are continual processes. It does make their condition and limits central to the future. The next chapter moves from the local access problem to measured planetary pressures and asks what the present evidence says about the world we are passing on.

The planet inside the account

4.1 A changing inheritance

It is possible to disagree about economic policy and still begin from measurements. The IPCC concluded in its 2023 synthesis that human activities had unequivocally caused global warming, with the global surface temperature in 2011–2020 about 1.1 °C above 1850–1900. It also assessed widespread changes in the atmosphere, ocean, ice and living world, together with unequal contributions and unequal exposure.[12] The number names a period and a comparison baseline; it is not a reading from this morning's thermometer.

The IPBES global assessment estimated that roughly one million animal and plant species were threatened with extinction. *Threatened* does not mean already extinct, and an estimate is not a census of every species. It is nevertheless a serious account of pressure on the living systems on which human activity also depends.[13] The International Resource Panel's 2024 outlook describes rising material use and a conditional path on which global extraction could increase by almost sixty percent from 2020 to 2060 without urgent concerted action.[14] That is a scenario under stated assumptions, not fate.

These assessments concern different things. Temperature is not biodiversity; extraction is not a local stock of clean water; a global average is not a patient's access to care. No honest account obtains one answer by adding the headline numbers together. Their shared message is that human activity can change the conditions it relies on, and that some changes last far longer than the sale that helped cause them.

4.2 The short return and the long consequence

Consider a company drawing water from a catchment. Its revenue can arrive this month, while damage to groundwater, soils or downstream ecology may develop over years. The company may have legal duties, insurance and public scrutiny; it is not necessarily blind to the future. Still, if the consequence is not observed or assigned to any responsible decision, the near-term return and the long-term condition can diverge.

Three clocks are operating at once. The transaction closes, the physical stock changes, and the supporting system responds or fails to regenerate. We cannot infer the third clock from the first. A forest takes time to regrow, a reservoir takes time to refill, and a damaged species relationship may not be recoverable on a planning horizon. Some resources circulate; others are dispersed or degraded. The physical form of the process matters.

The question for future generations is therefore not whether we can keep every present quantity unchanged. It is whether their world will retain the capacity to sustain lives and choices. A planet is not a machine with one dial called health. But the absence of a universal dial is no excuse to ignore measured stocks, flows, damage and renewal.

4.3 A projection is a conditional warning

The IPCC reports different projected outcomes under different emissions pathways. A pathway is not assigned a probability merely by appearing in a table; it describes a set of assumptions about future activity. The point of examining alternatives is that choices matter. A warning about a damaging future is useful precisely when that future can still be avoided.[12]

The same discipline should be applied to any new economic proposal. A diagram showing a balanced state does not establish that real actions will keep the world there. A model can prove a mathematical identity without proving that its variables cover every important harm. We will take those distinctions seriously, but we need not repeat them at every turn. The positive task is to design accounts that make consequential physical changes visible soon enough to guide action.

Read the year and the condition

The UNEP/IRP projection of almost sixty percent more extraction by 2060 refers to a scenario measured against 2020 and conditioned on a continuation without urgent concerted action. It does not say that 2060 extraction has already been observed, or that all resources increase by the same amount.

4.4 What is missing from the dashboard?

Suppose a district reports rising income while its aquifer falls, its clinic loses reliable service and its waste accumulates. The financial report may be true. The physical reports may also be true. What fails is the expectation that one

report substitutes for the others. At planetary scale the same error can be harder to see, because harms cross borders and generations.

We need a way to describe what an action does within an explicit physical boundary and how that change bears on continuing function. That does not require calculating the entire planet before opening a valve. In fact, such a calculation is impossible for an ordinary local action. The next question is more practical and more surprising: how can a large world behave consistently when no single participant computes its whole state? Physics offers examples of local relations that produce exact consequences, and those examples will lead us toward the account this book develops.

How does the world know which way to go?

5.1 No central calculator

When light crosses from air into water, its route bends. The light does not pause at the boundary while an invisible planner surveys every possible trajectory. When warmth spreads from a hot object into a cooler one, no administrator compares all the molecules. The behaviour follows local physical relations, and those relations can have consequences that are described by compact global statements.

This is the feature we are looking for, not a claim that societies should behave like photons. No person could calculate the complete state of the planet each time a tap is opened or a delivery truck moves. If a proposed physical account required such an all-knowing observer, it would fail at the first ordinary decision. We need to understand how a local change can be assessed without pretending that the entire world is locally known.

Physics provides two kinds of discipline. Conservation tells us that a carrier has not appeared from nowhere inside a closed account. A relation of motion or change tells us which transitions occur under specified conditions. Neither is the same as a moral rule or a social allocation policy. Their value for our purpose is structural: they show how to make claims precise enough that hidden creation, omitted boundaries and impossible action can be detected.

5.2 The bent path of light

Imagine a point above a flat water surface and another below it. In a single uniform medium, the straight segment is the shortest geometrical distance. Across the boundary, light can instead follow two straight segments joined at an angle. The route is longer in space than the direct line, yet the light travels at different speeds in the two media. Distance alone is not the relevant quantity.

In elementary geometrical optics the travel time of a candidate path γ is

$$T[\gamma] = \frac{1}{c} \int_{\gamma} n(\mathbf{r}) ds, \quad (5.1)$$

where c is the speed of light in vacuum, $n(\mathbf{r})$ is the refractive index and ds is a small piece of distance. The integral adds the time cost of all those pieces. Fermat's principle identifies the observed ray, in its domain of application, by stationary optical time: a small permitted displacement of the path produces no first-order change in T . The familiar phrase *least time* is useful for the elementary two-medium case, but stationarity is the more general statement.[16]

We can see the rule in a drawing with no mystery. Let the light enter the surface at a horizontal coordinate u . If its starting point is a horizontal distance a from the chosen origin and height h_1 above the surface, while its destination lies at horizontal coordinate b and depth h_2 , then the candidate time is

$$T(u) = \frac{n_1}{c} \sqrt{(u-a)^2 + h_1^2} + \frac{n_2}{c} \sqrt{(b-u)^2 + h_2^2}. \quad (5.2)$$

Differentiate with respect to the crossing point u . At a stationary path,

$$\frac{dT}{du} = 0 \implies n_1 \sin \theta_1 = n_2 \sin \theta_2, \quad (5.3)$$

which is Snell's law. The angles are measured from the normal to the interface. If $n_2 > n_1$, the ray bends toward that normal. A single rule accounts for the bending because the cost of another small distance is different on the two sides.

Why the straight line can lose

Take $n_1 = 1$ and $n_2 = 1.5$. If $\sin \theta_1 = 0.6$, stationarity requires $\sin \theta_2 = 0.4$. The path changes direction at the interface. It is not the shortest geometrical connection, because distance in the slower medium contributes more time.

5.3 What the analogy gives us, and what it does not

The optical integral teaches us to name the quantity being accumulated and the path over which it is accumulated. It also teaches caution. Light is not awarded a payment for finding a route; Fermat's principle is not a doctrine that people should choose the fastest path or that an economy should minimise travel time. It is a physical result for light under specified optical conditions.

Later in this book another path integral will appear. It has a different mathematical character. If a smooth scalar potential V is declared, integrating its gradient along a path between two states gives a difference of endpoint values. Such a gradient integral is path-independent while the optical time in Eq. (5.1) generally depends on which route is proposed. The similarity is the

discipline of summing local contributions; the distinction is essential to the mathematics.

Nor should the expression *the world knows* be taken literally. Local laws can generate orderly patterns without foresight. A photon does not inspect an economic objective, and molecules do not vote on the future. For human systems we must declare the actions, observations and decision rules ourselves. Physics will constrain and inform them, not silently select an institution.

5.4 From path to balance

A water movement between a reservoir and a treatment site also has a route. It may require a pump, electricity, maintenance and a pipe. Describing only the final delivery could hide the resource used along the way; counting the same transformation twice could invent an extra cost. The first task is therefore to keep the physical account closed: what left one place, what entered another, and what crossed the boundary?

But conservation is only the beginning. Even when an amount is conserved, useful energy can become dispersed; a system may remain energetically possible while losing its ability to perform a function. The next chapter examines the first and second laws of thermodynamics, then asks how a human body maintains function in a world where every activity consumes and transforms resources.

Energy, life and functional balance

6.1 The rule that prevents a free lunch

A pump lifts water because something supplies energy. Electricity reaches its motor through a network; that electricity was generated from an identified source. The pump cannot deliver indefinitely from an empty energy supply, however persuasive a contract or a price might be. This is the ordinary force of the first law of thermodynamics.

For a closed thermodynamic system under a familiar sign convention,

$$\Delta U = Q - W, \quad (6.1)$$

where U is internal energy, Q is heat entering and W is work done by the system. An open system also carries energy with matter crossing its boundary. The equation does not say that every useful form of energy remains available for reuse; it says that a claimed change requires a corresponding transfer or transformation. It is the scientific reason a perpetual-motion machine of the first kind cannot be built.[17]

The bookkeeping lesson is immediate. If a pump moves water while the account contains no energy source, either the physical description is incomplete or the pump has been granted impossible power. Recording the electricity as a flow does not decide whether pumping is wise. It prevents us from calling the action self-sustaining when it depends on something outside the picture.

6.2 Why conserved energy is not enough

The second law addresses a different question: which transformations are possible and what happens to the ability of energy to do useful work. For an isolated system, total entropy does not decrease. A real heat engine cannot convert every unit of heat from one reservoir into work while leaving no other change. Friction and mixing can preserve energy while dispersing it in forms less useful for the task at hand.[17]

The Earth is not an isolated box. It receives energy from the Sun and radiates energy to space. A living organism can build and maintain local order by taking in energy and matter and exporting heat and waste. The second law

does not forbid life; it explains why maintenance has a physical cost and why an account that shows only the organised result is incomplete.

At the clinic, a refrigerator preserves medicine by moving heat from its interior to the surrounding room while consuming electricity. Saying that the medicine remained cold without recording the power and the rejected heat would mistake a continuing service for a free property of the refrigerator. A durable economy likewise has to ask what flows sustain its useful arrangements.

6.3 The body does not maximise its temperature

Now consider a human body. Its temperature, blood pressure and concentrations of important substances vary while the body remains alive. Regulation does not mean setting every variable to zero, holding every amount perfectly fixed or making one quantity as large as possible. Too little and too much can both be dangerous. Physiology describes homeostasis as the maintenance of conditions compatible with function through interacting processes of sensing and response. Negative feedback is a central pattern: a departure triggers a response that tends to oppose it.[15, 18]

The point is clearer in a simple tank. Water enters and leaves, yet its level can remain within a useful region. If outflow increases, a sensor may detect a fall and a valve may admit more water. If the inlet fails, the level continues to fall despite the sensor. If the response is delayed, it may overshoot. The difference between a measurement, an available action and a successful response matters as much in a body as in an infrastructure network.

We sometimes speak as if growth or accumulation were an adequate objective for every system. The body shows why that cannot be a general physical rule. More heat is not always better; more glucose is not always better; more pressure is not always better. The same warning applies to a society that treats increased throughput or income as sufficient evidence of continuing health. A gain has to be assessed in relation to the conditions that make the system function.

6.4 A balance that remains active

Suppose the clinic's water tank contains ten units at the beginning and end of an hour. During the hour it receives four and supplies four. Another tank also begins and ends at ten but has both valves closed. Their equal snapshots conceal different service histories. This is why functional balance is not simply equality of two numbers: it can be a maintained process.

The stock account for the first tank is

$$x_{t+1} = x_t + q_{\text{in}} - q_{\text{out}}, \quad (6.2)$$

with each q measured in the same physical unit over the same interval. A constant stock means $q_{\text{in}} = q_{\text{out}}$ for that interval; it does not tell us whether the water was safe or whether a patient received it. Those are additional facts. At a larger scale, regeneration, consumption and disturbance have to be represented if we want a claim about continuing function.

Three separate tests

Can the movement physically occur? Does the represented state change in a desirable direction? Can the process continue under demand and disturbance? The three questions belong together, but no one answer proves the other two.

6.5 Where should the system return?

Negative feedback is intelligible only relative to a condition from which the system has departed. But what is that condition? A fixed number may be appropriate for one regulated variable and inappropriate for another. A body is not healthy because every variable is equal, and two towns need not hold equal stores. Nor does a mathematical minimum automatically cause physical motion toward itself.

We now need to separate a *reference* used for comparison from an *equilibrium* of a stated law of change and from the broader idea of *homeostasis* through time. Those distinctions will let us say what it would mean for an action to restore rather than merely increase something. The next chapter builds them before any new accounting unit is introduced.

Equilibrium is not a maximum

7.1 What is being kept in balance?

The body does not seek the highest possible temperature. The clinic does not become more secure merely by filling one tank without regard to the pressure in its pipes or the water left elsewhere. A living arrangement has several conditions of continued function. Some resources can be too scarce; some can be present in unusable places; some processes can run too quickly or consume the very reserve on which tomorrow depends.

This observation gives us a different kind of question from the one posed by a price or a sale. Instead of asking how large one return can become, we ask which physical conditions must be maintained and how a proposed action changes them. The answer cannot be a single universal target imposed on every place. A patient, a reservoir and an electrical network are not the same object. Their relevant variables and acceptable conditions must be declared for the case at hand.

Physiologists use homeostasis for the maintenance of conditions compatible with life through processes of regulation. The word does not mean motionless perfection. Breathing, circulation and temperature regulation continue while a person is well. In an economy, too, a stable stock might conceal a continuous flow of production, use, repair and replacement. A useful physical account must be able to distinguish a functioning flow from an identical-looking snapshot.

7.2 Reference, equilibrium and stability

To measure departure we first need a reference. Suppose a deliberately simple model has two water stores, A and B, with a total of twenty physical units. A reference of ten units in each place can be written

$$x^* = (10, 10). \tag{7.1}$$

The star means that this is a declared comparison state. It does not assert that ten is the correct real-world stock for a clinic, that the state will be

reached, or that it will remain in place. In a real design the reference would need a reason connected to use, safety and measurement.

An equilibrium makes a different claim. If a specified law of change maps a state x to its next state $T(x)$, then an equilibrium $\bar{x}$ satisfies $T(\bar{x}) = \bar{x}$. Under a rule that never acts, every state is an equilibrium, including a state in which store B lacks water. A mathematical fixed point is therefore not automatically a healthy condition.

Stability asks a further question: what happens after a disturbance? A system may be at equilibrium but move farther away when perturbed. It may return toward the reference, or it may return only while energy, materials and functioning equipment remain available. None of these behaviours follows from naming the reference. They require a declared dynamics and, ultimately, a test of how that dynamics behaves.

Three claims, three different questions

A *reference* is where we compare. An *equilibrium* is stationary under a specified rule. *Homeostasis* concerns continued function and response under change. A single calculation may inform all three, but it cannot make them identical.

7.3 A physically possible reference

Conservation places a useful limit on the discussion. The two stores together hold twenty units. Lossless transfers between them preserve that total. The reference (10, 10) is compatible with the total, whereas (12, 12) requires twenty-four units and cannot be reached by internal transfers alone. No better incentive or smarter calculation can supply those missing four units without an explicitly represented input.

Compatibility is not enough. A route may be absent or blocked. A source may lack enough available stock for a proposed transfer. The action may use electricity or lose water between the two measurements. We must therefore separate the question of where we would like a system to be from the question of what its physical network can actually do.

Now compare (18, 2) with (14, 6). Both contain twenty units. The latter is closer to (10, 10) in an ordinary geometric sense. But what does closer mean when two resources have different units, or when a four-unit shortage at B matters differently from a four-unit surplus at A? We need a declared measure of deviation before we can assign a number to improvement. Chapter 11 will

give each coordinate a reference and a scale; Chapter 12 will show why a Gaussian-shaped construction supplies a transparent, smooth measure.

7.4 Why the direction must depend on the state

The same transfer can help in one state and harm in another. Moving four units from A to B changes (18, 2) into (14, 6), bringing both stores toward the symmetric reference. Starting instead from (10, 10), the identical movement produces (6, 14), moving both away. An action name such as water delivery cannot carry a permanent positive sign independently of its starting state.

This is precisely why maximising every local transaction is not a satisfactory physical rule. A payment may reward the same delivery at both starting points; a field concerned with the represented balance must distinguish them. It must also account for the whole finite movement, because an action that initially helps can overshoot if carried too far. A local slope tells us a direction near the start, but the value of a completed action must include what happens as the state changes.

There is a caution here that strengthens rather than weakens the proposal. A mathematical measure of departure is not itself a law that opens valves or allocates water. It can inform incentives and expose consequences, but actions still require physical permission and an institution that decides what to do with the information. The problem is to build a trustworthy account first, then test whether a proposed way of using it improves continuing function.

7.5 The question that leads to a new account

Imagine that a community has two possible transfers. Both are feasible. One reduces an immediate shortage but consumes a scarce reserve; another produces less immediate delivery while leaving the stores better prepared for the next disturbance. A monetary invoice records payment, but it does not itself calculate the represented change in physical balance. An endpoint-only stock description can miss the energy carriers, losses and intermediate states needed to describe how either transfer happened.

We want an account whose sign and amount respond to what the physical action actually changes, under a stated reference and boundary. The account should not pretend that a desirable endpoint creates its own water, that a smooth mathematical minimum guarantees recovery, or that one stock coordinate describes every need. But it can make a crucial question answerable: did this accepted finite action reduce the declared burden of the represented system, and by how much?

What an Energy Balance Unit is

8.1 A physical consequence needs an account

The previous chapters have brought us to a practical question. The clinic cannot operate without payments, but a successful payment is not evidence that the water system remains capable of serving it. Conversely, water can be physically available while the people who need it cannot gain access. We do not have to abolish one description to improve the other. We need a second account that makes physical consequences visible at the point of action.

An Energy Balance Unit, or EBU, is the signed unit of that account. Given a declared physical potential V , an accepted finite action has the decrease of V between its complete before-state and after-state. If the represented state moves toward the declared reference, the EBU can be positive; if it moves away, it can be negative. Zero means that this particular potential did not change. It does not mean that no water moved, no energy was used, or no human work occurred.

This definition is deliberately connected to a physical event. A promise to deliver water is not the delivery. A payment for it is not the delivery. The account must identify the state before the action, the change that actually occurred, and the physical boundary within which that change is assessed. It is equally important to name what the account is measuring: a field built from specified carriers and functions, not a moral score for the person who acted.

8.2 The finite definition comes first

Let x^b be the complete represented state before an action and let $x^b + \delta$ be its complete successor under one fixed potential. Define

$$E = V(x^b) - V(x^b + \delta). \quad (8.1)$$

The minus sign makes a decrease in represented burden positive. No gradient approximation, motion law or actor payment is needed to define this finite comparison. The field, physical boundary and complete endpoints must be declared before the number has meaning.

8.3 Following the change along a path

Recall the bent ray from Chapter 5. Fermat's optical travel-time integral can depend on the spatial route. A fixed-potential EBU integral has a different structure: it integrates the differential of a supplied state function. Its role is to explain how changing local marginals add to the finite endpoint value already defined.

Choose an admissible piecewise C^1 path γ from x^b to $x^b + \delta$, and suppose V is C^1 on a neighbourhood of that path. The chain rule and fundamental theorem of calculus give

$$E = - \int_0^1 \nabla V(\gamma(s))^T \gamma'(s) ds. \quad (8.2)$$

Chapter 14 proves this identity carefully. A straight evaluation path need not be an observed physical trajectory; that interpretation requires a physical response model. The total is path independent under these assumptions because it is the difference of one single-valued potential.

One definition, a conditional path representation

Finite EBU is the before-minus-after potential difference. Under the regularity assumptions, a changing-field integral gives the same number. Neither expression grants physical permission, chooses a motion law or assigns ownership.

8.4 Where the rest of the physical process belongs

A water transfer may require pumping, filtration or another physical transformation. Those consequences do not become invisible merely because our first example follows water alone. If the pump consumes electricity, a fuller physical model records the energy carrier and its transformation. If water is lost, the lost amount enters a declared sink coordinate or crosses an explicit boundary. A potential defined on that fuller state can then evaluate all the consequences it is meant to value.

This is different from subtracting an unexplained “process cost” from the EBU equation. The general finite definition in this book remains

$$E = V(x^b) - V(x^b + \delta). \quad (8.3)$$

An engineering or institutional account may keep additional, separately named quantities. Such a quantity is not physical EBU by a change of

notation, and the same effect must not be counted twice. Our first water transfer is an intentionally lossless teaching world. Its simplicity lets us learn the equation; it does not assert that operating a real pump is free.

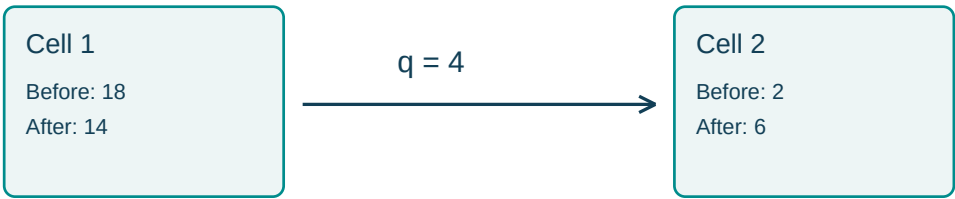
8.5 A four-unit transfer

A holds 18X, B holds 2X, and the declared reference is (10,10). Here X denotes one unit of the represented physical stock. Each location has scale 2X. The potential we will construct in Chapter 12 is

$$V(x) = \frac{1}{2} \left(\frac{x_1 - 10}{2} \right)^2 + \frac{1}{2} \left(\frac{x_2 - 10}{2} \right)^2. \quad (8.4)$$

Its value at (18,2) is 16. If four units are transferred from A to B, the state becomes (14,6) and the potential becomes 4. The exact EBU is therefore 12EBU. Four physical units arrived at B; twelve EBU express the decrease of this particular potential. They are different units describing different properties of the same event.

If eight units are transferred, the endpoint is the reference and the EBU is 16EBU. Transferring sixteen units instead reaches (2,18), with potential 16 once again, so its EBU is zero. The last movement was not empty: it simply overshoot far enough to cancel the initial gain in this account. At seventeen units the final potential is 20.25 and the EBU is -4.25 EBU. These calculations show why a fixed reward for each delivered stock unit cannot substitute for valuation along the full finite action.



Total stock: 20 before and after

Potential: 16 before, 4 after; field value: +12

Figure 8.1: A four-unit lossless transfer preserves twenty physical units while the declared potential falls from sixteen to four.

8.6 What would an actor receive EBU for?

Imagine two feasible actions from the same measured state. One restores a depleted supply without pushing another represented location into trouble. The other earns the same sales revenue while worsening the declared balance. A conventional invoice can look alike in both cases. The EBU account distinguishes their represented physical consequences. That distinction is the proposed incentive: useful restoration becomes visible and potentially rewardable, while depletion cannot be made to look like restoration merely by earning more money.

There is an important institutional step between a positive EBU calculation and a spendable personal balance. An institution would have to specify who may claim the action, how multiple contributors are treated, when a measured outcome becomes a receipt, and which future actions such a receipt could enable. A capacity design might allow demonstrated restorative value to support later physically permissible activity. That is a concrete use to investigate, not a theorem delivered by the integral. Later chapters keep a group result, an individual attribution and an institutional rule distinct.

The incentive also cannot be reduced to making every EBU number as large as possible. A well-functioning system may need continuing service and repair even when the represented state is already near its reference. An action may be essential despite a negative result in one field, and access to a clinic must not be made contingent on a person's ability to manufacture credits. The physical account can show a cost or benefit honestly; institutions must decide how essential provision, rights and obligations are protected. A proposed economic system has to be judged by its behaviour under demand and disturbance, not by the attractiveness of its arithmetic.

8.7 Permission and value are different questions

Before a transfer is accepted, the source must have enough usable stock, the route must exist and the declared physical constraints must hold. A positive hypothetical quote for an impossible transfer creates no water. A negative quote for an urgent, permitted service does not by itself forbid the service. Permission answers whether the physical change is allowed; EBU answers how the declared potential changes.

Likewise, neither the source's loss nor the destination's gain alone determines the group result when several actions occur together. The potential is evaluated for the actual combined state change. Its exact group total can be known without a justified division into personal claims. We will keep that distinction as the mathematics develops.

The immediate task is modest and powerful: build an account in which every symbol has a physical referent, every action has a boundary, and every signed value follows the whole declared finite change. We have named the unit and seen its consequence in one small world. The next chapter draws the world as cells and routes, so that a sentence such as four units reached B becomes an action we can test for physical possibility before calculating its value.

Cells, nodes, and edges

9.1 Why the map comes before the actor

A model of economic activity can begin by giving an actor a goal. The actor seeks a resource, maximizes a reward, or chooses a trading partner. Physical accounting must first establish a world in which these actions have a definite meaning. Where is the resource stored? Which transformations are available? Which places can exchange it? Otherwise a clever decision rule can deliver stock that was never withdrawn or use a route that the represented world does not contain.

The simplest useful map is a directed graph. A graph is a collection of places together with connections between them. A directed connection has an orientation. Here a place is called a cell or node, and a directed connection is called an edge. The graph describes available physical transformations; it does not prescribe which one an actor selects.

A cell need not be a square on a geographical grid. It may be one well-mixed tank, a warehouse, or a region for which a particular stock can be measured. Good boundaries keep important exchanges visible. Combining a reservoir and treatment plant into one cell may conceal a transformation between them. Dividing one well-mixed tank into a thousand cells may add notation without adding physical information.

9.2 Vocabulary before notation

Word	Meaning in the present model
Cell or node	One represented place carrying a scalar stock.
Index	A stable label used to identify a cell or an edge.
Edge	One declared directed transfer possibility.
Source	The cell from which an action withdraws stock.
Destination	The cell that receives stock.
Path	An ordered sequence of adjoining edges.
Topology	The pattern of available connections.
Support	The coordinates changed by a specified action.

An index is a label, not a size or a ranking. Writing $i = 1$ for A and $i = 2$ for B does not declare A more important. It fixes the order in which their numbers appear. That order must remain stable throughout a calculation and its record.

We write x_i for the stock at cell i . Thus $x_1 = 18$ and $x_2 = 2$ are two scalar statements. A scalar is one number with its declared unit. A vector collects several numbers in a fixed order.

9.3 A vector is an ordered physical record

The complete state of N cells is written

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \in \mathbb{R}^N. \quad (9.1)$$

The notation $\mathbb{R}^N$ means the set of ordered lists of N real numbers. Our physical domain is narrower: every stock is nonnegative, and their sum is the fixed total M . Writing the larger mathematical space is convenient for derivatives; it does not permit negative water in an executable state.

For A and B the state is $x = (18, 2)^T$. The superscript T means transpose. It turns a row into a column or a column into a row. It is neither a power nor a time label. A row multiplied by a matching column gives the dot product:

$$a^T b = \sum_{i=1}^N a_i b_i. \quad (9.2)$$

For $a = (2, 3)^T$ and $b = (5, 7)^T$, the result is $2 \cdot 5 + 3 \cdot 7 = 31$. Later, a marginal vector multiplied by a state change will add the local contributions to a change of potential. The operation remains this ordinary multiply-and-add procedure.

9.4 An edge is a directed possibility

Write an edge as $e = (i \rightarrow j)$. One unit sent on this lossless edge removes one stock unit from source i and adds one to destination j . The existence of $i \rightarrow j$ does not imply that $j \rightarrow i$ exists. A pipe may operate only in one direction. A permitted road route may differ from its reverse. If both directions are available, represent them as separate edges.

The graph also does not say how much will move. A road's existence does not fill a truck. The action contains a finite nonnegative quantity q_e , expressed in

stock units. Direction is carried by the edge; we do not silently use a negative quantity to create a reverse process that the graph has not declared.

Nor is an edge a universal relation of distance. Two distant tanks may have a direct pipe, while neighbouring tanks may have no connection. Distance can matter to delay, equipment, or an extended physical model. It does not enter this field through an assumed inverse-square law. The graph and the potential have explicitly defined jobs.

9.5 One column for one transfer

For edge $e = (i \rightarrow j)$, define the column S_e by

$$(S_e)_k = \begin{cases} -1, & k = i, \\ +1, & k = j, \\ 0, & \text{otherwise.} \end{cases} \quad (9.3)$$

Read the subscript outside the parentheses as “entry k of the column.” It does not multiply two symbols. The minus sign marks withdrawal, the plus sign marks delivery, and zero marks an unchanged coordinate.

The action’s state change is

$$\delta_e = S_e q_e. \quad (9.4)$$

For A to B, $S_e = (-1, 1)^T$. Sending four units gives $\delta_e = (-4, 4)^T$, so the successor is $(18, 2)^T + (-4, 4)^T = (14, 6)^T$. This is the calculation already performed in words. The column has made its signs reusable.

Read the column aloud

One lossless edge column says: withdraw one unit here, deliver one unit there, change nothing else directly. Multiplying by a finite quantity gives the physical increment. The column records where the change occurs; it does not choose the quantity or value the action.

9.6 From columns to a matrix

Place all edge columns side by side to form the incidence matrix,

$$S = [S_1 \ S_2 \ \cdots \ S_m]. \quad (9.5)$$

There is one row per cell and one column per edge. The letter m here counts graph edges; the symbol E is reserved elsewhere for an EBU value.

If $q = (q_1, \dots, q_m)^T$ collects finite quantities, then Sq is the sum of their state increments.

Consider a separately declared three-cell illustration: A is cell 1, a store is cell 2, and the clinic tank is cell 3. Edges run $1 \rightarrow 2$ and $2 \rightarrow 3$. Both are lossless. Then

$$S = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}. \quad (9.6)$$

For quantities $q = (4, 2)^T$, multiplication gives

$$Sq = \begin{bmatrix} -4 \\ 4 - 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ 2 \end{bmatrix}. \quad (9.7)$$

The store receives four and sends two, for a net gain of two. The net entries add to zero. This matrix product is physical bookkeeping, not a proof that the proposed actions are jointly feasible. If the store begins empty and the event rule forbids use of simultaneous incoming stock, its outgoing proposal still fails. Event timing is not determined by matrix notation.

9.7 A shared source has one stock

Now let A supply three destinations: the clinic, a food store, and a fire tank. In this new four-cell example,

$$S = \begin{bmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (9.8)$$

With quantities $(2, 1, 0.5)^T$, the state increment is $(-3.5, 2, 1, 0.5)^T$. Every source entry appears in the same row. The total withdrawal is 3.5, not three independent uses of the same stock.

This geometry matters twice. Physical feasibility must consider aggregate use of the source. Valuation must consider the combined change of any shared potential coordinate. Independently asking whether each action fits can miss a joint conflict. Independently valuing each action from the full initial state can also count the same improvement more than once. Those are distinct problems, and later chapters address them separately.

9.8 A path is not a single edge

A route from A through a depot to B is a sequence of edges. Its intermediate stock may matter. So may timing, physical availability, and the order in

which each action becomes possible. Combining the route into one arrow can be appropriate only if the model explicitly justifies the coarser transformation and retains every relevant boundary.

Suppose the depot begins with two units. A simultaneous event requests four units into the depot and three out. The endpoint has three units, so it is nonnegative. But the initial depot does not contain the three outgoing units. A model may allow synchronized passage through a real continuous pipe; another may require receipt before onward dispatch. Both can be specified. Neither follows from the positive endpoint alone.

The same caution applies more strongly to fuel, electricity, and delivered water. They are different carriers, and an energy conversion is not a lossless relocation of a homogeneous scalar. Their actions should form a linked physical chain with appropriate units and conversion accounts. The simple graph notation can teach how to record support, but it cannot erase the engineering of the chain.

9.9 Three maps that must stay distinct

The physical graph tells us which transformations exist. The potential's dependency structure tells us which coordinates each potential term reads. The action-interaction structure tells us which actions overlap in a way relevant to joint valuation or feasibility. These maps need not coincide.

In our separable potential, each local term reads one cell, so the potential has no cross-coordinate terms. Nevertheless two actions withdrawing from A both change A's local quadratic. They interact in the joint finite value. A diagonal potential therefore does not imply independent actions.

Conversely, two geographically close places may have no common action support. Their values can be independent within this model if no potential factor couples them. A richer model could add such a factor, but then the affected neighbourhood used in a local quote must include it. The general principle is simple: locate dependence from the actual equations, not from a picture or a metaphor.

9.10 Building a graph from a practical description

Begin with the carrier and its unit. Next choose cells whose stocks can be measured consistently. Name each transfer process and its direction. Assign stable row and column indices. For each lossless edge put -1 at its source, $+1$ at its destination, and zeros elsewhere. Keep a separate record of physical quantity limits, timing, and permitted information.

Do not place EBU values or reference stocks inside S . The incidence matrix does not depend on whether A currently has excess or deficit relative to its reference. It describes the change per unit of the named transfer. Stocks belong in x , references in x^* , scales in σ , and finite quantities in q . Separating these objects makes later changes auditable.

An edge record might state its identifier, endpoints, carrier, quantity unit, and physical validity conditions. A version identifier can freeze that declaration for an event. This helps prevent a route or parameter from changing halfway through an account. Versioning does not validate a bad physical definition; it makes the definition traceable.

9.11 Notation laboratory

For a four-cell transfer from cell 2 to cell 4, the column is $S_e = (0, -1, 0, 1)^T$. Its second entry is -1 , its fourth is $+1$, and its dot product with the all-ones vector is zero. A quantity $q_e = 3$ gives $(0, -3, 0, 3)^T$.

If there are N cells and m edges, the matrix has shape $N \times m$. The quantity vector has shape $m \times 1$, so the result has shape $N \times 1$. The inner dimensions match. This simple size check catches many transcription mistakes before any arithmetic is attempted.

For any cell i , the component formula is

$$(Sq)_i = \sum_{e: \text{dest}(e)=i} q_e - \sum_{e: \text{src}(e)=i} q_e. \quad (9.9)$$

It says to add all incoming quantities and subtract all outgoing quantities. A reader who finds the full matrix abstract can always return to this one-cell account. Matrix form is especially useful for proofs; component form is especially useful for tracing a real event.

9.12 Guided problem: build a returning route

A depot sends resource to a clinic and to a water plant. The clinic can return unused resource to the depot. In this illustration all transfers involve the same homogeneous carrier and are lossless. Index depot 1, clinic 2, plant 3, and edges $1 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 1$. The matrix is

$$S = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}. \quad (9.10)$$

For $q = (2, 1, 0.5)^T$, the change is $(-2.5, 1.5, 1)^T$. Its entries add to zero. The reverse edge is its own event component. It is not a negative number concealed in the first quantity.

Now suppose the starting state is $(5, 1, 0)^T$. The endpoint is $(2.5, 2.5, 1)^T$, with total six units before and after. The depot's outgoing total is three, and the clinic's outgoing quantity is one-half, each available from its initial stock. Under a source-availability rule excluding unreceived inflow, those particular source checks pass. Other physical constraints would still need their own declared checks.

9.13 Disconnected regions and unreachable references

The graph can impose conservation statements stronger than the global total. Suppose there are two disconnected regions and every edge begins and ends inside one region. Summing the incidence entries over either region gives zero for every edge. Each region therefore retains its own stock under the declared internal actions.

For a concrete four-cell illustration, let cells 1 and 2 form one connected pair and cells 3 and 4 another, with no edge between the pairs. Starting stocks are $(8, 2, 3, 7)$. The two regional totals are both ten. A reference $(5, 5, 5, 5)$ is compatible with these regional totals. A reference $(6, 6, 4, 4)$ has the same global total 20 but demands twelve units in the first region and eight in the second. It cannot be reached using only the declared internal edges.

This is an exact obstruction supplied by topology. No improvement in quote arithmetic removes it. A new cross-region route would change the physical graph and could remove the obstruction, but its construction is another physical and institutional question. The field should expose the remaining deviation without pretending that the actor has an unavailable action.

More generally, a cut is a partition of the cells into two sets. The only internal transfers that can change the total on one side are edges crossing that cut. Writing the sum of the component ledger on that side cancels every wholly internal edge. This simple cancellation is useful when auditing a region, even when its individual cells are numerous.

9.14 Renaming cells should not change the physics

If A and B exchange their numerical indices in a data file, every associated object must be reordered together: state, references, scales, and incidence rows. The physical result should be unchanged. A consistent permutation changes notation, not the model.

For example, the main state $(18, 2)$ and column $(-1, 1)$ become $(2, 18)$ and $(1, -1)$ when the two row labels are exchanged. The same four-unit transfer produces the reordered endpoint $(6, 14)$. This is the earlier endpoint $(14, 6)$

written in the new order. It must not be confused with physically reversing the action while keeping the original labels.

This distinction is elementary but important in large records. A matrix with correct numbers and the wrong identifier-to-row map can conserve total stock while moving it between the wrong places. Physical identity belongs in the record alongside arithmetic consistency. Stable identifiers make an account reproducible across software ordering choices.

9.15 Common questions and practice

Does S calculate the best route? No. It records effects of declared edge quantities. A route selector is another object.

Are all entries of a physical state allowed real numbers? The state is represented by real coordinates, but admissibility restricts them to the declared nonnegative fixed-mass domain and any further physical conditions.

Does a zero entry mean a place cannot be affected indirectly? It means this action column does not directly change that coordinate. Later actions or a richer potential can create other dependencies, which must be represented separately.

For practice, draw A connected to two destinations and add a return edge from one destination. Write all columns and compute their sum for quantities $(3, 2, 1)$. Your source increment should be -4 , the returning destination's net increment $+2$, and the other destination's increment $+2$. Explain how the result depends on the declared ordering of coordinates but not on the order in which the three columns are added.

Next remove the return edge. A proposal to return one unit is now unavailable even if it would improve the potential. This illustrates a central boundary: an algebraically attractive direction is not automatically an edge in the physical world.

What to carry forward

A cell is a state location. An edge is a directed physical possibility. Its incidence column records source withdrawal and destination delivery. The product Sq adds finite increments. Feasibility, timing, value, and action selection require their own declarations.

With the map established, we can audit quantities, units, and boundaries. The next chapter follows one physical change without letting a mathematical shorthand create or destroy stock.

Quantities, boundaries, and one physical change

10.1 The ledger underneath every valuation

Before asking whether an action improves the represented world, ask whether its physical account closes. A transfer cannot deliver more of the homogeneous resource than the declared source withdraws in a lossless model. A missing quantity cannot disappear because a potential decreases. An extra quantity cannot enter because an account has enough capacity to pay for it.

The main model has a finite number of nonnegative stocks and a fixed total M . It represents internal relocation of one homogeneous scalar. The physical state space is

$$\mathcal{X}_M = \left\{ x \in \mathbb{R}^N : x_i \geq 0, \quad \sum_i x_i = M \right\}. \quad (10.1)$$

This set is called a simplex. The unfamiliar name adds no new assumption: every coordinate is nonnegative and all coordinates share one total. In two cells with total 20, specifying A's stock determines B's stock as $20 - x_1$. The two numbers cannot vary independently while the total remains fixed.

10.2 Quantity is not rate

We use q for a finite source-side quantity. It has the same stock unit as x_i . If a flow rate j is constant over a duration Δt , the quantity is

$$q = \Delta t j. \quad (10.2)$$

The rate has units X/time; multiplying by duration gives X. Once the finite quantity has been formed, the action increment is $S_e q$. Multiplying it by Δt again would count time twice.

For example, a rate of eight units per hour acting for half an hour moves four units. The same quantity is moved by a rate of two units per hour for two hours, if the declared process is constant and no other event intervenes. The endpoint stock change is the same in this ideal example. Other physical

consequences, such as equipment heating or delay, may differ in a richer model and would need representation there.

If the rate varies, quantity is the integrated rate, $q = \int j(t) dt$. One cannot replace this integral by initial rate times duration without an assumption that makes that replacement valid. The finite account begins once the event's quantity has been established; it need not pretend every physical flow is constant.

Object	Unit	Example
Stock x_i	X	18 units in A
Rate j_e	X/hour	8 units per hour
Duration Δt	hour	one-half hour
Quantity q_e	X	4 units sent
Increment $S_e q_e$	X per coordinate	$(-4, 4)^T$

10.3 The exact internal-transfer identity

Let $\mathbf{1}$ be the column vector whose entries are all one. Its transpose sums a state: $\mathbf{1}^T x = \sum_i x_i$. Every lossless incidence column has one -1 and one $+1$, so

$$\mathbf{1}^T S_e = 0. \quad (10.3)$$

For a group of additive transfer increments $\delta_G = \sum_{a \in G} S_a q_a$, linearity gives $\mathbf{1}^T \delta_G = 0$. Hence

$$\mathbf{1}^T (x^b + \delta_G) = \mathbf{1}^T x^b = M. \quad (10.4)$$

This is an exact algebraic conservation identity under the declared transfer model. It does not require that the action lowers potential or that an actor acts wisely. A damaging relocation conserves the same total as a helpful one.

The identity also does not by itself ensure nonnegative stocks. A proposal to send 25 units from A's initial 18 has changes $(-25, 25)$, whose sum is zero, but its endpoint is $(-7, 27)$. Conservation and feasibility are separate conditions. The negative coordinate rejects the proposal under our physical domain.

10.4 Worked ledger for A and B

The starting state is $x^b = (18, 2)^T$. For $q = 4$, the source loses four and the destination gains four. The physical residual is

$$r_M = (14 + 6) - (18 + 2) = 0. \quad (10.5)$$

At $q = 8$, the state is $(10, 10)^T$, with the same zero residual. At $q = 17$, it is $(1, 19)^T$, again with zero residual. Yet their finite field values are different: 12, 16, and -4.25 , respectively, under the main potential.

This comparison is worth reading slowly. The physical total cannot determine the distributional potential. The potential cannot replace the stock ledger. They answer complementary questions about the same action. A programme that reports only one of them has left the other question unanswered.

10.5 The event boundary determines what is credited

The symbol x^b denotes the shared pre-action baseline. In an undisturbed illustrative event, it is simply the current state. In a later forcing-first model, an external event changes an earlier state x to x^b ; the actor then acts from x^b . Both sides of the actor's finite comparison use that same baseline.

Suppose an external conservative relocation moves the system from $(10, 10)$ to $(18, 2)$. Its total stock is still 20, but it has increased potential from zero to 16. An actor then transfers four back toward the reference, producing $(14, 6)$ and potential 4. The actor's field reduction is 12. The external increase of 16 belongs in the external account, not as an actor debit or credit created by relabelling time.

These are three declared states used in an algebraic illustration, not a simulated trajectory or scientific observation. Their purpose is to make the baseline visible. If an account instead compared the final potential 4 directly with the earlier reference potential zero and called the result the actor's effect, it would combine the external event and the action. The resulting -4 would answer another question.

10.6 No hidden creation by clipping

Suppose a proposed update produces $(-1, 21)$. Replacing -1 by zero may seem like harmless housekeeping. It changes total stock from 20 to 21. If the positive coordinate is also adjusted, a new redistribution has occurred. Either way, the operation is a physical or mathematical correction requiring an explicit account.

For the main model, infeasible actions are excluded before execution. We do not repair their successors silently. A broader model may include spill, saturation, external replenishment, or emergency corrections, but those processes must carry named increments or boundary entries. A state projection can be a useful numerical procedure; it is not automatically a law of physical conservation.

This principle is more general than water. An inventory programme that sets a negative medicine count to zero has not produced medicine, but it has changed its reported record. An account should expose the disagreement so that its cause can be investigated. Concealing it creates an apparently consistent state while destroying the information needed to explain the event.

10.7 What changes when a route loses usable stock

The central study model is lossless. To understand the boundary discipline, consider a separate accounting illustration with efficiency $\eta = 0.8$. A source sends $q = 5$, a destination receives four, and one unit enters a declared loss compartment ℓ . The extended increment is

$$(\Delta x_i, \Delta x_j, \Delta \ell) = (-5, 4, 1). \quad (10.6)$$

Its entries sum to zero. The usable-stock total falls by one, while the extended carrier account remains conserved. Alternatively, a model may record the one unit crossing an explicit external boundary. It may not leave the unit unnamed.

The loss coordinate need not enter the potential used in the lossless teaching model. A physical ledger and a valuation functional have different scopes. If a lossy world values the sink coordinate, its marginal enters the EBU calculation; if that coordinate is audit-only, its marginal contribution is zero. Neither choice authorizes an unexplained second charge under the name EBU. A richer valuation requires a newly declared potential over the relevant physical state.

A zero residual is necessary but incomplete

The numbers $(-5, 3, 2)$ also sum to zero, but they do not satisfy the declared $\eta = 0.8$ transfer rule. Check the source withdrawal, destination delivery, and loss relation separately. Conservation of a total cannot certify every component of a physical transformation.

This illustration does not change the lossless example used through the first calculation. It shows why a loss-aware calculation needs more than changing an efficiency coefficient in an edge formula. The physical transformation, sink or boundary, and the potential's scope must be specified first.

10.8 A homogeneous carrier is a strong simplification

A conserved scalar is not a claim that all useful things can be added in one physical unit. One litre of water and one kilowatt-hour of electricity cannot

become two units of a single conserved stock merely because both support a clinic. Their linked use requires separate carriers and a transformation relation with consistent dimensions.

The same applies to physical degradation. Energy can remain conserved while its usefulness changes. A stock of usable doses may fall when medicines expire although their material constituents remain. The carrier named by a coordinate must therefore be explicit: usable resource under a declared quality definition, elemental mass, energy, or another measured quantity. Its conservation statement applies to that declaration and boundary.

Our ideal transfers preserve the homogeneous scalar exactly. They are a clean setting for learning local finite accounting. Extending them to heterogeneous resource chains is scientific work, not a change of labels. It must preserve what is conserved, represent relevant changes of quality, and avoid an unlicensed conversion between different EBU accounts.

10.9 Net change and gross movement

Two opposite transfers can leave stocks unchanged while moving a substantial gross quantity. Suppose the two lossless directions between A and B are both declared, and an event sends three units each way. The increments $(-3, 3)$ and $(3, -3)$ add to zero. The endpoint is the initial state.

The stock ledger correctly reports zero net change. A separate action record correctly reports six units of gross source-side movement, three on each route. Neither description should erase the other. If the complete physical model includes process energy or equipment wear, those can change even when the represented stock increment cancels. The ideal cost-free scalar example records zero field value for the net event because its endpoints coincide; it does not make a claim about an actual pump's energy use.

This distinction prevents another kind of hidden correction. An auditor should not delete both action lines merely because their increments cancel. The lines identify what was attempted or executed, while the endpoint describes the resulting stock distribution. Physical feasibility may depend on gross use of a route or shared machine, so it too can differ between an empty event and a nonempty cancelling group.

10.10 Conservation under a change of accounting boundary

Imagine observing only B. A four-unit delivery raises its stock by four. Within this smaller boundary the delivery is an inflow. Now enlarge the boundary to include A. The same event is an internal transfer with zero total change. The physical event did not change; the boundary did.

Write this explicitly before comparing two ledgers. A local total can rise while the global total is conserved, because the local boundary has an incoming edge. A statement such as “stock increased” is incomplete until its region and carrier are named. EBU’s local potential difference can still be calculated, but a source contribution must not disappear just because the reader looks only at the destination’s tank.

For any region, add the stock changes of its cells. Every transfer with both endpoints inside contributes minus and plus the same quantity, cancelling internally. Edges entering the region contribute positively; edges leaving contribute negatively. This cancellation is the regional version of the all-ones-vector identity. It explains why enlarging a boundary changes which terms are called external without changing the complete physical account.

In a real study, changing the boundary can also change which consequences are represented and hence which potential terms are needed. It must therefore be recorded as a model revision, not as a way to make an inconvenient burden vanish from a report. A narrower account can be useful if its scope is stated honestly.

10.11 Measurement is not a new flow

Real gauges have finite resolution, calibration limitations, and timing uncertainty. Removing numerical error-controller material from the introductory model does not make these practical issues disappear. They belong in the interpretation of observed records and in later domain validation.

If a gauge reports 18 units, the underlying quantity may not be known with infinite precision. The algebraic example treats the input as exact so that its mathematical implications are visible. An empirical application must state how the input was obtained and what uncertainty accompanies it. It must not insert measurement uncertainty into the stock equation as though uncertainty were additional water.

A measured residual can reflect an omitted flow, incompatible timestamps, sensor bias, or computational rounding. A declaration of acceptable measurement agreement must precede judging a real record. Choosing a convenient allowance after observing an inconvenient mismatch would not establish conservation. The equations here specify the physical target against which such a measurement protocol would be assessed.

10.12 Guided problem: reconstruct a branching event

Consider a new three-cell state $(9, 3, 0)^T$, with lossless edges from cell 1 to cells 2 and 3. The finite quantities are two and four. The increments are $(-2, 2, 0)^T$ and $(-4, 0, 4)^T$. Their sum is $(-6, 2, 4)^T$, and the endpoint is $(3, 5, 4)^T$.

Check the ledger three ways. Cell by cell, source withdrawal is six and total delivery is six. By totals, the starting 12 equals the ending 12. By matrix structure, each column sums to zero, so every linear combination does too. The agreement is not three independent empirical tests; it is three views of one declared algebraic identity.

Now alter only the second quantity to eight. Aggregate withdrawal becomes ten, exceeding the source's nine. The formal endpoint is $(-1, 5, 8)^T$. The total remains 12, but the state is infeasible. Do not clip its first entry or report the eight-unit delivery as executed service. The example demonstrates why a joint source check belongs before an accepted event.

Finally convert rates correctly. If the valid quantities two and four were delivered over one-quarter hour at constant rates, the rates were eight and sixteen units per hour. If those same rates acted for half an hour, the quantities would double. A duration change can therefore alter physical feasibility even when the rates appear unchanged.

10.13 Practice with answers

A lossless edge carries a constant rate of three units per hour for two-thirds of an hour. The finite quantity is two units. Starting from $(7, 1)$, the endpoint is $(5, 3)$, and total stock remains eight. Explain why writing $x' = x + \Delta t S q$ with this already integrated $q = 2$ would be wrong.

In a separate extended loss ledger, a source sends ten units at $\eta = 0.9$. Destination delivery is nine and the loss-compartment increment is one. A record showing destination eight and loss two has zero total residual but violates the stated efficiency. Identify the missing component check.

For the main model, a no-action event has $q = 0$, so $\delta = 0$ and state is unchanged. If a real observer nevertheless sees a change, at least one premise of the no-action, no-external-change illustration does not describe the observation. The response is to locate the process or measurement issue, not to declare that zero caused a nonzero transition.

Declaring a reference and a scale

11.1 A comparison needs a declared centre

A tank reading of ten units is a physical fact within a measurement protocol. Whether it represents shortage, adequate provision, or excess cannot be read from the number alone. The same quantity can support a small clinic and be inadequate for a large hospital. A model that measures deviation must therefore declare what each stock is being compared with.

Write the reference for cell i as x_i^* . The star identifies a reference value; it is not multiplication and does not mean that the value has been proved optimal. The vector x^* collects all references in the same cell order as the state. In our main example it is $(10, 10)^T$.

The reference is a model declaration whose practical meaning requires domain justification. It might be informed by engineering function, an agreed service profile, or a research design. Algebra cannot decide which interpretation is appropriate. The important first step is to record that interpretation explicitly and keep it fixed during a comparison.

11.2 Reference, observation, and equilibrium

Three ideas are easily conflated. An observed mean summarizes data. A reference states the centre used by a model. A dynamic equilibrium is a state that remains unchanged under a particular evolution rule. These objects may coincide in a carefully justified application, but their symbols do not make them identical.

If a community's water distribution is persistently unequal, its historical mean distribution may reflect that inequality. Choosing a reference equal to that mean is a substantive decision. Conversely, choosing a more balanced reference does not prove that available routes or institutions can reach it. A reference makes the question precise; it does not answer every practical question associated with it.

For the Gaussian potential taught here, x^* is the unique zero of the full quadratic when all scales are positive. That mathematical minimum is a

statement about the chosen function. It does not prove that an actor process converges to the point, that a society should adopt it, or that all living functions are satisfied there.

Three sentences with different meanings

“The mean stock was ten” describes a dataset. “The reference is ten” defines a comparison. “Ten is an equilibrium” describes a specified dynamic law. Use the sentence supported by the available definition or evidence.

11.3 Compatibility with a conserved total

The first model preserves total stock M . We therefore choose references satisfying

$$x_i^* \geq 0, \quad \sum_i x_i^* = M. \quad (11.1)$$

This ensures that the reference lies in the same nonnegative fixed-mass domain as the physical states. In the main example, $10 + 10 = 20$. A reference of $(12, 12)$ would require 24 units, which a closed 20-unit world cannot supply.

The compatibility condition is simple but scientifically important. Without it, a positive minimum deviation may reflect an impossible reference rather than poor action choices. A model might intentionally study an incompatible target in another project, but it would need to explain that problem. We do not introduce it silently here.

Compatibility is weaker than reachability. If the only physical edge runs from B to A, then starting from $(18, 2)$ the direct transfer toward $(10, 10)$ is unavailable. If the graph is disconnected, each component preserves its own total under internal transfers. The reference must then also be compatible with those component totals to be reachable by those actions. Declaring a global total alone does not construct a route.

11.4 The scale tells us how to compare deviations

Let $\sigma_i > 0$ be a declared physical scale in the same units as x_i . The standardized deviation is

$$y_i = \frac{x_i - x_i^*}{\sigma_i}. \quad (11.2)$$

The numerator says how many stock units the cell differs from its reference. Dividing by the scale expresses that difference in units of the declared width. The result is dimensionless.

With reference ten and scale two, stock fourteen has standardized deviation two. Stock six has standardized deviation minus two. The signs distinguish above from below. Both are the same standardized distance from the reference. A quadratic potential will give them equal local burden because that symmetry is part of its definition.

The scale is not automatically a measurement error, tolerance, acceptable range, or statistical standard deviation. The symbol σ is familiar from probability, but notation is not evidence of a probability model. Here it controls the geometry of a declared potential. A later calibration could connect it to data; that connection would need to be stated and justified.

11.5 A scale is not a boundary

At $x_i = x_i^* + \sigma_i$, the standardized deviation is one. Nothing discontinuous happens there. There is no switch from safe to unsafe and no automatic prohibition. The potential and its derivative change smoothly through this point.

This distinguishes the Gaussian construction from a model that uses a flat acceptable interval with penalties outside it. The present field has one declared centre and positive curvature everywhere. Every nonzero local deviation contributes to its potential. Physical feasibility still includes nonnegative stock and any separately declared domain constraints; those are not manufactured by treating σ_i as a threshold.

If an application requires a hard storage capacity or a protected ecological condition, it must declare that physical or institutional rule separately. Calling a scale a safety certificate would give it a job its definition does not perform. Conversely, the lack of an introductory threshold term does not claim that real systems lack critical boundaries.

11.6 Worked standardization table

For one cell with $x^* = 10$ and $\sigma = 2$, the relevant values are

Stock x	Deviation $x - 10$	Standardized y	$\frac{1}{2}y^2$
2	-8	-4	8
6	-4	-2	2
8	-2	-1	0.5
10	0	0	0
12	2	1	0.5
14	4	2	2
18	8	4	8

Read the third column before the fourth. Standardization is a physical-unit conversion within the model; squaring and halving is the declared valuation shape. Separating these operations helps a reader locate where an assumption enters.

A deviation of four stock units is two scales, and its quadratic contribution is two. Doubling that physical deviation to eight makes the contribution eight, four times as large. This is a transparent consequence of the square. It is not an empirical theorem that doubling every real shortage multiplies its harm by four.

11.7 Different cells may have different scales

Consider a new illustrative reference (10,10) with scales (2,4). The main stocks remain (18,2). A is four scales above reference and B is two scales below it. Their contributions are eight and two, for total ten.

The field now assigns a smaller contribution to B's eight-unit deviation than before. No physical stock changed; the declared geometry changed. This example is a sensitivity calculation across two different models. It must not be inserted into one action record as though the actor caused the difference between potential 16 and potential 10.

Unequal scales can be useful when a stock deviation has different functional significance in different places. They can also conceal unjustified weights. A narrow scale steepens the field and makes changes at that cell count more strongly. Its provenance belongs in the model definition, not in an after-the-fact explanation of why a preferred action won.

11.8 Changing physical units must not change the answer

Suppose one stock unit represents one cubic metre, and we switch to litres. Every stock, reference, scale, and finite quantity is multiplied by 1000. The standardized deviation remains unchanged:

$$\frac{1000x_i - 1000x_i^*}{1000\sigma_i} = \frac{x_i - x_i^*}{\sigma_i}. \quad (11.3)$$

The potential and its finite difference are therefore unchanged. The marginal's numerical value changes because it is now per litre rather than per cubic metre. Multiplying it by a correspondingly converted quantity restores the same dimensionless product.

If an implementation changes stocks to litres but leaves σ numerically unchanged, it has changed the model, not just the unit. Its potential becomes

much steeper. This is a useful audit test because every piece of the declaration is visible. A physical unit conversion must transform all quantities with that unit consistently.

11.9 Why a reference must remain fixed during an action

An action comparison uses one potential before and after. If the reference or scale changes between the two evaluations, the result includes a model change. A zero physical increment can then appear to create value.

For example, take a cell with stock 14, reference 10, and scale 2. Its contribution is two. If an editor moves the reference to 14 without moving stock, the contribution becomes zero. The number fell by two, but no physical action supplied that reduction. The field definition changed.

References can be revised in a research programme or calibrated between declared epochs. Such revisions need an explicit version boundary and an account of the redefinition. They cannot be credited to an actor as though the same physical field had improved. The same principle applies to a scale revision or to adding and removing potential terms.

11.10 Changing cell granularity changes the declaration

Suppose one well-mixed tank has stock x , reference x^* , and scale σ . An editor divides it into two bookkeeping cells with equal half-stocks and half-references. It is tempting to copy the old scale into both cells and assume the sum of their potentials still means the same thing. It does not.

With copied scales, each deviation is half the original, and the two squared contributions sum to half the original potential. If each new scale is instead set to $\sigma/2$, each standardized deviation equals the old one, and their total potential doubles. Neither result is a physical consequence of moving water; it comes from changing the accounting representation.

If the halves are constrained always to remain equal and the aim is to preserve this particular aggregate quadratic exactly, choosing each scale as $\sigma/\sqrt{2}$ makes the two contributions sum to the original. That is a conditional algebraic mapping, not a universal rule for spatial refinement. Once the halves can vary independently, the finer model can represent an internal imbalance that the original one-cell state could not see.

The example shows why cell boundaries, references, and scales should be calibrated together. Merely increasing the number of cells is not a neutral way to gain apparent precision. A change of granularity may reveal useful

new physical structure, but it also changes the field's meaning and must be declared.

11.11 The reference is not a quota for each transaction

A ten-unit reference at B does not mean that every delivery should contain ten units. The reference is a state coordinate, whereas an action quantity is a change. If B already contains six, adding four reaches ten; if it contains nine, adding four reaches thirteen. The same delivery has different consequences at the two baselines.

Nor does a cell's reference assign ownership of that stock to a particular actor. The physical index identifies the represented place. Ownership and permission require their own maps and rules. Keeping those objects separate allows a potential to be studied without pretending that an equation has decided all institutional arrangements.

Finally, a reference at one time does not guarantee adequacy at every future time. A continuing-life application may have changing demands and carriers. Such extensions require a time-varying model or additional physical state. They cannot be simulated simply by shifting the reference after an outcome and reporting the resulting lower potential as earned improvement.

11.12 A reference document that a reader can audit

A useful declaration identifies the carrier, cell boundary, stock unit, reference value, scale, and provenance for each coordinate. It states the total-mass compatibility condition and whether the action graph can connect the relevant places. It also states the interpretation intended for the potential: a standardized model deviation rather than a complete measure of welfare.

For teaching, the provenance is openly illustrative. We choose (10,10) and (2,2) to make arithmetic readable. These are not estimated parameters or a registered scientific configuration. In an empirical application, the same fields would need evidence and review. The algebra does not change its standard of precision; the origin and adequacy of its inputs become additional questions.

The document should also distinguish fixed parameters from state variables. If carrying capacity, demand, or another functional characteristic is allowed to change physically, it may need its own represented coordinate. Quietly editing a reference whenever conditions change can make a poor process appear successful by moving its target after the fact.

11.13 Guided problem: asymmetric needs

For a separate two-cell exercise, declare reference (12, 8), scales (2, 2), total stock 20, and current state (18, 2). Both coordinates are six units from reference, or three scales. Potential is $\frac{1}{2}(3)^2 + \frac{1}{2}(-3)^2 = 9$.

A lossless six-unit transfer reaches (12, 8), the declared reference. An eight-unit transfer reaches (10, 10), leaving deviations (−2, 2), each one scale, and potential one. Equal raw stocks are therefore not the minimum of this changed model. The reference embodies different declared functional stocks at the two cells.

Do not conclude that either reference is ethically correct from the calculation. The example establishes the mathematical dependence of value on the declaration. An author who changes from equal to asymmetric references must explain the domain reason, rather than claim that the equations discovered it.

11.14 Guided problem: incompatible mass

Suppose total stock is 20 and someone proposes reference (12, 12). The reference lies outside the fixed-mass domain. With equal scales, the best balanced feasible point is (10, 10), but it still has a positive deviation from the proposed reference. In fact each coordinate differs by minus two and contributes one-half at scale two, for total one.

This is not a result showing that an actor failed to recover to a reachable reference. The reference itself demands more stock than the closed model contains. The correct first-model response is to require a compatible reference or explicitly change the physical boundary and problem. The example explains why compatibility is checked before controller performance is interpreted.

11.15 Misconceptions and practice

“Sigma is the noise level.” Only if a separate declaration connects it to measurement uncertainty. In this model it is a physical reference scale.

“One scale from the reference is a safe boundary.” No such boundary is implied. It is a location on a smooth quadratic.

“Gaussian means actual stocks follow a normal distribution.” The Gaussian shape motivates the potential geometry. An actual distribution requires a stochastic process and evidence or a mathematical derivation for that process.

“A compatible reference must be reachable.” Conservation compatibility removes one obstruction; graph structure and allowed actions may create others.

For practice, take reference 15, scale 3, and stocks 9, 15, and 21. Their standardized deviations are minus two, zero, and two; their local potentials are two, zero, and two. Now convert every physical quantity to a unit one-tenth as large. The numerical stocks and scale become ten times larger, while all three potentials remain unchanged.

For a three-cell reference (4, 6, 10) and total stock 20, check compatibility. Then imagine no edge connects the first two cells to the third, and the third begins with eight. The reference is globally compatible but unreachable by internal transfers within the two disconnected components. Explain the missing condition in words before attempting any controller design.

What to carry forward

The reference names the centre of comparison; the positive scale measures deviations in physical units. Neither is automatically an empirical mean, uncertainty bound, or equilibrium. Keep them fixed within an action account and compatible with the declared physical domain.

We can now assemble these declarations into a potential and see why a Gaussian landscape gives especially transparent finite mathematics.

Homeostasis as a Gaussian landscape

12.1 From comparison to a function

A function is a repeatable rule that maps an input to an output. Our input is a physical state. The output will be a nonnegative number measuring standardized deviation from the declared reference. Calling this number a potential gives it a mathematical role: its differences and derivatives can be evaluated consistently across actions.

For cell i , define

$$V_i(x_i) = \frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2, \quad \sigma_i > 0. \quad (12.1)$$

The global potential is

$$V(x) = \sum_{i=1}^N V_i(x_i). \quad (12.2)$$

The word global describes the argument of this mathematical function. It does not require every actor to read the whole state. A later locality identity shows exactly which terms a local action needs.

The factor one-half is a convenient normalization. It makes the derivative especially simple. Omitting it would create a consistently rescaled model if every dependent equation were changed, but mixing the two conventions would change numerical quotes. We retain the one-half throughout this edition.

12.2 Why the landscape is called Gaussian

Consider the unnormalized shape

$$g_i(x_i) = \exp \left[-\frac{1}{2} \left(\frac{x_i - x_i^*}{\sigma_i} \right)^2 \right]. \quad (12.3)$$

It has the familiar Gaussian bell shape, with its highest value at the reference. Taking the negative logarithm gives $-\log g_i(x_i) = V_i(x_i)$. The potential is therefore the relative negative-log geometry of that shape, with its minimum set to zero.

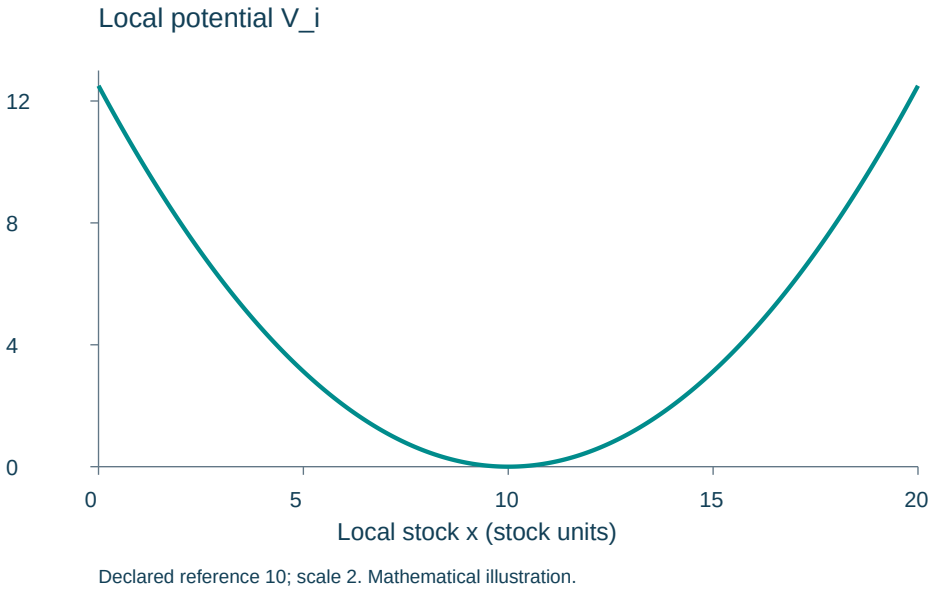


Figure 12.1: The one-coordinate quadratic penalty with reference ten and scale two. Its height is a declared deviation measure, not an amount of physical stock.

This construction is a definition, not a claim that nature universally follows a Gaussian law. No probability density has yet been fitted to stock observations. No random mechanism has been specified that makes the system's stationary distribution equal to g_i . The name explains the shape used for deviation; it does not supply missing dynamics or evidence.

On the fixed-mass nonnegative domain, coordinates are also constrained together. Even though the quadratic is a sum of one-coordinate terms, the actual stocks cannot vary independently: their sum is fixed. A product of Gaussian shapes in an unconstrained mathematical space is not automatically a product probability law for the physical system.

12.3 The bowl and its centre

Each local potential is a smooth quadratic bowl. It is zero at x_i^* , positive elsewhere, and symmetric about that centre. Moving one scale above or below the reference gives $V_i = 1/2$. Moving two scales gives $V_i = 2$. Moving four scales gives $V_i = 8$.

The symmetry is an explicit modelling choice. It does not assert that real deficit and excess have equal biological consequences. A clinic with too little water may face different problems from one with too much. The first Gaussian

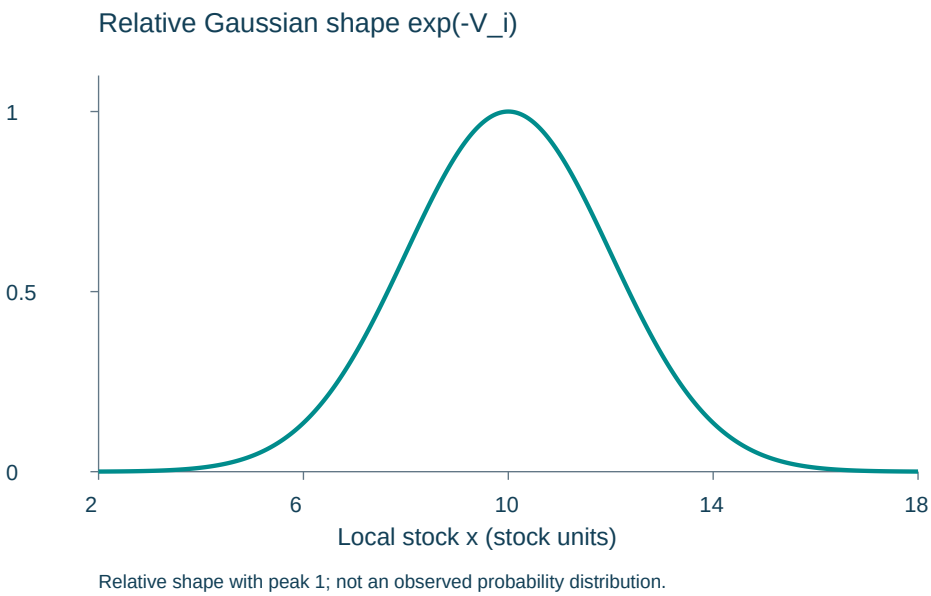


Figure 12.2: The exponential shape associated with the quadratic. This relative shape does not itself supply a probability model or prove that observations are normally distributed.

model isolates a transparent geometry for studying an accounting and action process. A later asymmetric or multi-coordinate field would need its own declaration and review.

Definition and immediate consequence

With fixed positive scales, $V(x) = \frac{1}{2} \sum_i ((x_i - x_i^*)/\sigma_i)^2 \geq 0$. Equality holds exactly when every represented coordinate equals its reference. This follows from a sum of nonnegative squares. It is not a statement that every unrepresented human or ecological need is satisfied.

12.4 Equal total, unequal potential

Return to the main world. At (18, 2), standardized deviations are (4, -4) and potential is 16. At (14, 6), they are (2, -2) and potential is 4. At (10, 10), both are zero. Each state contains the same total stock, 20.

The landscape distinguishes configurations of a conserved quantity. It does not create or destroy the quantity it evaluates. This is why a conserved physical ledger can coexist with a changing potential. The total-mass invariant

concerns a sum of stocks; the potential concerns squared departures from a reference.

The reflected state (2, 18) again has potential 16. It is different from (18, 2), although the scalar potential is equal. A single potential value does not reconstruct the state. If an account records only V , it loses the information about which place is above and which below its reference.

12.5 Reading the quadratic along the physical domain

For two cells with total 20, every physical state can be written $(r, 20 - r)$, with $0 \leq r \leq 20$. Under equal references ten and scales two,

$$V(r, 20 - r) = \frac{1}{2} \left(\frac{r - 10}{2} \right)^2 + \frac{1}{2} \left(\frac{10 - r}{2} \right)^2 \quad (12.4)$$

$$= \frac{(r - 10)^2}{4}. \quad (12.5)$$

This one-dimensional curve is the potential restricted to the physically allowed line. Its centre is $r = 10$. The two endpoints $r = 0$ and $r = 20$ each have potential 25. Every state in this particular closed world therefore has a finite potential between zero and 25.

The boundedness follows from the domain before an EBU-guided process is tested. A comparison between two action rules cannot claim success merely because both remain bounded in this finite simplex. Useful questions concern how often they occupy large-deviation states, whether they recover after forcing, whether they cycle, or whether affordability makes them refuse useful opportunities.

12.6 The marginal: a local slope

The local marginal is the derivative of the potential with respect to its stock:

$$\mu_i(x_i) = \frac{dV_i}{dx_i} = \frac{x_i - x_i^*}{\sigma_i^2}. \quad (12.6)$$

Below reference, the marginal is negative: adding a small amount lowers local potential. Above reference, it is positive: adding more raises local potential. At the reference it is zero.

Derive the formula rather than memorize it. Write $V_i = (x_i - x_i^*)^2 / (2\sigma_i^2)$. The reference and scale are fixed. Differentiating the square gives $2(x_i - x_i^*)$; the two cancels the one-half. The remaining denominator is σ_i^2 .

The potential is dimensionless in this model. The marginal therefore has units X^{-1} , or potential units per stock unit. It is neither a stock nor a complete

action value. Multiplying by a small stock increment gives the first-order potential change. A finite increment also encounters curvature, which we derive in Chapter 15.

12.7 A gentle derivative laboratory

For one cell with reference ten and scale two, $V(x) = (x - 10)^2/8$. Starting at 14, add a small increment h . Direct expansion gives

$$V(14 + h) - V(14) = h + \frac{h^2}{8}. \quad (12.7)$$

Divide by $h \neq 0$: the ratio is $1 + h/8$. As h approaches zero, the ratio approaches one. That limit is the marginal at 14. The formula also shows what the derivative leaves out for a finite change: the quadratic term $h^2/8$.

At stock six, the same calculation gives marginal minus one. Adding a small positive amount lowers potential. At ten, the marginal vanishes, but a finite nonzero displacement raises the potential by its square divided by eight. Zero initial slope is therefore not permission to move a finite distance without effect.

This is the bridge between an intuitive landscape and calculus. A slope is a nearby rate of change. It describes the tangent at one point. The finite EBU account compares two actual endpoints within the declared model.

12.8 The gradient and the Hessian

Collect the marginals into the gradient vector,

$$\mu(x) = \nabla V(x) = \left(\frac{x_1 - x_1^*}{\sigma_1^2}, \dots, \frac{x_N - x_N^*}{\sigma_N^2} \right)^T. \quad (12.8)$$

The symbol ∇ , read “nabla,” denotes the vector of partial derivatives. A partial derivative varies one coordinate while holding the others fixed in the mathematical function. Physically allowed directions can then be built from combinations that respect conservation.

The Hessian records derivatives of those slopes:

$$H = \text{diag} \left(\frac{1}{\sigma_1^2}, \dots, \frac{1}{\sigma_N^2} \right). \quad (12.9)$$

“Diagonal” means the off-diagonal entries are zero. Each local slope changes at the constant rate $1/\sigma_i^2$ with its own stock. The Hessian is positive definite because all scales are positive. For any nonzero vector d , $d^T H d = \sum_i d_i^2 / \sigma_i^2 > 0$.

The matrix form of the potential is

$$V(x) = \frac{1}{2}(x - x^*)^T H(x - x^*). \quad (12.10)$$

It compresses the sum of standardized squares. It does not replace their physical interpretation. The row-vector, matrix, and column-vector multiplication expands back into the same local contributions.

12.9 Ellipses, distance, and unequal scales

In an unconstrained two-coordinate drawing, points with a fixed potential satisfy

$$\frac{(x_1 - x_1^*)^2}{\sigma_1^2} + \frac{(x_2 - x_2^*)^2}{\sigma_2^2} = 2V. \quad (12.11)$$

These curves are ellipses. With equal scales they are circles in the chosen physical coordinate units. A larger scale stretches the contour along its coordinate: a larger physical deviation is needed to produce the same contribution. This is the geometric meaning of scale weighting.

The physical fixed-mass states occupy only a slice of this drawing. For two cells they lie on a line of constant sum. The field evaluated along that line is the restriction of the same quadratic, not a second potential. A picture of a circular bowl should therefore not suggest that every direction in the plane is a permitted physical motion.

The quantity $\sqrt{2V}$ is the Euclidean length of the standardized deviation vector. It can be described as a distance in these standardized coordinates. It is not geographical distance, and it is not itself the signed action value. Taking its square root changes finite differences and derivatives. We retain V as the declared potential even when the distance picture helps intuition.

12.10 Diagonal geometry does not mean independent outcomes

The absence of off-diagonal Hessian entries says that the declared local marginal of one coordinate does not directly depend on another coordinate in the unrestricted function. Physical conservation still couples stocks, and the graph couples possible actions. If A sends to B, both coordinates change together.

Two actions that share A can therefore interact in their combined value. Their increments overlap in A's squared deviation. This does not require a correlated multivariate Gaussian or a non-separable potential. It follows from applying a separable nonlinear function to a sum of increments.

A richer potential might use a symmetric positive-definite matrix with off-diagonal entries to represent declared cross-coordinate dependence. That would be another model. Its affected-factor neighbourhood, calibration, and interpretation would require review. We do not introduce it merely to obtain an attractive interaction effect. The diagonal model already has enough structure to distinguish local potential geometry from action overlap.

12.11 A bounded domain gives a bound before any experiment

Because nonnegative stocks sum to M , every coordinate lies between zero and M . Its absolute deviation from a fixed reference cannot exceed the larger of x_i^* and $|M - x_i^*|$. Hence

$$V(x) \leq \frac{1}{2} \sum_i \frac{\max\{(x_i^*)^2, (M - x_i^*)^2\}}{\sigma_i^2}. \quad (12.12)$$

For finitely many cells and fixed positive scales, the right-hand side is finite. It is a convenient bound, not necessarily the tight maximum because each coordinate cannot attain its individual extreme simultaneously under the common total.

This distinction matters to scientific interpretation. If a future comparison asks whether EBU prevents unbounded growth of this potential, the physical domain has already answered that question for every admissible policy. A meaningful study must ask something that the domain has not guaranteed: recovery after a shock, frequency of substantial deviation, distribution across cells, cycling, or refusal. The theorem is useful because it narrows the question honestly.

12.12 Convexity without mystery

A quadratic bowl with positive curvature is convex. Geometrically, the straight chord joining two points on its graph lies on or above the graph. Algebraically, the tangent at a starting point underestimates the potential at a finite displaced point by a nonnegative quadratic amount.

This will matter for valuation. If the initial slope predicts an improvement, multiplying that slope by a large action quantity overstates the finite field improvement. The potential curves upward relative to its tangent. Exact endpoint comparison automatically includes that curvature.

Convexity is a property of the potential, not a promise about action choices. A random actor can choose a direction that raises a convex potential. A capacity rule can permit cycles. A graph can block a path toward the minimum. A

simple bowl is mathematically helpful precisely because these separate mechanisms remain visible; the shape does not decide them by itself.

12.13 Why the potential is not a controller

Imagine drawing this landscape on paper and placing a marker at (18,2). The drawing tells us the value at every admissible state. It does not move the marker. To move it, one must specify a physical action and an actor or dynamic rule that chooses that action.

A gradient-flow law would be one possible added rule. A largest-value selector would be another. An earlier research proposal asked what happens under a separately declared random choice among feasible and affordable options; that proposal is not the governing definition or an active study authorization in this edition. These mechanisms can use the same potential while producing different behaviour. This book does not introduce an optimizing or restoring law just because the picture resembles a bowl.

Even the phrase “downhill” should be read carefully. It identifies a decrease of the declared potential along a direction. It does not make the system slide there by itself. The distinction protects the later experiment from becoming a demonstration of a recovery rule placed into the plant in advance.

12.14 Do not replace the function while keeping its name

The bounded function $1 - \exp(-V)$ can be formed from the quadratic. It is not the same potential. Its derivative multiplies the quadratic gradient by $\exp(-V)$, and its finite differences give different values. Large deviations become compressed toward one.

Such a transformation might be studied under another declared model, but it cannot be substituted merely because both expressions use a Gaussian exponential. The active definition is the quadratic negative-log potential itself. Field, marginal, curvature, and finite account must all be derived from that same definition.

Likewise, dropping a coordinate, adding a weight, or changing a scale changes the functional. Precise notation prevents an appealing diagram from concealing a substantive amendment. Every quantitative claim in the main example traces back to Equations 12.1 and 12.2.

12.15 Guided problem: three cells

In a new three-cell illustration, choose reference $(4, 6, 10)$, scales $(2, 2, 4)$, and state $(2, 8, 10)$. Total stock and reference total are both 20. Standardized deviations are $(-1, 1, 0)$, so $V = 1$.

The marginals are $(-1/2, 1/2, 0)$. Moving a small lossless quantity from cell 2 to cell 1 has direction $(1, -1, 0)^T$. Its initial potential derivative is $(-1/2)(1) + (1/2)(-1) = -1$, so the direction initially lowers potential. A two-unit transfer reaches the reference, giving exact field reduction one. Initial slope times quantity would predict two, twice the available reduction.

This calculation uses a new reference and scale declaration, identified openly. It reinforces the same mathematical structure without quietly changing the parameters of A and B. The reader can check each coordinate independently before adding the total.

12.16 One potential value, several physical descriptions

The scalar potential deliberately compresses a state vector. That compression is useful for finite identities, but it loses information. In the symmetric two-cell fixture, the same value four occurs at $(14, 6)$ and $(6, 14)$. These states place the shortage at different locations. A clinic at B experiences them differently even though the total standardized squared deviation is equal.

In a larger graph many more states can share one value. They can differ in which cell is far from reference, whether an available edge points toward relief, and how much stock a prospective actor can access. Equal potential does not imply equal feasible menus, equal service, or equal future behaviour.

For this reason, a future study should retain the state and relevant distributional diagnostics alongside V . The finite account uses the scalar because its changes have a precise meaning. Interpretation should not discard the coordinates that gave it that meaning. A contour of equal potential is a mathematical level set, not a class of socially or physically interchangeable worlds.

12.17 Practice and reflection

With reference ten and scale two, calculate local potential and marginal at stocks 4, 10, and 16. The potentials are 4.5, 0, 4.5; the marginals are $-1.5, 0, 1.5$. Explain why a nonnegative potential can have a negative derivative.

For the main two-cell state, verify $\mu = (2, -2)^T$ and $H = \text{diag}(1/4, 1/4)$. Then explain why equal total stock does not imply equal potential, and why

equal potential does not imply equal state. Use $(18, 2)$, $(10, 10)$, and $(2, 18)$ to support the answers.

Finally, write one statement that this potential establishes and one statement it leaves open. A valid pair is: “Its value is zero exactly at the declared reference” and “The proposed actor process may or may not spend much time near that reference.” The first follows from the definition; the second needs a specified process and further analysis.

From a marginal to an edge field

13.1 What the word field means here

A map of local potential values is not yet an action account. To understand a proposed transfer, we need to know how the potential changes in the direction of that transfer. The marginal at each cell tells us how its local contribution responds to a small addition. The edge field combines the source and destination responses with the physical signs of the action.

In this book, the edge field is a directional derivative of a declared potential. It is sometimes called a force by analogy with potential-based physics. Its unit here is potential per stock, not newtons. We derive its formula from the action column and the chain rule; no gravitational or electromagnetic law is assumed.

This careful use of the word field matters. It gives us a useful mathematical structure without claiming that the potential is a newly discovered physical interaction. A stock distribution and a declared Gaussian reference determine the values. An actor or physical law still has to specify what happens next.

13.2 The direction of one lossless transfer

For an edge $e = (i \rightarrow j)$, the state after a finite quantity q is $x^b + S_e q$. Its source coordinate is $x_i^b - q$; its destination coordinate is $x_j^b + q$. To find the initial response, imagine changing q by a very small amount while holding the initial baseline fixed.

The source changes at minus one stock unit per unit sent, and the destination changes at plus one. The chain rule gives

$$\left. \frac{d}{dq} V(x^b + S_e q) \right|_{q=0} = -\mu_i(x_i^b) + \mu_j(x_j^b). \quad (13.1)$$

Define the edge field with the opposite sign, so a burden-reducing direction has a positive field:

$$f_e(x^b) = \mu_i(x_i^b) - \mu_j(x_j^b) = -\mu(x^b)^T S_e. \quad (13.2)$$

This sign convention is the bridge to signed EBU. The potential change is negative when a small action improves the field. The corresponding field improvement is positive.

13.3 Deriving the equation one term at a time

The source contribution to the derivative is $\mu_i dx_i/dq$. Because $dx_i/dq = -1$, this is $-\mu_i$. The destination contribution is $\mu_j dx_j/dq = +\mu_j$. Every unaffected coordinate has derivative zero and contributes nothing. Adding these terms gives the potential derivative; negating it gives f_e .

At the main state (18, 2), the marginals are 2 and -2 . Removing a small amount from A lowers its local potential, because A is above reference. Adding that amount to B also lowers its local potential, because B is below reference. Thus

$$f_e = 2 - (-2) = 4. \quad (13.3)$$

The two beneficial endpoint effects add. An infinitesimal sent quantity dq changes potential to first order by $-4dq$.

The field does not mean that every finite quantity earns $4q$. It is the starting slope. As A loses stock and B gains it, both marginals move toward zero. The exact finite calculation must include that change.

A small quantity and its changing slope

Starting at (18, 2), sending $q = 0.1$ gives exact field reduction $4(0.1) - (0.1)^2/4 = 0.3975$. The initial-slope estimate is 0.4. Their difference, 0.0025, comes from curvature. It is a known algebraic term in this model, not an unexplained numerical discrepancy.

13.4 Four endpoint patterns

The signs of the endpoint marginals help interpret the direction, but the difference determines the result. An above-reference source has positive marginal. A below-reference destination has negative marginal. Together they give positive field. If both are below reference, removing source stock is locally harmful while adding destination stock is locally helpful; their magnitudes compete.

If both are above reference, withdrawal helps the source while delivery worsens the destination. Again, the field compares the two effects. A below-reference source sending to an above-reference destination has negative field because both endpoint contributions raise burden.

Source	Destination	Initial effect
Above reference	Below reference	Both changes lower local burden.
Below reference	Below reference	Source harm competes with destination relief.
Above reference	Above reference	Source relief competes with destination harm.
Below reference	Above reference	Both changes raise local burden.

These statements concern the declared potential. They do not decide whether the transfer serves an urgent need or violates another constraint. An application can require additional measurements of service and feasibility.

13.5 Raw stock is not the field

Two cells can contain the same stock and have different marginals because their references or scales differ. In a separate two-cell illustration, let both stocks be eight, references be ten and six, and scales both two. The marginals are $-1/2$ and $+1/2$. Sending from the second cell to the first initially lowers potential, although the raw stocks are equal.

The reference totals and stock totals in this example are both 16, so the comparison is compatible with the closed domain. The field expresses deviation relative to function as declared by the model. It is not simply a pressure to equalize every raw number.

Unequal scales provide another distinction. Keep the main stocks and references but change scales to (2, 4) for this sensitivity illustration. Marginals become 2 and $-1/2$, so the field is 2.5, rather than four. The physical action column is unchanged. The model’s valuation geometry has changed, and the record must identify that new declaration.

13.6 Field along a finite path

For the Gaussian potential, the marginal changes linearly with stock. Along $x^b + S_e q$,

$$\mu_i(q) = \frac{x_i^b - x_i^* - q}{\sigma_i^2}, \tag{13.4}$$

$$\mu_j(q) = \frac{x_j^b - x_j^* + q}{\sigma_j^2}. \tag{13.5}$$

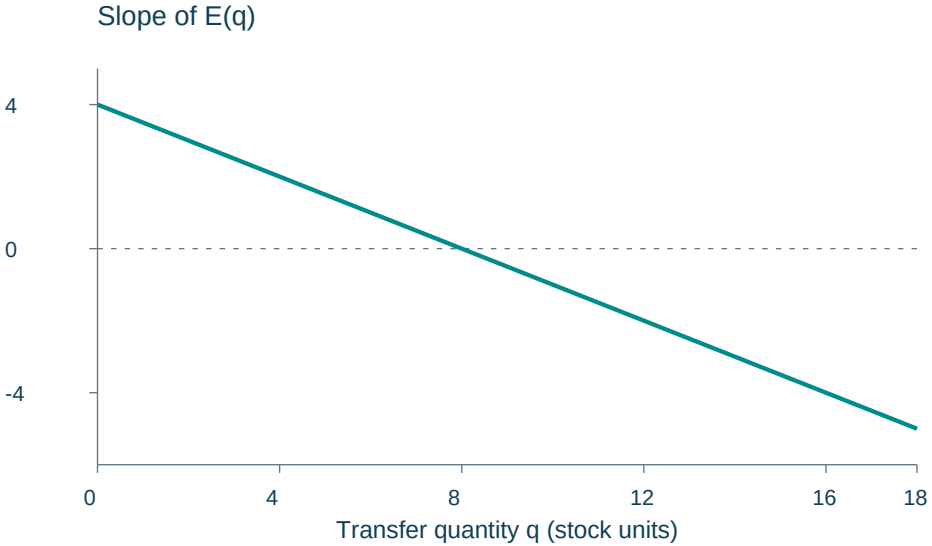
Their difference is

$$f_e(q) = f_e(0) - q \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (13.6)$$

The slope of the field along the action is negative because both positive curvature contributions add. In the main example this becomes $f_e(q) = 4 - q/2$.

At $q = 0$, the field is four. At $q = 4$, it is two. At $q = 8$, it is zero. Beyond eight, it is negative. These are field values at points along the same declared path, not different schedules chosen after the result. The whole function follows from the initial state, reference, scales, and action direction.

The quantity eight reaches the reference for both cells in the symmetric fixture. That coincidence is special to the compatible two-cell setup. In a larger system, equality of marginals on one edge does not imply that every coordinate equals its reference.



The slope crosses zero at $q=8$; it does not select an action.

Figure 13.1: The instantaneous field along the running transfer falls as the quantity increases. Integrating it, rather than multiplying its starting value by every finite quantity, gives the exact finite difference.

13.7 Zero field deserves careful reading

A zero edge field says the initial directional derivative is zero. It does not say the direction has zero finite cost. In the Gaussian model the curvature along every nonzero transfer direction is positive. Therefore a finite move from a

point with $f_e = 0$ increases the potential, unless the net physical increment is zero.

At the reference (10,10), both marginals vanish. Sending two units gives (8,12). Each endpoint is one scale from reference and contributes one-half, so total potential rises to one. The initial field was zero; exact finite EBU is minus one. The difference is fully explained by the quadratic term.

A zero field can also arise from equal nonzero endpoint marginals in a larger or differently constrained world. It means there is no first-order improvement along that edge. It does not certify that the complete represented world is at its minimum or that its real needs are satisfied.

13.8 The graph form of the field

Collect the edge fields into a vector f . With the incidence matrix S , the compact expression is

$$f = -S^T \mu. \quad (13.7)$$

The transpose turns each edge column into a row that selects the source and destination marginals. If there are N cells and m edges, S^T has shape $m \times N$, so the result has one field value per edge.

For a separately declared three-cell chain with marginals $(2, 0, -1)^T$ and edges $1 \rightarrow 2$, $2 \rightarrow 3$, the fields are two and one. The field vector reports the two directional slopes. It does not choose the first edge, allocate stock, or decide whether both actions can occur together.

This is an example of local computation with global notation. The formula is compact for a proof, but each edge needs only the relevant endpoint marginals under the separable model. A manager may assemble the whole vector for diagnostics; that does not authorize an actor to read a global score or a future outcome.

13.9 Reverse directions and cycles

If the reverse lossless edge exists, its column is $-S_e$ and its field at the same state is $-f_e$. At the main baseline, A-to-B has field four and B-to-A has field minus four. At a later state their values must be reevaluated. A reverse action performed after a forward action does not use the old reverse marginal.

Around a directed cycle of lossless edges evaluated at one common state, marginal differences cancel. For a three-cell cycle,

$$(\mu_1 - \mu_2) + (\mu_2 - \mu_3) + (\mu_3 - \mu_1) = 0. \quad (13.8)$$

This is an identity for initial fields at that state. It does not claim that arbitrary unequal finite quantities around the cycle have zero value, nor that a real cycle uses no energy. The complete physical state, including any affected energy carrier or sink, determines the finite account.

If equal simultaneous quantities circulate around a lossless closed cycle, the net stock increment can be zero. The field component is then zero because the endpoint equals the baseline. Actual circulation may use energy, time, or equipment in a fuller model. Those carriers must be represented before claims about their burden are made.

13.10 The field around a shared source

Consider a separate three-cell fixture with state $(14, 3, 3)$, reference $(8, 6, 6)$, and equal scales two. Both state and reference sum to twenty. The marginals are $(1.5, -0.75, -0.75)$. The source-to-each-destination field is therefore 2.25.

The equality of the two starting fields says the destinations are symmetric under this declaration. It does not allocate a source quantity to either actor. If one actor sends two units, the source marginal falls by one-half and its destination marginal rises by one-half. Recomputing the other edge at the resulting state gives a smaller source contribution even though its own destination has not changed.

This is why several actors cannot each treat a shared initial field as an unlimited personal resource. Their actions use one physical coordinate. The correct joint finite value includes that shared curvature. A simultaneous attribution convention can divide a joint value under declared path semantics, but it cannot create several independent copies of the source's starting state.

An experiment should therefore distinguish equal local fields, equal finite quotes, equal physical permissions, and equal selection probabilities. They coincide only under additional symmetry conditions. For example, unequal quantity menus can produce unequal group frequencies even when endpoint parameters are identical. These are protocol properties, not new forces in the potential.

13.11 A field can be useful even when it is not followed

A thermometer remains informative when nobody turns the heater on. Similarly, the derivative of a potential can diagnose the initial effect of a candidate action even when the actor chooses differently. The value's explanatory role does not require an automatic law of obedience.

This makes negative outcomes interpretable. If a permitted and affordable random action moves uphill, the field account can record the increase without treating it as a malfunction of the equation. The question for later research is how the complete policy changes the distribution of such events over time. That question needs the actor process, forcing, and account rules in addition to the local derivative.

13.12 An explicitly separate loss-aware boundary illustration

For a declared lossy transfer, the source changes by $-q$ and the usable destination by $+\eta q$. If the local potential covers those two usable-stock coordinates, the chain rule would give $f = \mu_i - \eta\mu_j$. If a loss coordinate enters the potential, its derivative contribution must also be included. The general formula is always minus the gradient dotted with the complete action direction.

For example, with $\mu_i = -1$, $\mu_j = -3$, and $\eta = 1/4$, the two-coordinate expression is $-1 + 3/4 = -1/4$. Source harm exceeds delivered relief. A lossless expression would give two and answer the wrong physical question. This illustration changes the transformation, so it is outside the main lossless domain and does not authorize an implemented lossy study.

The lesson is structural. Derive the field from the actual declared state change. Do not derive a convenient lossless slope and then add an arbitrary adjustment elsewhere to compensate. Physics determines the directional derivative's coefficients.

13.13 Where the Onsager connection comes from

Lars Onsager's two 1931 papers studied coupled irreversible processes, including heat conduction, electrical transport and diffusion. Their reciprocal relations connect response coefficients under microscopic-reversibility assumptions. The theory concerns thermodynamic forces and fluxes in a linear-response setting; it is not a general statement that every system follows a gradient of any function one chooses [19, 20].

The useful idea for EBU is to distinguish a driving difference from the strength of response. Think of two otherwise similar routes with the same endpoint marginal difference. One route can carry a large flow; the other responds slowly. The field difference and the route's responsiveness describe different things. This is why the original EBU architecture introduced an Onsager-type force–mobility–flux structure rather than treating the marginal itself as a transported quantity.

13.14 Force, mobility and flux in the EBU model

For a lossless edge, the already derived field force is $f_e = \mu_i - \mu_j$. The earlier EBU controller used

$$J_e = M_e[f_e - \theta_e]_+, \quad M_e > 0, \quad \theta_e \geq 0. \quad (13.9)$$

Here J_e is a flow rate and M_e is mobility: the declared response per unit force. The positive-part operation $[a]_+ = \max(a, 0)$ permits flow only in the declared edge direction and only above the activation threshold θ_e . This edge activation threshold is not an Allee threshold or a lower/upper homeostatic stock band. It belongs to this optional controller, not to the Gaussian potential.

In the dimensionless Gaussian normalization, force has units X^{-1} , mobility has units X^2/time , and their product has units of stock per time. A finite duration gives quantity $q = \Delta t J_e$. Physical permission still has to check that quantity; high mobility cannot create stock that the source does not hold.

At the main baseline the force is four. As a separate controller illustration, choose mobility one-half and activation threshold zero. The requested rate is two stock units per time unit. Over half a time unit it requests one stock unit. The exact finite field value of that quantity is $15/4$, not the initial-force estimate four. Mobility sets the response rate; the finite equation values the complete accepted change. These illustrative choices are not a registered experimental configuration.

13.15 Why borrow this structure, and where does it stop?

The structure makes a conditional descent argument possible. If there is no external drive and the physical evolution is exactly $\dot{x} = SJ$, then the chain rule gives $\dot{V} = -\sum_e f_e J_e$. For the declared threshold-mobility response, every active product is nonnegative, so that particular continuous-time law cannot increase the potential. Chapter 33 derives this carefully and explains the extra conditions needed for finite timesteps.

This is a reason to study the controller, not evidence that economic actors obey it. The EBU potential is not automatically thermodynamic free energy, and its decrease is not automatically measured entropy production. Nor does using diagonal mobilities demonstrate Onsager reciprocity for a real coupled social or ecological system. The original physical analogy motivates a mathematical architecture; the declared model still has to stand on its own.

13.16 Why the actor law remains an explicit choice

It would be possible to propose a flow proportional to positive field, or to choose whichever edge has the largest field. Either choice would add a con-

troller. It might be useful for a different research question, but it would not follow from Equation 13.2 alone.

A separately declared prospective actor model can keep the valuation rule distinct from the actor's choice. Feasible candidate actions can receive positive, zero, or negative values. A separately declared capacity policy can filter affordability. Random selection among remaining options is another declared step. Replacing that selection by largest field or largest finite EBU would change the experiment's subject.

This distinction also prevents an easy but circular success claim. If a plant is given autonomous restoring flow, observing movement toward the reference does not by itself demonstrate an economic accounting mechanism's contribution. The field may inform analysis without being a hidden force that physically compels the actors.

13.17 Guided problem: equal field, unequal finite behaviour

Two separate hypothetical edges have the same initial field four. Edge A connects cells with scales two and two, so its curvature along quantity is $1/4 + 1/4 = 1/2$. Edge B connects cells with scales one and one, so its curvature is two. Their initial tangents match, but their fields fall at different rates as quantity increases.

For a quantity of two, the exact field reductions, derived fully next chapter, are

$$F_A(2) = 4(2) - \frac{1}{2}(1/2)(2)^2 = 7, \quad (13.10)$$

$$F_B(2) = 4(2) - \frac{1}{2}(2)(2)^2 = 4. \quad (13.11)$$

Both calculations assume compatible declared states that give initial field four and permit the quantity. The example isolates curvature; it is not a recommendation between two real services. It shows why the starting field cannot replace a finite schedule.

Now ask what would happen at quantity four. Edge A gives $16 - 4 = 12$, while B gives $16 - 16 = 0$. The same initial field is consistent with very different finite values. A complete quote requires the potential along the action direction, not merely one slope sample.

13.18 A sign routine for every new example

When a formula feels abstract, first state the physical change: the source loses and the destination gains. Next ask how each change affects its local potential. Translate those answers into derivative signs. Only then combine the terms.

For an above-reference source, $\mu_i > 0$, so its negative stock increment contributes a negative potential change. For a below-reference destination, $\mu_j < 0$, so its positive stock increment also contributes a negative change. The negative of their sum is positive field. This routine is slower than memorizing a sign, but it is much harder to misuse when the carrier or action direction changes.

The routine also exposes ambiguous action descriptions. If one cannot say which coordinate decreases, it is too early to calculate the field. A transformation involving several inputs and outputs needs a correspondingly complete column before the dot product has physical meaning.

13.19 Practice with solutions

In the main model, calculate the field after sending four units from A to B. The endpoint is (14, 6), marginals are (1, -1), and the field is two. After a total of eight units, both marginals and the field are zero. After a total of twelve, marginals are (-1, 1), and the field is minus two.

For a separate compatible state (9, 7), reference (10, 6), and equal scales two, marginals are (-1/4, 1/4). The field from the second cell to the first is one-half. A transfer in that direction initially improves the potential even though the sending cell has the smaller raw stock.

Finally explain why a reported field of $4X^{-1}$ cannot be entered as four stock units in a physical ledger. Its unit is different and its meaning is a derivative. To form a first-order field change one multiplies by quantity; to form exact finite EBU one evaluates the complete finite difference under the declared physical state and boundary.

What to carry forward

The edge field is the negative directional derivative of the declared potential. For a lossless transfer it is $\mu_i - \mu_j$. It combines endpoint slopes, changes along finite motion, and supplies no actor policy by itself. Exact finite value requires the curvature that the starting slope omits.

Why the whole path has one endpoint value

14.1 Many small changes, one complete comparison

A pipe can transfer water steadily or in pulses. During the movement the local marginal changes: the source becomes less full and the destination less empty. If we keep using the first marginal, we will count later units as though the earlier ones never arrived. The integral solves precisely this problem. It follows the changing marginal along the entire represented path.

There are two objects here. The finite definition compares states, $E = V(x_0) - V(x_1)$. The path theorem explains when a continuous calculation gives that same number. The theorem does not define a new kind of EBU or add a charge for traveling between the endpoints. It connects the finite account to local differential information.

Think of measuring elevation along a walk. Adding all upward and downward elevation changes gives the difference between the final and initial heights. A long detour changes the sequence of readings, but not that height difference. Distance walked and effort expended are different quantities. In an EBU application, pumping energy and wear likewise need their own physical representation; they cannot be inferred from the endpoint change of an inventory potential.

14.2 The assumptions that make the connection exact

Theorem 14.1 (Fixed-field path identity). Let V be a single-valued C^1 function on an open neighbourhood of an admissible piecewise C^1 path $\gamma : [0, 1] \rightarrow \mathcal{X}$, with $\gamma(0) = x_0$ and $\gamma(1) = x_1$. Hold the potential and its parameters fixed throughout the comparison. Then

$$E = V(x_0) - V(x_1) = - \int_0^1 \nabla V(\gamma(s))^T \gamma'(s) ds. \quad (14.1)$$

The same statement holds intrinsically on a smooth constrained state manifold, using its differential dV along a piecewise smooth path.

Proof. On each smooth segment the chain rule gives $\frac{d}{ds} V(\gamma(s)) = \nabla V(\gamma(s))^T \gamma'(s)$. Integrating gives the potential at that segment's end

minus the potential at its start. Summing over segments cancels the intermediate endpoints. Negating the result gives the stated sign convention. $\square$

The proof is short because the important work is in the declaration. We have supplied a state function, the path lies in its domain, and the derivative is regular enough for the fundamental theorem of calculus. We have not started with an arbitrary vector field and tried to prove that a potential exists. That distinction removes a frequently misplaced topological condition: *no simply connected domain is required when the single-valued potential is already supplied*. A hole in the domain does not invalidate the chain rule for a path on which V is defined.

If instead someone gives only a vector field and asks whether it is a gradient, global compatibility becomes a separate problem. Vanishing local curl alone can fail on a domain with holes. That is not the problem solved by defining EBU through an already declared V .

14.3 What the local differential says

Write

$$dE = -dV = -\nabla V(x)^T dx. \quad (14.2)$$

This is an infinitesimal relation along the fixed field. It says how a tiny represented displacement contributes to the full comparison. It does not say that a finite displacement can be priced by one derivative sample. For an edge increment $dx = S_e dq$, it becomes $dE = f_e(x) dq$, where $f_e = -\nabla V^T S_e$.

A useful analogy is a meter whose reading changes while a tank fills. Multiplying the first rate by the full duration works only when the relevant rate stays constant. Integrating changing readings is not an optional correction. It is the calculation needed to describe the whole event.

The path parameter s is not necessarily clock time. Replacing it by a smooth increasing parameter changes both the velocity and the integration element, leaving their product and the integral unchanged. Thus the mathematical account need not depend on how quickly an interpolation is traversed. A physical consequence that does depend on speed, such as heating or spoilage, requires a state and action model that actually represents it.

14.4 The same endpoints do not mean the same operation

Two pumps may move the same quantity between the same tanks while using different amounts of electricity. If the model records only tank stocks, the

two endpoints match and the tank EBU matches. That is an exact statement about an incomplete boundary for comparing the pumps' total effects. Add the energy carrier and other relevant changes, and the complete endpoints can differ.

This is why path independence is useful without being a claim that process is irrelevant. It isolates what one account measures. A water-state potential need not be a complete engineering cost functional. Calling its gradient a force does not give it units of newtons or turn its integral into mechanical work. A particular thermal normalization will later connect potential, energy and probability under explicit physical assumptions; it is not presumed here.

The distinction can benefit a future EBU system in two ways. First, it prevents a bookkeeping detour from manufacturing additional value for the same complete change. Second, it makes omissions identifiable: if a supposedly equivalent route uses extra energy, the account must show where that physical difference enters. Better maintenance or route choices could follow from such information, provided the boundary and measurements are adequate.

14.5 An integral checked against the endpoints

For the familiar two-cell transfer, $V(q) = 16 - 4q + q^2/4$ and $f(q) = 4 - q/2$. A four-unit movement therefore has

$$E = \int_0^4 (4 - q/2) dq = 16 - 4 = 12. \quad (14.3)$$

Direct evaluation gives the same answer from $V(0) = 16$ and $V(4) = 4$. A two-stage physical execution gives seven for the first two units and five for the next two. The changing contributions sum to twelve because their baselines join.

The agreement is more informative than a numerical coincidence. If a software quote returns sixteen, it probably kept the first field value for the whole movement. If it returns fourteen by adding two same-baseline two-unit quotes, it probably treated two counterfactual beginnings as one physical history. The endpoint identity gives an independent way to recognize both mistakes.

14.6 A path that is declared is not necessarily observed

A straight segment between the endpoints is often convenient. On a convex stock domain it stays within that domain. But it is a physical trajectory only if the declared mechanism supplies that timing and simultaneous flow. Endpoint measurements alone cannot prove that every intermediate point occurred.

For the total, every admissible path satisfying the theorem gives the same result. For a decomposition among children or actors, the chosen path can matter. We will therefore calculate the joint outcome and its interaction structure before treating a path split as an attribution rule. A convenient interpolation must remain visible as a convention.

14.7 Regularity is part of the theorem

The C^1 assumptions above are sufficient and easy to check. A weaker formulation can use absolute continuity of $V \circ \gamma$ together with its almost-everywhere chain rule and an integrable derivative. Merely having a derivative almost everywhere is not enough: singular functions can change while their derivative is zero almost everywhere. Appendix A explains the corresponding issue for repeated finite differences.

For physical work the practical lesson is to distinguish an abrupt event from a smooth path. A jump still has a well-defined endpoint EBU whenever both potentials exist. It need not inherit an ordinary smooth-path description of the actual mechanism. We can keep the exact finite account without pretending that all events are differentiable.

The connection worth retaining

A finite EBU is defined by a state-potential difference. A regular path under the same field integrates to that difference. Path independence protects the total comparison; it neither erases unrepresented process effects nor selects an individual entitlement.

The next chapter returns to finite actions. With a quadratic potential we can carry out the integral algebraically, keeping the entire curvature correction instead of approximating it away.

Finite changes and curvature

15.1 The whole movement matters

A slope describes the landscape at one point. A finite action travels a distance through that landscape. If the slope changes along the way, its starting value multiplied by the distance cannot give the exact endpoint difference. The Gaussian model makes this distinction unusually transparent because the omitted term can be written exactly.

Return to the two-cell state (18, 2). The starting edge field is four. A four-unit transfer therefore has initial linear estimate $4 \cdot 4 = 16$. Its exact field reduction is twelve. An eight-unit transfer has linear estimate 32 but exact reduction sixteen. At sixteen units the linear estimate is 64, while the exact reduction is zero. No measurement noise or numerical integration is needed to explain these differences. They follow from the curvature of the declared potential.

This chapter retains the original book's lesson about overshoot while deriving it directly for finite quantities. We are not choosing a timestep for a restoring controller. We are evaluating a declared physical increment exactly.

15.2 Expand one square first

For one coordinate, write its initial deviation as $a = x_i - x_i^*$ and its physical increment as d . Then

$$(a + d)^2 = a^2 + 2ad + d^2. \quad (15.1)$$

Dividing by $2\sigma_i^2$ gives

$$V_i(x_i + d) - V_i(x_i) = \frac{x_i - x_i^*}{\sigma_i^2} d + \frac{d^2}{2\sigma_i^2}. \quad (15.2)$$

The first term is marginal times increment. The second is the curvature contribution. It is nonnegative and vanishes only when $d = 0$, given a finite positive scale.

This is exact expansion, not an approximation truncated after a convenient term. A quadratic has no cubic or higher terms to omit. That fact makes

the Gaussian domain a clear introductory setting: the finite correction can be derived with the algebra of a square before using vector calculus.

15.3 Add the coordinates

Let δ be any declared finite state increment. Sum the one-coordinate expansions:

$$V(x^b + \delta) = V(x^b) + \mu(x^b)^T \delta + \frac{1}{2} \delta^T H \delta. \quad (15.3)$$

The dot product $\mu^T \delta$ is the initial linear contribution. The quadratic form $\delta^T H \delta$ equals $\sum_i \delta_i^2 / \sigma_i^2$. The word “form” simply describes this matrix expression producing a scalar.

Define the finite EBU as $E = V(x^b) - V(x^b + \delta)$. In the calculations of this chapter we also write $F = E$ when emphasizing the field integral. Then

$$F = -\mu(x^b)^T \delta - \frac{1}{2} \delta^T H \delta. \quad (15.4)$$

The sign reverses because improvement means lower potential. A beneficial initial direction makes the first term positive. Curvature subtracts from that initial linear estimate.

Exact quadratic finite-change identity

For fixed references, fixed positive scales, and the declared additive Gaussian potential, every finite increment satisfies Equation 15.3. Therefore exact field reduction equals the initial linear improvement minus one-half the squared standardized increment. Physical feasibility is a separate requirement on the increment and event.

15.4 Specialize to one lossless edge

For $i \rightarrow j$, the increment is $\delta_i = -q$, $\delta_j = q$, with every other entry zero. Substituting into the exact identity gives

$$F_e(q) = q(\mu_i(x_i^b) - \mu_j(x_j^b)) - \frac{q^2}{2} \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (15.5)$$

Name the positive directional curvature

$$k_e = \frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} > 0. \quad (15.6)$$

Then the schedule is the simple parabola $F_e(q) = f_e(0)q - k_e q^2/2$. This notation makes the roles visible: $f_e(0)$ is the initial benefit per stock unit, while k_e tells how quickly that benefit falls as more is sent.

For the main fixture, $f_e(0) = 4$ and $k_e = 1/2$. Thus

$$F(q) = 4q - \frac{q^2}{4}. \quad (15.7)$$

The physical source limit for this isolated lossless transfer is $0 \leq q \leq 18$. The algebraic parabola can be evaluated outside that interval, but such values are not physically executable points of the declared event.

15.5 Read the complete schedule

q	A	B	After-potential	Field reduction
0	18	2	16	0
2	16	4	9	7
4	14	6	4	12
6	12	8	1	15
8	10	10	0	16
12	6	14	4	12
16	2	18	16	0
17	1	19	20.25	-4.25
18	0	20	25	-9

Every row is an independent counterfactual from the same baseline, not the next step of a trajectory. That distinction is essential. The $q = 6$ row means send six units from (18,2), not send six after the $q = 4$ row has already happened.

Potential initially falls as the action moves toward the reference. It reaches its minimum at eight. Further movement crosses the reference and climbs the other side. The total field reduction remains positive until sixteen because the final potential is still below the original 16. Positive total value therefore does not imply that every part of the path is downhill.

15.6 The maximum of a schedule is not a selector

Differentiating the cost-free field schedule gives

$$F'_e(q) = f_e(0) - k_e q. \quad (15.8)$$

If $f_e(0) > 0$, its unconstrained vertex is at $q_v = f_e(0)/k_e$. In the main fixture this is eight. The positive root other than zero is $q_0 = 2f_e(0)/k_e = 16$.

These quantities describe the function. They do not instruct the actor to choose q_v . A physical cap may make the vertex unavailable. A service obligation may require another quantity. Naming the maximum allows us to understand the geometry without quietly installing optimization as a physical law.

If the initial field is nonpositive, every positive quantity has negative field reduction in this quadratic lossless model. At zero initial field, curvature alone gives $-k_e q^2/2$. At negative initial field, both terms are negative. This is a conditional mathematical statement about one fixed direction, not a claim that the action must be morally prohibited.

15.7 An integral gives the same finite difference

The field along the path is $f_e(s) = f_e(0) - k_e s$. Integrating from zero to the finite quantity gives

$$\int_0^q f_e(s) ds = f_e(0)q - \frac{1}{2}k_e q^2 = F_e(q). \quad (15.9)$$

This is the fundamental theorem of calculus in a particularly simple form. The area under the changing field equals the endpoint potential reduction. It is not a different competing definition of single-action value.

For $q = 4$, the field falls from four to two. Its average is three, so the integral is $3 \cdot 4 = 12$. For $q = 8$, the average is two and the integral is sixteen. For $q = 16$, the field falls from four to minus four; positive and negative signed areas cancel.

The integral uses a mathematical path through state space. For an isolated linear transfer whose physical model realizes constant-rate motion, that path can also describe the modeled physical progression. If only endpoints have been declared, the path is an interpolation used to derive the identity. A mathematical identity alone does not establish which physical timing occurred.

15.8 Average, initial, and final slopes

For $q > 0$, the average field is

$$\bar{f}(q) = \frac{F_e(q)}{q} = f_e(0) - \frac{1}{2}k_e q. \quad (15.10)$$

It lies halfway between the initial and final fields because the field is linear along this quadratic path. The exact total is quantity times this average, not quantity times the initial value.

At the main $q = 12$ point, initial field is four, final field is minus two, average field is one, and total reduction is twelve. The final field is already negative even though the total remains positive. This distinction helps interpret an action that has passed the potential minimum: the action as a whole still improves on its starting state, but its last increments undo some earlier improvement.

An intensity diagnostic such as $F(q)/q$ can be informative, but it is not the total finite value. Ranking by intensity and ranking by total answer different questions. Neither ranking is the action-selection rule of the prospective random programme unless explicitly adopted.

15.9 At the reference every nonzero net change costs field value

At $x^b = x^*$, the gradient is zero. Equation 15.4 becomes

$$F = -\frac{1}{2}\delta^T H \delta. \quad (15.11)$$

It is negative for every nonzero net increment. This follows from positive definiteness of H . The reference is a strict minimum of the potential.

The phrase “net increment” matters for groups. Several nonzero actions can cancel so that their sum is zero. In the declared state the group’s endpoint then equals its baseline and its EBU is zero. It would be false to infer negative EBU merely from the presence of nonzero child actions. If their operation changes another represented physical carrier, that carrier belongs in the complete state and the potential must be evaluated there.

The negative sign does not itself forbid departure. A clinic may need to draw from a well-placed reserve to treat a patient. Physical permission and service are separate questions from the potential difference. An institution that makes a negative EBU unaffordable adds a rule of its own; the geometry neither requires nor justifies that rule.

15.10 Two coordinates with unequal scales

Use a separate sensitivity example with the main stocks and references but scales (2, 4). Initial potential is ten, field is 2.5, and directional curvature is $1/4 + 1/16 = 5/16$. Therefore

$$F(q) = \frac{5}{2}q - \frac{5}{32}q^2. \quad (15.12)$$

The vertex remains eight in this two-cell compatible setup, because a transfer of eight reaches both references simultaneously. The maximum is ten, the

entire starting potential. At four units, the reduction is $10 - 2.5 = 7.5$, and the after-potential is 2.5.

The unchanged vertex should not be generalized to all networks. Here there is only one mass-conserving degree of freedom, and both deviations are opposite amounts. In a multi-cell world an action can improve one pair while leaving other deviations untouched. Different scales then affect tradeoffs among available directions and finite quantities.

15.11 Several actions introduce cross terms

For two additive increments δ_a and δ_b , the net increment is their sum. Expanding its quadratic term gives

$$(\delta_a + \delta_b)^T H (\delta_a + \delta_b) = \delta_a^T H \delta_a + 2\delta_a^T H \delta_b + \delta_b^T H \delta_b. \quad (15.13)$$

Consequently the joint field reduction differs from the sum of standalone reductions evaluated at one baseline by

$$F_{ab} - F_a - F_b = -\delta_a^T H \delta_b. \quad (15.14)$$

This previews the group chapters. The cross term is present even though the potential is separable. The overlap occurs because both actions change a shared coordinate.

If both remove stock from the same source and change disjoint destinations, their source components have the same sign, giving a positive inner product and a negative correction. If one adds to a coordinate while the other removes, that shared contribution can have the opposite sign. No universal assertion of positive synergy follows from the existence of a group.

This is a comparison of fixed increments at the same baseline. If each hypothetical subset reruns a resolver and gets new quantities, it defines another set of comparisons. The simple fixed-increment expansion must not be used to describe it without checking those new dependencies.

15.12 Worked shared-source correction

Use a new three-cell declaration: state (14, 3, 3), reference (8, 6, 6), scales two at every cell, and lossless edges from the source to both destinations. Initial potential is

$$\frac{1}{2} \left(3^2 + (-1.5)^2 + (-1.5)^2 \right) = 6.75. \quad (15.15)$$

Each actor proposes quantity two. Alone, either action has initial field 2.25 and directional curvature $1/2$, so its standalone reduction is $2.25(2) - (2)^2/4 = 3.5$. The two standalone values sum to seven.

Together the increment is $(-4, 2, 2)$, giving endpoint $(10, 5, 5)$. Standardized deviations there are $(1, -0.5, -0.5)$, so potential is 0.75. Exact group reduction is $6.75 - 0.75 = 6$, not seven.

The cross-term calculation gives the same correction. The action increments are $\delta_a = (-2, 2, 0)$ and $\delta_b = (-2, 0, 2)$. With $H = I/4$, their weighted inner product is one, entirely from the common source. Thus $F_{ab} = 3.5 + 3.5 - 1 = 6$. No potential coupling between different cells was needed.

This group is physically possible under an ordinary frozen-source availability rule because aggregate withdrawal four does not exceed fourteen. Physical feasibility therefore does not solve the separate overcounting problem. Each check has its own job.

15.13 Worked cancellation correction

At the main baseline, suppose both directed lossless edges exist. Let action A send four from A to B, and action B send four in the reverse direction. The reverse standalone action is not feasible from B's two units under a frozen-source rule, so its same-baseline value must not be presented as an admissible physical comparator there.

To make the comparison legitimate, declare a separate baseline $(10, 10)$, with the same reference and scales. Both four-unit standalone transfers are now physically feasible. Each alone raises potential from zero to four, so each has field value minus four. The joint cancelling group has net increment zero and field value zero.

The same-baseline correction is therefore $0 - (-4 - 4) = 8$. It is positive because the hypothetical standalone endpoints each depart from the reference while the group endpoint does not. This does not identify an extra eight units to issue in addition to the group value. It is a diagnostic difference between two different accounting comparisons.

The explicit baseline change in this example is essential. Without it, an appealing algebraic illustration would depend on a standalone action that cannot be performed in the declared physical world. Mathematical formulas must retain the admissibility assumptions of the comparisons they claim to represent.

15.14 A fuller physical account changes the state

The schedule above is exact for the declared lossless two-stock world. It does not include electricity consumption, wear or a wasted carrier. To include one of those effects, declare its physical coordinate, the action's complete

increment and a potential over the chosen complete state. Then calculate $E = V_{\text{pre}} - V_{\text{post}}$ again. The answer may change because the world represented by the model has changed, not because an arbitrary term was appended to the definition of EBU.

An invoice, operating cost or policy penalty can be kept in an institutional account with its own meaning. Calling such a number EBU would hide the distinction this book is trying to make. In a lossy transfer, for example, the difference between source withdrawal and destination delivery must first appear in a declared sink or cross a boundary; only then is it possible to discuss what the potential values and what it leaves as audit-only.

15.15 Why dividing the action does not erase curvature

Take the main four-unit transfer and split it into two consecutive two-unit actions under one fixed potential and no intervening external change. The first action goes from potential 16 to nine, giving seven. The second begins at the new state (16, 4) and goes from nine to four, giving five. Their sum is twelve, the unsplit four-unit field reduction.

If both pieces were instead valued independently from the original state, each would claim seven, for a total of fourteen. That is a different calculation and overstates the actual four-unit endpoint reduction. The second piece has been allowed to reuse the first piece's initial field.

This example distinguishes sequential telescoping from independent same-baseline claims. The correction is not an arbitrary penalty for using two names. It is the mathematical consequence of evaluating the second physical change where it actually begins. Simultaneous groups require a joint value and a separately declared attribution convention; those are developed later.

15.16 Guided problem: derive an entire schedule

Use a separate lossless two-cell example with total 16, state (12, 4), reference (8, 8), and equal scales two. Initial marginals are one and minus one, so the field is two. Directional curvature is one-half. The exact schedule is

$$F(q) = 2q - \frac{q^2}{4}, \quad 0 \leq q \leq 12. \quad (15.16)$$

At $q = 2$, field reduction is three; the endpoint (10, 6) has potential one, compared with initial four. At $q = 4$, the reference is reached and reduction is four. At $q = 8$, the endpoint is (4, 12) and reduction is zero. At $q = 10$, reduction is $20 - 25 = -5$.

Now perform the same exercise by endpoint evaluation alone. This independent form of arithmetic verifies signs and normalization. Finally integrate the field $2 - q/2$ from zero to each quantity. The three methods agree because they are three expressions of the same quadratic identity, not three separate scientific findings.

15.17 Practice and reflection

For the main schedule, solve $F(q) = 12$. The two roots are four and twelve. Both provide the same total field reduction but leave different endpoint distributions, $(14, 6)$ and $(6, 14)$. Explain why a value alone cannot identify what physically happened.

Calculate the difference between the linear estimate and exact field reduction at $q = 3$. It is $q^2/4 = 2.25$; the linear estimate is twelve and the exact reduction is 9.75. This difference is a curvature contribution, not an optional audit adjustment.

At the reference, let $\delta = (-1, 1)^T$ with scales two. The quadratic term is $\frac{1}{2}(1/4 + 1/4) = 1/4$, so the field value is minus one-quarter. Explain how a zero gradient and a negative finite value coexist without contradiction.

Permission before value

16.1 A favourable number cannot create stock

The potential tells us what a specified state change would mean in the declared account. It does not make that change possible. A source holding three units cannot supply an isolated four-unit lossless transfer; a road that is closed cannot deliver medicine because the delivery would be useful. Before an EBU quote can describe an executable action, a separate physical question must be answered: *can this complete transformation occur now?*

For one isolated lossless edge $i \rightarrow j$, with nonnegative stock and no other bound, the answer requires $0 \leq q \leq x_i^b$. This simple interval already separates value from permission. From the state (18, 2), a transfer of seventeen units is possible under that physical rule even though it raises the potential and has negative EBU. A request for nineteen units can be entered into the quadratic formula, but it lies outside the event's physical domain. A mathematical value for an impossible endpoint is a counterfactual, not a permit.

16.2 A group can fail even when each child passes

Suppose one source holds eight units and two actions each request five. Either action can pass an isolated source check. Their simultaneous group cannot pass a frozen-source rule because its aggregate request is ten. For that rule the necessary condition is

$$\sum_{a: \text{src}(a)=i} q_a \leq x_i^b. \quad (16.1)$$

The same principle applies to shared pump capacity, receiving storage and any other physical limit that a group uses together. Conservation of the final total does not settle these limits.

Now imagine a chain in which the middle node begins empty, receives four units and sends four out during one event. Its final stock is zero, not negative. Yet a warehouse that must receive and inspect stock before dispatching it may refuse this event, while a continuously operating pipe could allow a declared flow-through process. The endpoint is identical; the physical timing rule is

not. We cannot settle the difference by choosing the more attractive EBU calculation.

Three questions, not one

Physical feasibility asks what can occur. EBU asks how the complete accepted change alters the declared potential. Institutional authority asks who may decide and on what terms. A mathematical answer to one question does not silently answer the other two.

16.3 A finite menu is a study design, not a law of nature

A research implementation may offer only a finite list of candidate quantities, even if the represented resource is continuously divisible. That makes the study reproducible; it does not prove that omitted quantities are physically impossible. A menu should be generated without looking ahead to which result will favour a chosen actor policy. Otherwise an outcome can be made to look like a property of EBU when it is really a property of the menu that was offered.

An admissible set also does not rank its members. Choosing the largest EBU, choosing uniformly among identified plans, or prioritizing an urgent delivery are different decision rules. Each can be studied, but none is hidden inside the definition of a potential. In particular, an essential negative-EBU action is not physically prohibited just because it is negative.

16.4 What a permission record must say

For an accepted action, record the predecessor state and its epoch, route or other physical action identifier, requested and accepted quantities, shared resources, governing constraints and the resulting complete increment. If a request was reduced, the rule that reduced it belongs in the record. If only endpoints were checked, the record must not imply that transient path constraints were checked as well.

This discipline leaves us with a physically meaningful candidate for valuation. The next chapter can now calculate its exact EBU without asking the potential to perform the job of a route, a stock certificate or a human decision-maker.

The exact local signed EBU quote

17.1 The action finally has a number

We have taken care to identify a physical state, a reference, a potential and an action that can actually occur. Only now is it safe to attach a number to the action. The number is not the quantity delivered, the price paid or the initial slope of the potential. It is the complete change in the declared potential between the accepted predecessor and successor.

Let x^b be the state immediately before an action, let q be its finite source-side quantity, and let S_e be the declared increment per unit of that quantity. Under one fixed field V ,

$$E_e(q) = V(x^b) - V(x^b + S_e q). \quad (17.1)$$

Positive E_e means the represented potential fell. Negative E_e means it rose. Zero means the two endpoints have equal potential; it does not prove that the action was empty. The complete increment and the version of V belong beside the number on any trustworthy quote.

The finite definition

For a fixed, declared potential, exact finite EBU is before-potential minus after-potential for the complete accepted physical change. It is not defined by the local gradient approximation, a monetary price or an extra charge invented after the event.

17.2 Why a local computation can give the whole answer

Suppose the potential is a sum of factors, $V = \sum_{\alpha} \phi_{\alpha}$, each with a declared set of input coordinates. An action changes only some of those coordinates. In $V(x^b) - V(x^b + S_e q)$ every factor whose inputs are unchanged cancels exactly. Thus a local evaluator needs the changed factors and their complete inputs, not a new calculation over every untouched place in the system. This is the precise mathematical answer to the opening question of how a local action can be assessed without a global recalculation each time.

Local evaluation does not license incomplete observation. If a factor couples the source to a downstream or environmental coordinate, that entire factor must be reevaluated. If an action changes an energy carrier as well as water, the complete physical increment must say so. Computational locality follows from a declared factorization; it is not a promise that the physical world has no long-range dependencies.

17.3 The Gaussian expression is a theorem below the definition

For the quadratic potential of Chapter 12, write $H = \text{diag}(1/\sigma_i^2)$ and $\mu(x^b) = H(x^b - x^*)$. Exact expansion of the squares gives

$$E_e(q) = -q \mu(x^b)^T S_e - \frac{q^2}{2} S_e^T H S_e. \quad (17.2)$$

The first term uses the initial slope. The second tells us how the slope changes over the finite action. No small- q approximation has been made. For a lossless transfer from i to j , $S_e = -e_i + e_j$, and the expression becomes

$$E_e(q) = q(\mu_i - \mu_j) - \frac{q^2}{2} \left(\frac{1}{\sigma_i^2} + \frac{1}{\sigma_j^2} \right). \quad (17.3)$$

This edge polynomial is a corollary for this physical increment and this quadratic potential. It is not the general definition of EBU. A lossy route, a valued sink or a coupled potential changes the specialized formula while leaving Equation 17.1 intact.

17.4 One calculation with three independent readings

Return to the lossless two-cell state (18, 2), reference (10, 10) and scales (2, 2). A four-unit transfer reaches (14, 6). Direct evaluation gives $V_{\text{pre}} = 16$ and $V_{\text{post}} = 4$, hence $E = 12$.

The Gaussian expansion gives the same result: the initial edge field is 4 per stock unit and the directional curvature is $1/2$, so $E = 4(4) - (1/2)(4^2)/2 = 12$. Along the linear evaluation path the field falls from 4 to 2; its average is 3, and integrating over four units again gives 12. These are three derivations of one number, not three independent claims of value.

The agreement is useful for checking software, but it does not establish that four units were available, that a pump ran, or that a clinic was served. Physical permission comes from the action model; occurrence comes from the event record; service comes from a separately declared outcome condition.

17.5 The cost of an omitted carrier

If a pump consumes electricity and that consumption matters to the claim, one may extend the physical state with an energy coordinate and declare the pump's full transition. The potential over that extended state then evaluates its chosen consequences. If some loss crosses the model boundary, the boundary crossing must be explicit. By contrast, adding a number called "process burden" to the right of Equation 17.1 would create a different account. It cannot silently redefine the physical EBU.

There is room for engineering cost ledgers and institutional charges. Their units, provenance and decision role should be written alongside EBU, not confused with it. That separation lets a reader ask which question each number answers and prevents the same physical loss from being counted twice.

17.6 From a quote to a record

Before execution, a quote names a possible predecessor, potential version, action increment and quantity. After execution, an event record must say which quantity was actually accepted and which complete successor occurred. The recorded EBU is recalculated from that pair. If the world changed between quotation and execution, the old quote is an historical proposal, not the current baseline.

The number can then be attributed to actors only under a declared rule. An exact group total does not by itself decide who owns it or what anyone may spend. We will first follow the physical event into its record, then return to the more difficult case of several actions sharing one field.

From a pre-action schedule to a measured record

18.1 A promise is not yet a physical event

An operator proposes to send four stock units from cell 1 to cell 2. Before anything moves, we can calculate what that action would mean under the frozen model. Afterward, a meter may show that only three units moved. The proposed quantity, permitted quantity and observed quantity answer different questions. Keeping all three is part of a useful account.

The preceding chapter gave an exact value for a finite action. Here we ask what information makes that equation reproducible across time. The object of interest is still a physical change. A message requesting a transfer does not move the water. A signed agreement does not prove delivery. A correct calculation of a planned successor does not prove that the successor occurred.

The original book called this progression a quote-to-settlement lifecycle. We retain that teaching idea, while separating a physical value record from any later rule that changes a spendable balance. This chapter specifies the information relationships needed to understand an account. It does not adopt a complete operating protocol for the prospective Gaussian study.

Three quantities in one story

The request says what an actor asks to do. The accepted domain says what the declared rules permit. The measured record says what the evidence supports as having occurred. A scientifically useful record never silently substitutes one for another.

18.2 Freeze the whole function, not one attractive point

For the recurring two-cell example, let

$$x^b = (18, 2), \quad x^* = (10, 10), \quad \sigma = (2, 2).$$

A lossless transfer of quantity q changes the state to $(18 - q, 2 + q)$. Its exact schedule is

$$E(q) = 4q - \frac{q^2}{4}. \quad (18.1)$$

If the accepted domain is $0 \leq q \leq 4$, the quoted value at full execution is $E(4) = 12$. The equation also specifies every smaller accepted quantity. At $q = 3$ it gives $12 - 9/4 = 9.75$. That is the value of the three-unit event on this schedule.

Multiplying the full quote by the fraction executed would give $12(3/4) = 9$, which is different. The first three units and the fourth unit do not have the same marginal effect. We can see this without calculus: the three-unit successor is $(15, 5)$, its potential is $25/4 = 6.25$, and $16 - 6.25 = 9.75$. The direct endpoint calculation agrees with the function.

This is why a pre-action account should identify its whole domain and formula. A few selected values may illustrate the formula, but they do not replace it. A table can represent an exact schedule only if its interpolation rule is also justified. Connecting points with straight lines usually changes a curved function.

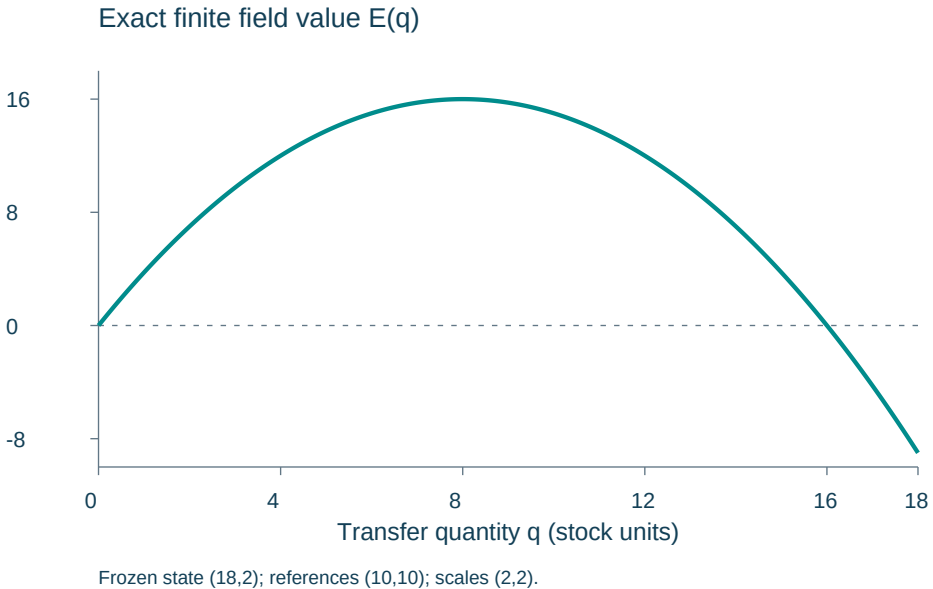


Figure 18.1: The full finite-value curve for the running two-cell example. A positive initial slope does not make every finite quantity beneficial.

18.3 The information held fixed

To reproduce the schedule, the reader needs the complete baseline state, coordinate ordering, action map, reference vector, positive scales, physical boundary, units and accepted domain. A version label should identify the equation and those definitions. A timestamp alone is insufficient: two records at the same time can describe different fields.

The scope must also say whether q is quantity or rate. Here q is a quantity. If a meter reports a rate j over duration Δt , then $q = \Delta t j$ when the rate is constant. A variable rate requires its time integral. Replacing a reported average rate with an instantaneous endpoint reading can change the quantity, even if the numerical values look similar.

For a group, the record needs the accepted vector, the joint successor and the path convention if child attributions are displayed. A child's measured shortfall can change the group's field value and the attributions of other children. It cannot always be handled by independently scaling one child's old number. Chapter 24 explains the mathematical reason.

18.4 The shared baseline removes an external change

Suppose an external, lossless redistribution changes the physical state from $(16, 4)$ to $(18, 2)$ before the actor's epoch. Under our fixed potential,

$$V(16, 4) = 9, \quad V(18, 2) = 16.$$

The external change increased represented deviation by seven. The actor then transfers four units, ending at $(14, 6)$ with $V = 4$. The actor's field value is $16 - 4 = 12$.

Using the earlier state as the actor's baseline would give $9 - 4 = 5$. It would mix the external worsening with the actor's improvement. Neither number is arithmetically mysterious; they answer different questions. The external record is $+7$ in potential change, while the action record is $+12$ in avoided potential. Together they explain why the complete interval changed potential by $4 - 9 = -5$.

Write the general relationship with pre-forcing state x , frozen post-forcing baseline x^b and action successor y :

$$D_{\text{ext}} = V(x^b) - V(x), \tag{18.2}$$

$$E = V(x^b) - V(y), \tag{18.3}$$

$$V(x) - V(y) = E - D_{\text{ext}}. \tag{18.4}$$

The subtraction makes the attribution boundary explicit. Calling the external change a disturbance does not require it to increase burden. It can also reduce burden, in which case D_{ext} is negative.

No-action neutrality

At the shared baseline, $E(0) = V(x^b) - V(x^b) = 0$. An idle actor receives no field reduction from an external event merely because that event preceded the observation. This is an algebraic statement within the declared comparison.

18.5 An epoch has a beginning and an expiry

Imagine a pump quote issued at 14:00 for a short movement of water. At 14:03 another pipe fails. If that failure changes a required precondition, the original quote may no longer describe the available action. A protocol must say whether the action can continue, whether its authorization expires, or whether a new epoch is opened.

What cannot be justified by the original formula is silently choosing the most favourable baseline after seeing the outcome. An actor might prefer the high-burden state immediately after failure; an institution might prefer the lower-burden state before it. Reproducibility requires the comparison rule to be specified before those preferences can select it.

A new epoch is not an admission that earlier arithmetic was wrong. It records that the facts or applicable model have changed. A useful archive keeps the earlier proposal, its expiry reason and the new proposal. This lets a later reviewer distinguish a legitimately revised opportunity from an outcome-dependent rewrite.

The exact timing and expiry mechanics belong to a registered protocol. We need not invent them here to understand the boundary. The general requirement is consistency: the physical interval, state commitment, accepted domain and measurement claim must refer to the same event.

18.6 Shortfall, overexecution and missing evidence

For a measured quantity within the valid domain, the committed schedule can be evaluated at that quantity when its other assumptions remain satisfied. Our three-unit example is such a case. It does not require a new potential or a proportional approximation.

If the accepted domain ended at four but the meter reports five, the physical report must retain five. A rule limiting accepted quantity cannot erase an extra unit from the reservoir. At the same time, the earlier commitment did not authorize five. Describing the physical consequence and deciding the institutional response are separate tasks.

The mathematical function may be evaluable beyond its accepted domain, but mathematical evaluability is not permission. A retrospective diagnostic can explicitly calculate the consequence of overexecution under the frozen field. It must not be mislabeled as an authorized receipt or a precommitted positive credit.

If the sensor is missing or inconsistent, “not established” is a possible outcome. Zero is a measured or derived value, not a universal substitute for uncertainty. Crediting the full request would invent execution; charging an arbitrary negative amount would invent a physical burden. Any sanctions or compensation need their own declared basis.

18.7 Identity prevents counting the same change twice

One transfer can appear in the source meter, the destination meter, a pump controller and a transporter’s report. Those records can corroborate one another. They do not automatically describe four actions. A stable event identity connects them to one physical transformation.

Conversely, a chain can contain several physical transformations under one invoice. Fuel use, electricity generation, delivery and water pumping must retain their physical boundaries where they are in scope. A commercial reference number can link the chain without collapsing its quantities or converting its resource units into one scalar.

Repeated actions can be distinguished by their execution intervals and predecessor states. Splitting an actual continuous movement into bookkeeping pieces must not let each piece claim the same original improvement. If the pieces are sequential, each uses the preceding piece’s successor. If they are simultaneous, the group account must reconcile their joint effect.

18.8 A hash protects a record, not a sensor

A digital fingerprint can help establish that the equation and inputs shown after an event match the committed bytes. It does not establish that the water existed, that a meter was calibrated, or that the declared reference was appropriate. Integrity of a file and truth of a measurement are different properties.

For a physical account, the evidence chain can include device identity, calibration records, location, time interval, unit conversion, independent observations and documented corrections. The appropriate set depends on the domain. A teaching tank does not prescribe a surveillance system for a hospital.

Records should explain uncertainty without burying it in the potential's scale. In the Gaussian model, σ_i defines the reference geometry. A sensor's uncertainty concerns knowledge of x_i . Treating those as the same parameter would change both the meaning of the field and the meaning of confidence in the observation.

18.9 Corrections preserve the earlier account

Suppose a later calibration finds that a meter overstated delivery. A correction should identify the earlier record, the evidence for the correction and the authorized correction procedure. Deleting the earlier line conceals what participants actually knew and acted on at the time.

A physical reversal is different. Returning water from B to A is a new action with a new predecessor state. It may reverse the earlier endpoint change, but it is not a correction to the fact that the first action occurred. Its value must be computed from the complete physical transition under the applicable fixed field.

Versioning the potential is also different from correcting a measurement. If new evidence justifies a different reference, the same unchanged state can acquire a different potential. That is a model revision. It needs a separate reconciliation term when accounts span the change; it is not physical restoration by the actor.

18.10 A complete worked action record

Consider the three-unit shortfall again. A readable record contains the following facts:

Item	Declared or observed content
Physical object	Lossless A-to-B transfer of one homogeneous stock
Baseline	(18, 2)X, ordered A then B
Field	Reference (10, 10)X; scales (2, 2)X
Domain	Accepted quantity between zero and four stock units
Schedule	$E(q) = 4q - q^2/4$ under the fixed field
Observation	Three stock units transferred; endpoint (15, 5)X
Calculation	Before 16; after 6.25; signed field value 9.75
Status	Illustrative arithmetic; no real event or spendable credit claimed

The record is longer than the number 9.75 because it explains what the number means. Someone can check the arithmetic without knowing the actor’s income, preferences or occupational status. Whether that person is entitled to compensation remains a separate question.

18.11 Guided problem: a changed observation

Keep the same frozen field and accepted domain. The source reports that 2.5X left A; the destination reports that 2.5X arrived. Calculate the successor, its potential and the exact value. Then compare the result with scaling the full four-unit quote by the delivered fraction.

The successor is (15.5, 4.5)X. Each endpoint differs from its reference by 5.5X in magnitude. Thus $V = 2(1/2)(5.5/2)^2 = 7.5625$, and the field reduction is $16 - 7.5625 = 8.4375$. Equivalently, the schedule gives $4(2.5) - (2.5)^2/4 = 8.4375$.

Proportional scaling would give $12(2.5/4) = 7.5$. It is smaller because it assigns the fourth unit’s lower marginal contribution to earlier units as well. The endpoint equation avoids that substitution. The two meters agree on conservation in this illustration; agreement alone does not establish their calibration in a real application.

18.12 Practice and reflection

First, evaluate $E(1)$, $E(2)$ and $E(4)$. The answers are 3.75, 7 and 12. Explain why doubling quantity from one to two does not double value. Next, suppose an external event takes (18, 2) to (16, 4) before a two-unit action. The new baseline has $V = 9$; the action ends at (14, 6) with $V = 4$, so the action value is 5. The external potential change was -7 . Neither should be replaced by the other.

Finally, write an example of a missing measurement, a physical reversal and a field-version change. For each, name the record that must be retained and the new information that must be added. The purpose is to recognize distinct events before assigning numbers to them.

What to carry forward

A schedule is a conditional account over a declared domain. Verified execution supplies the point to evaluate. External changes, expired conditions, missing evidence and later corrections remain visible. The equation determines the field difference; an institution must separately define the consequences of recording it.

The next chapter follows these signed values into the real-needs examples that motivated the book. Understanding a positive number is incomplete until we can also explain a zero or a negative number without confusing the action with the person it serves.

Positive, zero and negative EBU in real needs

19.1 The sign belongs to an action in a state

The same stock transfer can arrive where it is urgently needed, where stock is already well placed, or where delivery displaces something else. The name of the product does not determine the sign of its EBU. The declared before-state, complete physical transformation and fixed potential determine the calculation.

This statement is sometimes uncomfortable because everyday language uses “valuable” for several different things. A treatment can be valuable to a patient. A delivery can have a high market price. A redistribution can lower a particular stock potential. These statements may all be true, but they do not express the same relationship.

The examples in this chapter keep the physical account visible alongside service and human need. All numbers are invented teaching inputs. They are not clinical guidance, a valuation of a life or a claim that medicine can be represented adequately by two stocks.

Read the sign with its noun

Positive EBU means the declared fixed potential fell across the complete represented transition. It does not mean a good person, an adequate wage, a complete cure or a socially optimal decision. Negative values are equally specific.

19.2 One delivery, three starting states

Use two storage locations with compatible references $(10, 10)X$ and scales $(2, 2)X$. A delivery sends $2X$ from the first to the second, with perfect quantity retention in this ideal illustration. Any energy use would require an additional represented carrier and is outside this two-stock example.

Start first from (18,2)X. The potential is 16; the successor (16,4)X has potential 9. The delivery's field value is +7. The clinic still has less than its declared reference, so a positive result does not mean all relevant needs have been met.

Now start from (11,9)X. The potential is $1/4$. The same delivery produces (9,11)X, also with potential $1/4$. Its field value is zero. Two physical stocks changed, and medicine reached the clinic, but the chosen quadratic assigns equal deviation to the endpoints.

Finally start from (10,10)X. Potential is zero. The same delivery produces (8,12)X, whose potential is 1. The field value is -1 . This is the simplest example of a permitted physical change that moves the modeled system away from its reference.

Before	After	V before/after	E
(18, 2)	(16, 4)	$16 \rightarrow 9$	+7
(11, 9)	(9, 11)	$1/4 \rightarrow 1/4$	0
(10, 10)	(8, 12)	$0 \rightarrow 1$	-1

The common action description is not enough to reproduce any of the three answers. The starting state is indispensable. So is the fixed field: using different references would ask a different question.

19.3 A zero is not an empty event

In the middle example, the source and destination exchange their deviations. The endpoint states differ, although their potentials agree. A scalar function assigns one number to a state containing several numbers; equal potential generally does not imply equal state.

This matters for interpretation. One endpoint may have a different available route, future need or exposure to failure. Those differences can change later opportunities even when current V is equal. They cannot be recovered from the scalar alone. A complete action record keeps the state, not merely its potential.

Zero can also arise when the total increment of a group cancels. Two opposing ideal transfers may leave the net stocks unchanged. That does not establish that real pumps consumed no electricity. If pumping energy is in scope, its transformation must be represented through the appropriate carrier or boundary. A richer state can therefore give a different EBU from the ideal two-stock calculation without adding an unrepresented charge to the equation.

19.4 The full quantity can reverse the initial sign

Return to (18, 2)X. A small A-to-B transfer improves the Gaussian field, but the exact schedule is

$$E(q) = 4q - q^2/4.$$

At $q = 8$, the endpoint is the reference and $E = 16$. At $q = 16$, the endpoint is (2, 18) and $E = 0$. At $q = 17$, the endpoint is (1, 19), whose potential is $81/4 = 20.25$, giving $E = -4.25$.

The positive initial slope did not promise that every amount in that direction would be beneficial. The transfer passed through the reference and continued until the full action increased deviation. A finite calculation includes the entire movement. A slope is a nearby statement.

One could design a controller that chooses $q = 8$ to maximize this particular schedule. That is an additional selection rule, not part of the definition of EBU. A different actor might choose an amount for a deadline, a random menu or another objective. This book teaches the measurement without silently building a maximizer into the word “field.”

19.5 An essential action can have negative field value

Suppose a patient needs a dose at the second location, while the simplified inventory reference currently places equal stock at both locations. The example starting at (10, 10) says a two-unit transfer has field value -1 . It does not tell us whether the patient should receive treatment.

The potential may not represent clinical timing, patient condition, dose compatibility, expiry or access. A reader who turns -1 into automatic refusal has supplied a social rule and a claim of model completeness that the equation never contained. A reader who changes the result to positive because the patient needs help has replaced the declared physical comparison with a different judgment.

An honest account can state both: the represented inventory deviation increased by one, and the delivery met a separately described urgent need. If the scientific model is inadequate for the intended decision, its scope must be revised through explicit evidence and governance. The old event remains described by the field actually used.

An action is not a person

Do not say that a patient “is a burden of one EBU.” Say which delivery action changed which represented state under which field. A human need and a person’s claim to care are not coordinates in this two-tank demonstration.

19.6 Feasibility, value and service are separate tests

A movement can have positive hypothetical value while being physically impossible. If A contains only one unit, an instruction to remove two cannot be made valid by a favourable clinic term. Conservation and source availability constrain the action before its potential difference is treated as an executable opportunity.

A feasible action can fail a service requirement. A delivery arriving after a deadline may preserve a stock balance while failing its intended use. Time can be part of the represented state, a constraint or a separately reported outcome, but it cannot be omitted and then assumed satisfied.

A negative action can be feasible and successful on a service dimension. The three dimensions therefore form a small report, not a single ranking:

Physical status	Field status	Service interpretation
Feasible	Positive	Can improve the field while leaving demand unmet
Feasible	Zero	Can move goods without changing the chosen potential
Feasible	Negative	Can meet a need while worsening represented deviation
Infeasible	Any hypothetical sign	No executed candidate follows from the sign

The report avoids calling every dimension “success.” A later scientific comparison must declare which outcomes matter and how conclusions are drawn. Choosing that interpretation after seeing results would change the question being tested.

19.7 A repair needs a represented physical consequence

A mechanic repairs a pump. A water-stock model that observes only the current litres may show no immediate change in x . Under that model, the stock field effect is zero. This is not evidence that repair is useless; the model

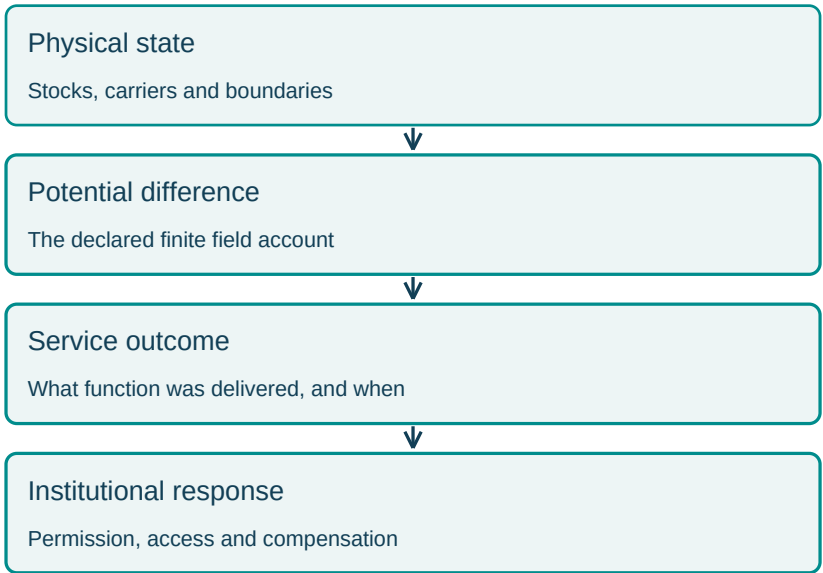


Figure 19.1: Physical stock, potential and an action account answer different questions. Their records must be linked without treating them as the same quantity.

lacks the equipment condition or other capacity coordinate needed to describe the benefit.

One option for future research is to represent a certified physical equipment state. That requires a definition, unit, measurement method and justified contribution to a potential. Merely writing “pump repaired” in a field does not supply these. Another option is to report the repair in a linked physical record while observing its later effects without issuing an invented current field reward.

The distinction also applies to diagnosis. Finding a leak provides useful information. If no physical coordinate in the model changes, a separate claim of positive physical field reduction needs an information or service model that actually defines such a change. Professional effort deserves consideration in compensation, but occupation and effort do not replace the state difference.

19.8 Five layers in an infrastructure decision

Consider building a shared cold store. First describe construction: materials, energy, land, transport and waste, with appropriate carriers and boundaries.

Second describe what capability has been established and how it is measured. Third identify who has authority to approve the construction. Fourth determine who bears its institutional responsibilities. Fifth specify how workers and providers are compensated.

These five questions can refer to one project without becoming one equation. A high future service expectation does not erase present construction burden. Conversely, a negative present physical account does not show that investment should never occur. Long-term planning can consider future outcomes, but its forecast is not an observed action receipt.

A changed route or machine can also alter future action maps and efficiencies. That is a versioned physical and modeling change. The old route's receipts should remain interpretable under the old description; the new route's receipts should identify the new description. Rewriting old efficiencies would make improvement impossible to audit.

19.9 The rural clinic as a linked case

The clinic needs medicine, refrigeration, water and a working backup supply. In ordinary conversation these form one service. Physically they contain different carriers and transformations. The medicine transfer changes dose stocks; refrigeration transforms energy while maintaining temperature; water delivery changes water stocks; repair changes equipment condition when that state can be measured.

Linking these records preserves dependence: medicine can become unusable without refrigeration, and a clinic can cease functioning without water. Adding their numerical EBU values without a declared common calibration would hide differences of meaning. A small positive water account cannot automatically compensate for an unrepresented failure of medicine storage.

The teaching scalar is useful because it allows complete arithmetic. It is not useful if we forget why it is small. For a real clinic, the scientific task includes deciding which states and dependencies are necessary for the intended claim. The potential must be evaluated as a model of that task, not as a universal vocabulary of health.

19.10 Recurring burdens can inform redesign

Suppose repeated emergency deliveries are physically expensive. The records can motivate questions about regional storage, local production, repair skills or transport reliability. They cannot by themselves prove which investment will succeed. Comparing a new store with existing deliveries requires consistent service, time horizon, boundary and uncertainty treatment.

This is a place where monetary and physical records can complement one another. A financially inaccessible improvement might reduce physical burden. A financially attractive improvement might shift burden outside the current boundary. An institutional study must examine access and distribution alongside physical consequences.

Such questions preserve the original book's ambition while making its status clear. EBU may provide information useful for adaptation. Whether people can act on it, whether benefits are captured, and whether outcomes improve are empirical and institutional questions. A sequence of exact receipts is evidence about the recorded events, not automatic proof of a just transition.

19.11 Guided problem: different needs at equal potential

Use the same field and compare $(14, 6)$ with $(6, 14)$. Both have potential 4. From each, consider a two-unit first-to-second transfer. The first ends at $(12, 8)$ with potential 1 and has value +3. The second ends at $(4, 16)$ with potential 9 and has value -5.

The starting potentials were equal, but the local directions were different. This shows why recording only V loses decision-relevant state information. Now suppose the second location in both cases has a separate urgent deadline. That service fact does not change the arithmetic. It changes the complete interpretation and may reveal that the current reference model needs additional representation.

19.12 Practice and reflection

For $x^b = (12, 8)$, calculate a two-unit and a four-unit transfer. The initial potential is 1. The first transfer reaches $(10, 10)$ and gives +1; the second reaches $(8, 12)$ and gives zero. State a reason why the zero-value action might still have a separately measured service outcome.

Then describe a useful maintenance action whose immediate benefit is missing from a stock-only model. Identify the missing physical quantity, possible evidence for it, and the institutional question that would remain even after it was measured. Do not assign a number to the benefit without defining the new field.

Finally rewrite the sentence "This household caused five negative EBU" as an action-specific report. A good answer identifies the device or transport event, model version, physical cause, service delivered and measurement status. It avoids converting a receipt into a judgment of the household's worth.

Local actions and exact global identities

20.1 The apparent paradox

An action affecting only two cells can be valued using data from those cells. Yet we have described its field effect as a change in a potential summed across a whole network. How can a calculation that does not read the other locations give the complete result?

The answer is cancellation. Under the declared separable potential, every untouched location contributes exactly the same term before and after the action. When those terms are subtracted, they disappear. The local calculation is not a statistical estimate of the remaining world; it is the part of an exact difference that survives.

This distinction is central to the project. It explains how a local action account can be mathematically complete within a declared model. It also identifies the boundary of that claim: the model's potential must include the relevant effects, and the action's evaluation must include every term that actually changes.

20.2 Support means the coordinates changed

Let δ be an action's finite increment. Define its write support by

$$W = \{i : \delta_i \neq 0\}. \quad (20.1)$$

For a positive lossless transfer along $i \rightarrow j$, this set contains i and j . A group can change more coordinates. A proposed edge of zero quantity changes none, even though its name still refers to two endpoints.

Support is an algebraic description of the state change, not a prediction of every future consequence. Water arriving at B may enable a later transfer to another place. That later event has its own increment and baseline. It does not automatically appear in the present endpoint difference.

For the Gaussian model,

$$V(x) = \sum_{i=1}^N V_i(x_i), \quad V_i(x_i) = \frac{(x_i - x_i^*)^2}{2\sigma_i^2}.$$

Each local term depends on one coordinate and fixed local parameters. This is what makes endpoint support sufficient for the present model.

20.3 The cancellation proof, one line at a time

Write the complete difference as a sum:

$$V(x^b) - V(x^b + \delta) = \sum_{i=1}^N \left[V_i(x_i^b) - V_i(x_i^b + \delta_i) \right]. \quad (20.2)$$

For an index outside W , the increment is zero. Its summand is therefore $V_i(x_i^b) - V_i(x_i^b) = 0$. Removing those zero summands gives

$$V(x^b) - V(x^b + \delta) = \sum_{i \in W} \left[V_i(x_i^b) - V_i(x_i^b + \delta_i) \right]. \quad (20.3)$$

Define V_{loc} as the sum over W . Equation (20.3) says that its before/after difference is the same as the global difference. The proof does not need the values of the unchanged terms.

Exact local-to-global cancellation

Under one fixed separable potential, with a correctly declared action increment, the sum of the changed local terms equals the complete potential difference. Every unchanged term cancels. This equality concerns the declared model; it does not prove the model contains every consequence that matters.

Notice what the proof did not use. It did not require equal references, equal scales or a Gaussian shape. The same cancellation works for any fixed sum of one-coordinate functions. Gaussian geometry provides convenient explicit formulas for the changed terms; additivity provides locality.

20.4 A five-cell numerical account

Take five cells with reference 10X and scale 2X each. Let

$$x^b = (18, 2, 12, 8, 10)X.$$

The corresponding potential terms are $(8, 8, 1/2, 1/2, 0)$, so the global potential is 17. Transfer 4X from cell 1 to cell 2. The successor is $(14, 6, 12, 8, 10)X$, with terms $(2, 2, 1/2, 1/2, 0)$ and total 5.

The global reduction is $17 - 5 = 12$. The local reduction is $(8+8) - (2+2) = 12$. Reading the last three stocks would add no information needed for this action

value, provided the declared state transition really leaves them unchanged and the potential has the stated separable form.

The whole network still has potential 5 after the transfer. The action improved its own affected region; it did not certify that all cells reached their references. Local correctness and system-wide completion are separate propositions.

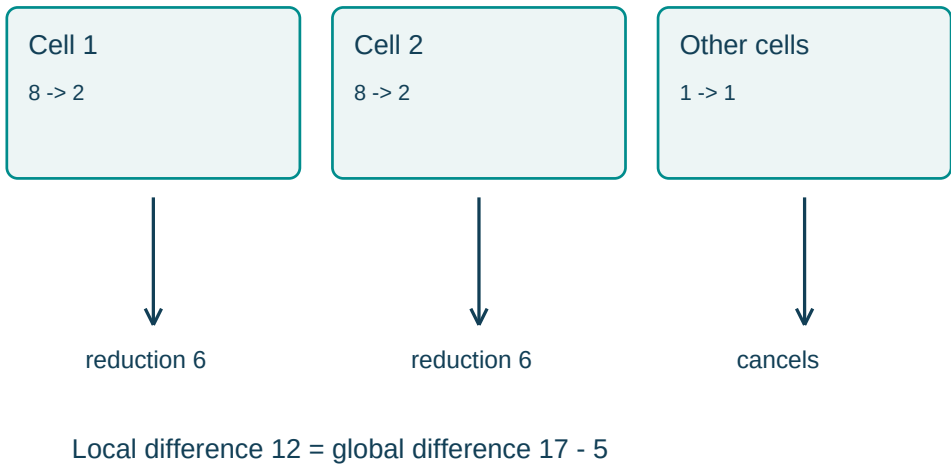


Figure 20.1: The unchanged cells jointly contribute one unit to both global totals and cancel. The local calculation includes both changed endpoints.

20.5 Global refers to the declared boundary

The adjective “global” can sound larger than the mathematics. Here it means all coordinates in the declared model. It does not mean all of Earth, all future time, every ecosystem, every financial consequence or every dimension of wellbeing.

Suppose the modeled network contains only five tanks. Equation (20.3) gives the exact change of that five-tank potential. If the pump also consumes electricity outside the represented boundary, the identity remains true for the tanks while the physical description is incomplete for a claim about the whole operation.

The proper response is to identify the missing carrier and intended scope. A linked electrical account can preserve the physical transformation. A calibrated extension might later place several effects in one declared functional. Neither extension follows merely from calling the five-tank sum global.

20.6 When a neighbour must also be read

A future potential might contain a factor depending on two coordinates, such as a declared interaction $w_{ij}(x_i, x_j)$. Changing x_i can then change that factor even if x_j stays fixed. Reading only the one-coordinate term at i would miss part of the difference.

To express the general rule, suppose

$$V(x) = \sum_{f \in \mathcal{F}} \phi_f(x_{D_f}), \quad (20.4)$$

where D_f is the set of coordinates used by factor f . The affected factors satisfy $D_f \cap W \neq \emptyset$. Every other factor cancels. Evaluation needs the union of coordinates read by the affected factors, which can be larger than the write support.

This is a conditional extension of the same cancellation argument. It does not adopt a new coupled potential for the present book's examples. It explains how to check locality if a later model needs such a potential.

20.7 A counterexample to endpoint-only reading

Consider an explicitly different teaching functional,

$$\tilde{V}(x) = \sum_{i=1}^3 V_i(x_i) + \frac{1}{2}(x_2 - x_3)^2,$$

with dimensionless numerical stock coordinates for this illustration, references $(5, 5, 3)$ and unit scales. Move one unit from cell 1 to cell 2, starting from $(6, 4, 3)$. The successor is $(5, 5, 3)$. Both the reference and physical state have total stock thirteen.

The separable part changes from $1/2 + 1/2 + 0 = 1$ to zero, a reduction of 1. The interaction changes from $1/2(4 - 3)^2 = 1/2$ to $1/2(5 - 3)^2 = 2$, an increase of $3/2$. The complete field reduction is therefore $1 - 3/2 = -1/2$.

An endpoint calculation that retained only V_1 and V_2 would report +1, including the wrong sign. A correct local calculation reads x_3 because the interaction factor involving x_2 depends on it. The direct write support is $\{1, 2\}$; the information neighbourhood is $\{1, 2, 3\}$.

The example also shows why the potential's dependency list is part of its scientific declaration. Locality cannot be imposed independently of the equation. Read too little and a changed term disappears. Read more than necessary and a selector may gain information that its design did not intend to expose.

20.8 Three different graphs

The physical graph connects locations by possible actions. The factor graph connects variables used together in the potential. The action-interaction graph connects action increments whose joint value contains a cross term. These graphs can differ.

Our main potential is separable, so its coordinate Hessian is diagonal. Nevertheless, two actions can both withdraw from A. Their increments overlap in the A coordinate, producing a quadratic cross term. A diagonal potential therefore does not imply independent action values.

Conversely, two physical edges can be far apart in a drawing but belong to the same modeled factor if the declared functional couples their variables. Geographic distance alone is not a reliable definition of a mathematical neighbourhood. The relevant question is dependency, followed by whether that dependency is physically justified.

Keeping the three graphs separate helps a reader avoid a common shortcut: “local” does not mean every kind of interaction is absent. It means the required information and support have been specified.

20.9 Global audit and actor information

A scientific evaluator may calculate full V to compare it with the sum of local effects. This is a useful arithmetic and implementation check. It can reveal a missing endpoint, duplicated factor, stale baseline or sign error.

That capability does not determine what an actor may inspect when selecting an action. If a future study defines local actors, its input boundary must be enforced independently of the evaluator’s larger view. The evaluator can observe a result without providing that result as a decision signal.

Likewise, an EBU-blind random proposal mechanism should not quietly read global potential to choose which candidates to offer. Local valuation and local selection are different design properties. Demonstrating the former by cancellation does not establish the latter.

This chapter uses paper calculations. It does not claim that a particular future software harness has already enforced every information boundary. A software claim needs its own implemented interface, tests and source identity.

20.10 Local computation does not choose governance

The theorem shows that some calculations can use small neighbourhoods. Different devices could compute them, and independent auditors could reproduce

them. These possibilities do not settle who may certify references, own sensors, approve model changes or resolve a dispute.

A centrally governed system can calculate local differences. A distributed institution can perform occasional global audits. Neither arrangement is made inevitable by a sum of potential terms. Political authority must be described in its own language rather than inferred from a graph drawing.

Privacy raises a related issue. A valid physical calculation may need quantities and dependencies while requiring little personal information. Whether a public record must identify an individual is an institutional decision. Mathematical support tells us which variables the calculation needs; it does not demand unlimited disclosure.

20.11 Aggregation can erase the effect

Suppose someone replaces A and B by a single cell containing their total 20X. The internal four-unit transfer leaves that total unchanged. If the aggregate potential depends only on the total, it reports zero change. The fine account reports a reduction of 12.

There is no algebraic contradiction. The aggregate description discarded distribution between the two locations, precisely the information on which the fine potential depends. An aggregate account that is intended to preserve that effect needs additional state, such as internal deviation, or a justified summary of its potential.

This matters for hierarchical systems. A building can expose boundary flows to a district, but a district total alone does not reproduce every room's temperature or water distribution. A sound hierarchy must state what each level retains and which questions cannot be answered after aggregation.

The introductory locality theorem is not an automatic theorem of lossless compression. It gives a starting point for designing summaries and a test for checking them: does the summary preserve the specific field difference the application claims to compute?

20.12 Guided problem: determine the necessary reads

Take six independent Gaussian terms. An action changes coordinates 1 and 4. Which terms must be evaluated? Only V_1 and V_4 , using their before and after coordinates and fixed parameters. Every other term cancels.

Now add a factor $w_{45}(x_4, x_5)$ and another $w_{56}(x_5, x_6)$. The action changes x_4 but not x_5 . It changes w_{45} , so x_5 must be read. It does not directly change

w_{56} , whose two inputs remain the same. The necessary coordinates are 1, 4 and 5 for this declared instantaneous action.

The answer would change if the transformation also updated x_5 or if another factor depended on x_4 . This is why one must inspect the actual action map rather than propagate “affected” indefinitely through a diagram. Future influence and present algebraic dependence are different things.

20.13 Practice and reflection

In the five-cell numerical example, replace the four-unit transfer by two units. The first two potential terms change from (8, 8) to (4.5, 4.5); their reduction is 7. The remaining terms still sum to 1, so the global potential changes from 17 to 10, also a reduction of 7.

Next, draw a physical path of four locations. Identify one transfer’s write support. Add a hypothetical pair factor connecting one of its endpoints to a neighbour and write the resulting information neighbourhood. Explain why a causal prediction about the neighbour’s later behaviour would require more than this factor list.

Finally, explain the aggregation counterexample in words a new reader can understand. A good answer says that total quantity was preserved while internal distribution changed. It does not say that conservation failed or that the local-to-global theorem was approximate.

What to carry forward

Local exactness follows from the cancellation of unchanged factors. Its scope comes from the potential and action dependencies. A complete local calculation can equal a global model difference while leaving empirical completeness, selection, privacy and governance as separate questions.

The factor extension in the next chapter makes the reading neighbourhood explicit for coupled fields. We then follow complete transitions through time before constructing the coalition landscape.

Beyond isolated coordinates

21.1 What the simple field has taught us

The two-stock examples made one fact unusually clear: an action is valued by what it does to a declared physical state, not by the name of the action or the intention of its actor. In a real service network, however, an energy carrier can support a water transfer, a store can protect a medicine shipment, and a damaged route can change the meaning of available stock. We need a way to keep those relationships visible without calculating the condition of the entire planet after every action.

The Gaussian used so far has one term per coordinate. That separability is a useful starting point, not a requirement of exact EBU. The finite definition only asks for one sufficiently defined potential, the same field before and after, and a physical transition represented completely enough for the claim at hand. We can therefore enlarge the field while retaining the identity that made local accounting possible.

21.2 Factors, not necessarily separate cells

Write the potential as a sum of declared factors,

$$V(x) = \sum_{\alpha \in \mathcal{F}} V_{\alpha}(x_{D_{\alpha}}), \quad (21.1)$$

where D_{α} is the set of physical coordinates on which factor α depends. One factor might describe water availability at a reservoir, another a coupled relation between water and pumping energy, and another a temperature-dependent condition of medicine storage. Their formulas, units, references and data sources must be specified; the symbols alone do not confer physical meaning.

Suppose an action changes only coordinates in a write set W . A factor whose dependency set does not meet W has exactly the same value before and after. Consequently,

$$E = V(x) - V(x + \delta) = \sum_{\alpha: D_{\alpha} \cap W \neq \emptyset} [V_{\alpha}(x_{D_{\alpha}}) - V_{\alpha}((x + \delta)_{D_{\alpha}})]. \quad (21.2)$$

This is the precise sense in which the calculation can be local. We evaluate every factor affected by the action, including coupled factors that read unchanged neighbouring coordinates, while unchanged factors cancel. It is not a promise that a poorly chosen global factor can be evaluated from two nearby sensors. Locality is a property to establish for the declared factor graph.

To see the distinction, take three dimensionless coordinates and choose

$$V(x_1, x_2, x_3) = \frac{1}{2}(x_1 - x_2)^2 + \frac{1}{2}x_3^2. \quad (21.3)$$

An action adds q to x_2 and changes neither x_1 nor x_3 . Its exact EBU is

$$E(q) = q(x_1 - x_2) - \frac{1}{2}q^2. \quad (21.4)$$

The x_3 term cancels, but x_1 must be read even though it is not written. This is how a nearby dependency enters a local account. Replacing the first term by a function of x_2 alone would make the calculation easier only by changing the scientific model. The example also shows finite curvature again: the initial slope is $x_1 - x_2$, whereas the value of a whole action includes $-q^2/2$.

Finite locality

For a fixed factorized potential and a represented state increment, all factors untouched by that increment cancel exactly. The result is an algebraic theorem. Whether the chosen factors faithfully represent the physical consequence remains a scientific question.

21.3 What coupling changes

Near a reference x^* , a quadratic extension can be written

$$V(x) = \frac{1}{2}(x - x^*)^\top H(x - x^*), \quad (21.5)$$

with a symmetric positive-semidefinite matrix H on the declared state domain. Nonzero off-diagonal entries represent interactions in the chosen field. For a finite increment δ , direct expansion gives

$$E = -\nabla V(x)^\top \delta - \frac{1}{2}\delta^\top H\delta. \quad (21.6)$$

No small-action approximation appears here. If H depends on state, the same exact endpoint definition still works, but this particular quadratic expansion does not.

Coupling also changes the reading neighbourhood. An action may write one coordinate while its value depends on another coordinate because a factor

reads both. The calculating node needs the relevant neighbouring state or a trustworthy authenticated summary. It does not need every unrelated coordinate merely because those coordinates exist in the world.

This distinction matters for governance. A mathematical proof that unaffected factors cancel is not a proof that every participant has access to the required measurements, that private data should be disclosed, or that a distributed network will reach agreement. The data protocol and the institution each require their own design.

21.4 Coupling prepares the joint-action question

When several actions share a factor, their joint value must be evaluated on the complete combined endpoint. Later chapters construct the coalition table and separate its interactions before considering any division among actors. For now the important lesson is that a factor's complete dependencies determine what must be read.

21.5 The limit of a single number

A physical system may carry water, energy, medicine and ecological function in unlike units. Declaring a scalar V requires a scientifically and institutionally justified relation among represented effects. Where that relation is not justified, a vector of separate accounts is more honest. One can still ask whether each physical balance closes and whether each service requirement is met. What one cannot do is add unlike consequences under an unstated exchange rate and call the result a law of nature.

The question of substitution is particularly important near essential limits. An extra unit of one abundant carrier may not replace a missing unit of another carrier that performs a specific function. A potential can encode such dependence through a coupled factor or a hard feasibility constraint, but it has to say which. A hard physical impossibility cannot be cured by a sufficiently favourable value elsewhere in the potential. Conversely, a scientific claim that two effects are exchangeable needs a measured basis; it cannot be read from the fact that a scalar sum is convenient.

There is a practical way to test the meaning of a proposed factor. Hold its input coordinates fixed while changing something it claims to represent. If the represented function changes but the factor cannot respond, the model is incomplete for that claim. If the factor responds to a coordinate that cannot be observed or verified at action time, the proposed local evaluator is incomplete. These are design tests, not failures of the algebra. The algebra is deliberately transparent enough to make them visible.

The factor view therefore extends the exact mathematics without pretending that model construction is finished. We now know how to calculate a consequence in a fixed field. The next question follows complete events through time: how do their signed records add when every successor becomes the next predecessor?

Action receipts and exact closure

22.1 What a physical receipt records

A shop receipt records a commercial exchange. An EBU action receipt names a physical predecessor, an accepted transformation, its successor and the version of the potential used to compare them. The two kinds of receipt may coexist, but one does not validate the other. A payment cannot prove that a reservoir recovered, and a positive EBU does not establish who was paid.

The preceding factor chapters showed why a complete change can often be evaluated from affected factors alone. We now connect several correctly evaluated changes. The central fact is simple: the state after one action is the state before the next. Its potential appears once with each sign, and cancels.

22.2 Follow the actual chain

Consider a sequence of authoritative states under one fixed potential,

$$x_0 \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_m. \quad (22.1)$$

For each complete transition define

$$E_r = V(x_{r-1}) - V(x_r). \quad (22.2)$$

The arrows are physical or modeled events, not an arbitrary sort order for files. If several actions occur together, one arrow can be their joint transition, retaining the identities of its children. A later allocation of that group's total is another question.

Summing the consecutive differences gives

$$\sum_{r=1}^m E_r = V(x_0) - V(x_1) + V(x_1) - V(x_2) + \cdots + V(x_{m-1}) - V(x_m) \quad (22.3)$$

$$= V(x_0) - V(x_m). \quad (22.4)$$

Equivalently,

$$\sum_r E_r + V(x_m) - V(x_0) = 0. \quad (22.5)$$

This is exact telescoping for any state function V . Gaussian geometry is not needed. Neither is an actor willing to improve the state. The intermediate states still matter: without them we cannot test whether each action was physical, whether its recorded value was correct or whether an event is missing.

What closure proves

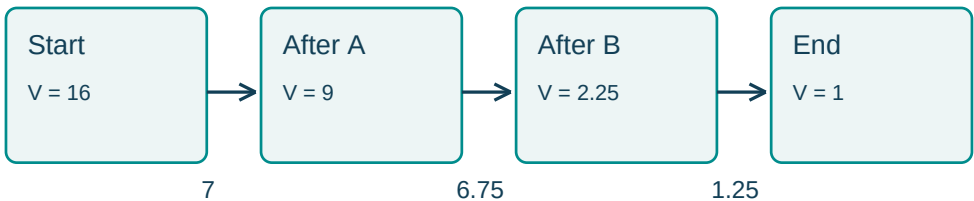
Correctly linked, complete transitions under one fixed potential have a total EBU equal to their endpoint potential reduction. The identity checks accounting consistency. It does not prove fair ownership, useful service or eventual recovery.

22.3 One sequence, not three fresh beginnings

Use the two-cell potential with reference (10,10) and scales (2,2). Three successive lossless transfers carry the state through

$$(18, 2) \rightarrow (16, 4) \rightarrow (13, 7) \rightarrow (12, 8), \quad V : 16 \rightarrow 9 \rightarrow \frac{9}{4} \rightarrow 1.$$

The action values are 7, $27/4$ and $5/4$, which add to $15 = 16 - 1$. The middle transfer starts from (16,4), not from (18,2). Reusing the initial slope for it would quote a different imagined action, not the second event in this sequence.



Action values: $7 + 6.75 + 1.25 = 15$

Endpoint difference: $16 - 1 = 15$

Figure 22.1: Three consecutive finite differences telescope to fifteen. Every action uses the preceding endpoint as its baseline.

Suppose a record assigned the first and third lines to one actor and the second to another. Their historical contributions would be $33/4$ and $27/4$, adding to the same 15. This partition uses an explicit owner rule; the potential did

not choose the owners. Individual accounts can remain signed. There is no physical theorem requiring a negative contribution to be prohibited or prefunded.

22.4 Why the negative line must remain

Let an action take potential from 1 to 4, then another take it from 4 to 0. Their EBU values are -3 and $+4$. The total is $+1$, exactly the endpoint reduction. Reporting only the positive line would manufacture a four-unit improvement from a one-unit net change. A society may decide how to support an essential negative action, but it must not remove the action's physical sign from history to make that decision easier.

22.5 External events need their own line

Now let an external event take x_t to x_t^b , followed by an actor action from x_t^b to x_{t+1} . Define the external potential change by

$$D_{\text{ext},t} = V(x_t^b) - V(x_t).$$

The actor's EBU is $E_t = V(x_t^b) - V(x_{t+1})$, so

$$\sum_t E_t = V(x_0) - V(x_N) + \sum_t D_{\text{ext},t}. \quad (22.6)$$

An external worsening can create deviation that a later actor reduces. An external improvement can remove deviation without actor contribution. Both signs belong in the record. For example, if an external redistribution moves potential from 0 to 4 and an actor then reduces it to 1, the actor records 3, not the -1 obtained by comparing the wrong endpoints.

If the potential itself changes between events, the calculation needs an explicit field-change line too. Comparing values produced by two different rulers as though they were one fixed V breaks Equation 22.5. We return to that harder question after the fixed-field theory is complete.

22.6 Balance is not carrier conservation

In a lossless transfer, physical stock is conserved while V changes. States $(18, 2)$ and $(14, 6)$ each contain twenty stock units; their potentials are sixteen and four. A twelve-EBU reduction did not create or destroy twelve litres. Equation 22.5 is a reconciliation of values, not a conservation law for matter or energy.

An auditor therefore checks both ledgers. First follow the physical states, routes, carriers and boundaries. Then verify potential versions, event identities and signed EBU lines. A residual can expose an omitted event, duplicate receipt, stale baseline or arithmetic error; it cannot by itself reconstruct the missing physical cause. The next chapter uses this discipline to examine closed loops and action splitting.

Closed cycles, splitting and external change

23.1 Can damage and repair mint value?

Imagine moving water away from a useful distribution and then returning it. The return movement may have positive EBU. If an actor reports only that movement, it looks like a renewable source of credit. The first movement, however, raised the represented potential and had the opposite sign. A complete account follows both events from their actual baselines.

For an undriven sequence that begins and ends at the same complete state $x_0 = x_m$ under one fixed potential, Chapter 22 gives

$$\sum_{r=1}^m E_r = V(x_0) - V(x_m) = 0. \quad (23.1)$$

The theorem does not need a Gaussian. It needs the complete chain and the same state function at every comparison. If a transition is a simultaneous group, its exact group total is one line in the chain. The equation still does not dictate how that line is divided among people.

At reference (10, 10) with scales (2, 2), transfer two units one way. The new state (8, 12) has potential one, so the action is worth -1 . Transfer the same amount back and the return action is worth $+1$. The positive restoration is real within the account; the claim of positive *net* creation from the complete loop is false.

23.2 The represented state must really return

A battery can recover its displayed charge while losing health. A tank can recover its volume while losing quality. A vehicle can return to a depot after using fuel. If those changes are part of the physical claim, a loop in one displayed coordinate is not a loop in the complete state. A carrier or boundary record must show what happened, and a potential over a fuller state may give a different result.

Equal potential is weaker still than equal state. States (18, 2) and (2, 18) have the same Gaussian potential but very different source stocks and future options. Their direct finite EBU is zero; the physical state did not return. An

audit of future feasibility needs the actual coordinates, not just their scalar value.

23.3 Dividing an action cannot divide away its consequence

From (18, 2), an unsplit four-unit transfer ends at (14, 6) and reduces potential by twelve. Split it into two genuine sequential two-unit actions. The first moves potential $16 \rightarrow 9$ and records seven. The second begins from that new state, moves $9 \rightarrow 4$ and records five. Thus $7 + 5 = 12$. Calling both pieces seven would reuse the original state twice.

This is refinement invariance of a state-function difference:

$$[V(x) - V(y)] + [V(y) - V(x^b)] = V(x) - V(x^b).$$

It does not claim that physically stopping and restarting a pump always has the same complete effects as running it continuously. Different energy use, heat or wear would change the complete physical transition if those carriers were represented. The identity says that a consequence cannot be made to disappear by changing how its record is sliced.

23.4 An aggregate cycle need not restore every actor's history

Suppose a return loop contains an action attributed to one owner and its reverse attributed to another. The total EBU is zero, but the two historical accounts can change by opposite amounts. A three-cell example from the physical foundation begins at (4, 4, 4): $B \rightarrow C$ has EBU -1 assigned to B , and the reverse $C \rightarrow B$ has EBU $+1$ assigned to C . The state returns; the owner vector changes by (0, -1 , $+1$). This is an existence example, not a claim that every possible attribution rule has such loops.

The distinction matters because exact group closure is sometimes mistaken for a fairness theorem. It is not. Aggregate EBU is a difference of state values, while a person's historical attribution can depend on the path of events and on the declared ownership rule.

23.5 External change can recreate work

If a disturbance moves a system away from its reference and an actor restores it, positive actor EBU may recur. That does not violate Equation 23.1: the interval contains an external event. In the notation of Chapter 22, the action total is

$$\sum_t E_t = V(x_0) - V(x_N) + \sum_t D_{\text{ext},t}.$$

If repeated disturbances increase potential by four and subsequent actions reduce it by four, the positive actor entries are balanced by real external changes. A request or desire alone is not a physical sink. To claim a recurring demand caused a stock change, a model must represent consumption, transformation or boundary exchange.

Nor may an actor-created disturbance be silently renamed an external one. The equation tells us what entries are needed; provenance and causal review tell us whether they were classified honestly. The mathematics cannot reconstruct an omitted event from an apparently convenient total.

23.6 Changing the ruler is a third kind of event

At the same state $(14, 6)$, scales $(2, 2)$ give potential four, while scales $(4, 4)$ give potential one. No water moved. The three-unit difference came from changing the field. If old and new values are joined without a field-change entry, intermediate potentials will not telescope as though one ruler had been used throughout.

This is not an argument against improving a model. It is a demand that learning about the world not be disguised as an action upon the world. Later we ask a harder question: whether EBU values earned under different physical fields can nevertheless share one stable physical interpretation.

Finally, no cycle equation proves recovery. On a fixed finite-mass domain a continuous Gaussian potential is bounded even if the system stays trapped far from its reference. A motion law, actor rule and disturbances must be specified before attraction or long-run homeostasis can be tested. The next chapter first examines the more immediate complication: several actions sharing one nonlinear comparison at the same time.

Several actors on one nonlinear field

24.1 Known quantities do not make value linear

Two couriers each deliver two units from A to B. We know their identities, quantities and directions. Yet valuing each delivery as though the other did not exist gives a different sum from valuing the combined four-unit movement. The difficulty is not missing quantity data. It is the curvature of the potential.

Physical increments can add linearly while the value of their sum remains nonlinear. This distinction is already present in a diagonal Gaussian potential. It does not require a mysterious nonlocal force or a new physical interaction between the couriers.

This chapter retains the original shared-source lesson and updates its arithmetic. A later chapter makes the common-path decomposition explicit. A decomposition that closes mathematically is useful, but its interpretation must be stated: path accounting, physical realization, causal contribution and institutional entitlement are different claims.

24.2 The two-courier counterexample

Use the repeated state $(18, 2)X$, references $(10, 10)X$ and scales $(2, 2)X$. Let the increments be

$$\delta_A = (-2, 2)X, \quad \delta_B = (-2, 2)X.$$

Either courier alone would produce $(16, 4)X$, reducing potential from 16 to 9. The two same-baseline singleton values therefore sum to $7 + 7 = 14$.

Their actual joint increment is $(-4, 4)X$, producing $(14, 6)X$ with potential 4. The joint field value is 12. The difference of two is not a rounding error or an unobserved physical loss. It is the discrepancy between a joint finite difference and a sum of different counterfactual finite differences.

A counterfactual sum is not the joint account

Each singleton value asks what would happen if that child acted alone at the frozen baseline. The joint value asks what happens when the accepted children act together. On a nonlinear field these are different comparisons, even when the child quantities are known exactly.

If the couriers actually act sequentially, the values are 7 and 5, which sum to 12. A physically meaningful sequence supplies different baselines. If execution is simultaneous, inventing that sequence afterward gives an allocation convention, not an observed intermediate history.

24.3 The exact group formula

Let the accepted increments be δ_a , and define $\delta_G = \sum_{a \in G} \delta_a$. Under fixed Gaussian geometry,

$$E_G = -\mu(x^b)^T \delta_G - \frac{1}{2} \delta_G^T H \delta_G \quad (24.1)$$

for the complete represented physical transition. Here $H = \text{diag}(1/\sigma_i^2)$ and $\mu(x^b) = H(x^b - x^*)$.

The same-baseline singleton sum uses δ_a separately in each quadratic term. Expanding the joint square gives

$$E_G - \sum_a E_a^{(0)} = - \sum_{a < b} \delta_a^T H \delta_b. \quad (24.2)$$

The superscript (0) marks the common frozen baseline. It does not denote an observed first step. The right-hand side is a signed cross-term diagnostic for these fixed increments and this fixed potential.

For the two couriers, $H = \text{diag}(1/4, 1/4)$ and $\delta_A^T H \delta_B = 2$, giving $12 - 14 = -2$. The shared coordinates and directions determine the sign.

24.4 The discrepancy can have either sign

Convexity does not make every group discrepancy negative. Consider opposing increments from the reference state (10, 10):

$$\delta_A = (-2, 2), \quad \delta_B = (2, -2).$$

Each action alone increases potential to 1, so each singleton value is -1 . Their joint increment is zero and the ideal group field value is zero. Thus $E_G - (E_A^{(0)} + E_B^{(0)}) = 2$.

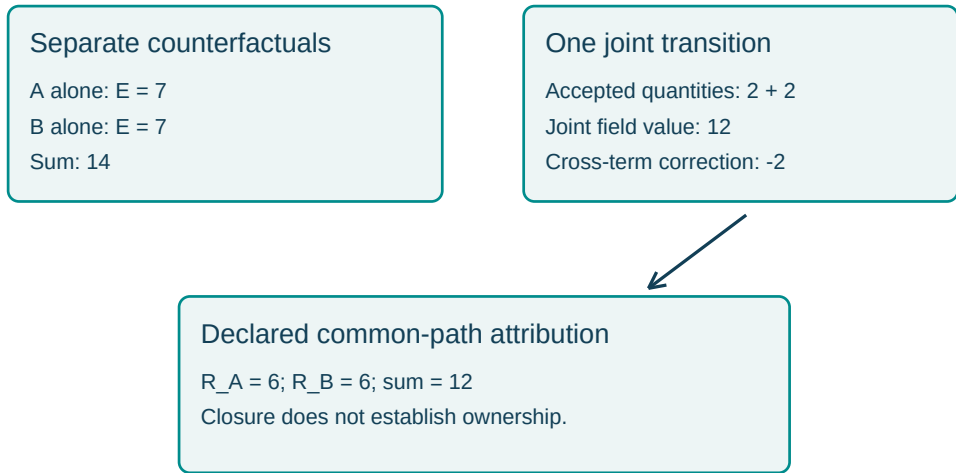


Figure 24.1: Two separately evaluated two-unit actions each have value seven at the frozen baseline. The joint four-unit action has value twelve, not fourteen. A later declared straight-path decomposition assigns six to each.

The cross product is -2 , agreeing with equation (24.2). Two same-direction increments gave a negative discrepancy; opposing increments give a positive one. The physical feasibility of the combined flows must still be declared. Endpoint cancellation does not automatically permit unlimited circulation or establish that the real process is free.

For disjoint coordinate supports under diagonal H , the cross product is zero. The field values then add. Shared physical constraints can nevertheless couple the permitted quantities. An arithmetic additivity result does not remove a common pump-capacity or source-availability constraint.

24.5 Three comparisons that must stay separate

The group-versus-singletons discrepancy compares simultaneous execution with two or more counterfactuals that each start alone. A parallel-versus-sequential comparison uses a complete sequential history. An institutional allocation compares ways of distributing a given joint number.

With fixed additive increments and the same endpoint, each physically feasible sequential order and the group have the same total field value. A feasible joint endpoint alone does not guarantee feasible intermediate states in every order. The singleton sum can still differ. Thus a nonzero singleton discrepancy does not establish that simultaneity created a physical efficiency gain.

If a resolver changes quantities when actions are replayed sequentially, or a delay changes the state, the sequential endpoint can differ. The comparison then has to identify the ordering, resolver, timing and endpoint. Reporting an “interaction benefit” without naming this comparator leaves its meaning incomplete.

24.6 Grouping follows dependencies

A shared coordinate often requires joint valuation. Shared constraints can also require joint physical resolution even when the eventual increments happen to have disjoint support. Relevant dependencies include overlapping execution or reservation intervals, common capacities, and effects on another action’s feasibility or completion.

A conservative grouping design collects connected dependent actions within a compatible accounting boundary. This prevents an action indirectly linked through a third child from being treated as independent without examination. It is a design rule requiring declared supports and timing; it is not established merely by two actors belonging to the same organization.

Every group should preserve its child records. Joint evaluation does not erase who sent what, which edge was used or which evidence supports the quantities. It acknowledges that the group’s nonlinear field total must be computed together.

24.7 The next question is the complete interaction landscape

The two-action cross term is the first member of a larger hierarchy. Before dividing a joint value among actors, we will examine all admissible subsets, subtract lower-order contributions and identify what truly belongs to a higher order. Chapter 43 later returns to the common-path allocation and its limits.

The full group value already includes every interaction under its declared response. An interaction diagnostic must never be issued again as additional value on top of that total. The next chapter defines the common-baseline table that makes those comparisons precise.

A landscape of possible groups

25.1 Before synergy, specify what is being compared

Suppose a clinic needs a delivery, refrigeration and a repaired access route. The three activities may be useful together in a way that their isolated descriptions miss. Yet the phrase “useful together” is incomplete until we specify the alternatives. Does the delivery have the same quantity when refrigeration is absent? Does a failed route reduce the delivered amount? Does the baseline change between attempts? These choices define the comparison.

A coalition table makes the alternatives explicit. It does not assume that actions are additive. Each permitted group is assigned its complete endpoint by a declared physical response rule. We can then ask how the values of the groups fit together. The endpoint rule might be a fixed sum of increments, an engineering resolver, or a well-posed flow. The arithmetic to come works at the level of the resulting table.

25.2 The complete Boolean cube

Definition 25.1 (Coalition landscape). Let $N = \{1, \dots, n\}$ index distinct actions. For every $S \subseteq N$, declare an endpoint x_S under the same initial condition x_0 , field V , boundary and comparison protocol. Set

$$E(S) = V(x_0) - V(x_S). \quad (25.1)$$

For the normalized fixed-baseline landscape require $x_\emptyset = x_0$, so $E(\emptyset) = 0$.

A group is a subset, so its label does not specify an order. With two actions there are four entries: neither, A alone, B alone, and both. With three there are eight. The mathematical structure is the Boolean lattice, ordered by set inclusion. The word Boolean means that each action is absent or present in the table; it does not require the physical state to be binary.

All entries must refer to one field and comparator rule. Changing a reference halfway through the table changes the question. So does replacing A’s fixed quantity in one row with a resolver-dependent quantity in another without saying so. A resolver-dependent table is legitimate when declared, but it is a different landscape from the fixed-increment table.

25.3 One baseline, several counterfactual endpoints

The table is a family of comparisons, not a record asserting that the world simultaneously visited all its endpoints. An observed group can supply one row. The remaining rows may come from repeated controlled interventions, a justified model, or separate physical observations with a defensible common-baseline design. Their provenance must be stated.

For EBU this is an important separation of duties. A joint action record needs its before and after states. An interaction study needs enough alternative groups to distinguish individual and joint effects. One cannot infer a three-way interaction merely by measuring the final state after three activities. The lower-order rows are part of the information needed to know what was additional.

In practical terms, this can keep a promising project from being credited for a complementarity that has never been measured. It can also reveal a real complementarity that a list of separate prices or work orders obscures. Both possibilities matter for decisions about shared infrastructure.

25.4 A complete two-action table

Return to two couriers each moving two units from $(18, 2)$ under the fixed reference $(10, 10)$ and scales $(2, 2)$. All four alternatives are physically permitted in this teaching domain:

Group	Endpoint	Potential	$E(S)$
$\emptyset$	$(18, 2)$	16	0
$\{A\}$	$(16, 4)$	9	7
$\{B\}$	$(16, 4)$	9	7
$\{A, B\}$	$(14, 6)$	4	12

The duplicate singleton endpoints are not an error: distinct couriers can have equal physical increments. Their identities are distinct, while their effects under this declaration coincide. The table says the joint value is twelve and the sum of isolated comparisons is fourteen. It does not say that twelve plus a further interaction reward may be issued. The joint value already includes the interaction.

25.5 Conditional addition is a change of baseline in the table

For disjoint groups A and B , define the conditional increment

$$E(B \mid A) = E(A \cup B) - E(A) = V(x_A) - V(x_{A \cup B}). \quad (25.2)$$

This is an exact difference between table entries. If the declared physical execution of B after A actually takes x_A to $x_{A \cup B}$, it also represents that live sequential action. Without this compatibility it is only a table contrast. A sequential resolver may instead reach another endpoint.

For two actions, the discrepancy between B after A and B alone is

$$E(B \mid A) - E(B) = E(AB) - E(A) - E(B), \quad (25.3)$$

using $E(\emptyset) = 0$ and writing AB for $\{A, B\}$. In the courier example, B after A gives five rather than seven, a difference of minus two. The interaction measures how the context changes the increment.

An actual sequential endpoint $y_{A \rightarrow B}$ gives total $V(x_0) - V(y_{A \rightarrow B})$. Comparing it with the simultaneous group yields

$$E(AB) - E_{A \rightarrow B} = V(y_{A \rightarrow B}) - V(x_{AB}). \quad (25.4)$$

This is zero when the complete endpoints match, even though the same-baseline interaction can be nonzero. The distinction prevents a curvature correction from being advertised as proof that simultaneous organization is more efficient.

25.6 When the full table cannot be physically constructed

Two actions can each be permitted alone but impossible together because they share a source or a machine. Conversely, an action may be possible only with another action supplying a necessary service. If a required row is physically inadmissible, the full Boolean interaction is not an observed physical quantity on that domain.

There are several honest responses: restrict the study to a fully admissible motif; declare a modeled extension and label its unobserved rows; or work on the actual feasible partially ordered set. These are different scientific objects. Filling an impossible row with zero does not recover the original comparison: zero is a value, not a marker for missing permission.

For a finite feasible poset P , inversion uses its own incidence function μ_P , determined by

$$\mu_P(a, a) = 1, \quad \sum_{a \leq c \leq b} \mu_P(a, c) = 0 \quad (a < b). \quad (25.5)$$

One must inspect the intervals rather than transplant Boolean signs. There is a useful nuance: if the feasible family is downward closed, every interval $[A, B]$ with $A \subseteq B$ in that family is still a full Boolean interval. Its signs remain $(-1)^{|B \setminus A|}$. There may nevertheless be no feasible top group on which to ask the intended high-order question.

25.7 What a system gains from declaring the alternatives

A coalition table turns an ambiguous assertion about cooperation into a reproducible question. It identifies whose absence or presence is being varied, what stays fixed, and what physical endpoint each variation supplies. This can help distinguish shared-source competition from genuine complementarity, and prevent an institution from paying multiple times for the same represented improvement.

The benefit to society would depend on how the information is used. A table could help explain why a water repair and an energy repair belong in one investment decision. It could equally show that an apparently attractive bundle has no extra physical benefit under the chosen model. Neither outcome ranks the people involved. The next chapter develops the exact subtraction that extracts those relationships.

Finding the part that only a group contributes

26.1 Subtract what has already been explained

When two actions act together, their total can include what each would do alone and a further correction for being combined. With three actions, subtracting only the singleton values is insufficient. Some of the difference may already be explained by pairs. A genuine three-way coefficient is what remains after every lower-order contribution has been accounted for exactly once.

This idea is easier to understand as bookkeeping than as new terminology. If a sum contains individual terms and pair terms, those terms must be removed before naming the remainder “three-way.” Möbius inversion provides the general rule for that subtraction. The name refers to a well-established mathematical construction, not an additional physical force discovered by EBU.

26.2 The interaction coefficient

Definition 26.1 (Boolean Möbius coefficient). For a complete finite coalition table $E : 2^N \rightarrow \mathbb{R}$, define

$$m_E(T) = \sum_{S \subseteq T} (-1)^{|T|-|S|} E(S). \quad (26.1)$$

In particular $m_E(\emptyset) = E(\emptyset)$. For the normalized landscape the empty coefficient is zero.

The sign alternates as entries are added or removed. A singleton has $m_E(\{i\}) = E(i) - E(\emptyset)$. A pair has

$$m_E(ij) = E(ij) - E(i) - E(j) + E(\emptyset). \quad (26.2)$$

Thus the two-courier coefficient is $12 - 7 - 7 + 0 = -2$. It is a negative correction to the same-baseline singleton sum. It is neither a two-unit physical loss nor a two-unit charge added to a twelve-unit receipt.

For three actions the alternating sum is

$$m_E(123) = E(123) - E(12) - E(13) - E(23) \\ + E(1) + E(2) + E(3) - E(\emptyset). \quad (26.3)$$

The singleton signs become positive because each singleton was subtracted twice inside the three pair terms. Inclusion–exclusion restores the correct count.

26.3 Why all the pieces reconstruct the whole

Theorem 26.2 (Boolean inversion). For every $S \subseteq N$,

$$E(S) = \sum_{T \subseteq S} m_E(T). \quad (26.4)$$

The coefficients are unique.

Proof. Insert the definition of $m_E(T)$ and collect the coefficient of one fixed $E(A)$ with $A \subseteq S$. That coefficient is $\sum_{A \subseteq T \subseteq S} (-1)^{|T|-|A|} = (1-1)^{|S|-|A|}$. It is zero unless $A = S$, when it equals one. This proves reconstruction. Uniqueness also follows recursively: the empty value fixes the empty coefficient, and the value at each larger set fixes its coefficient after all proper-subset coefficients are subtracted. $\square$

No derivatives, dynamics or probability laws appear in this proof. Any finite table has the decomposition. The physical quality of an EBU interaction therefore comes from the quality of its declared coalition endpoints and potential, not from the sophistication of the transform applied afterward.

26.4 Three equal actions reveal what pairs already explain

Let three couriers each carry one unit on the same ideal edge from (18, 2). All groups are admissible. A group of size r has value $4r - r^2/4$, so the singleton, pair and triple values are $15/4$, 7 and $39/4$.

Each pair coefficient is $7 - 15/4 - 15/4 = -1/2$. The triple coefficient is

$$\frac{39}{4} - 3(7) + 3\left(\frac{15}{4}\right) = 0. \quad (26.5)$$

Yet the triple value differs from the singleton sum by $39/4 - 45/4 = -3/2$. All of that discrepancy is the sum of the three pair corrections. Calling the entire $-3/2$ a three-way interaction would confuse the size of the group with the order of its unexplained interaction.

This distinction matters when discussing cooperation. A large project can contain many pair dependencies without containing a nonzero high-order coefficient. Conversely, a modest three-action motif can contain a truly three-way dependence that no pair comparison reveals. A system should report the order it measured, not the number of participants in its photograph.

26.5 Interaction is not necessarily beneficial

The term synergy is often used for a positive extra benefit. We will use “interaction” for the signed coefficient and, where synergy appears, keep the sign explicit. A positive coefficient means the group value exceeds the prediction from its proper-subset coefficients under this table. It does not establish that the total group value is positive, that the group is physically efficient, or that everyone benefits.

For example, two opposite transfers at a reference state can each have negative standalone value while their joint endpoint is unchanged. Their positive pair coefficient corrects the sum back to zero. It does not create a positive group outcome. Similarly, a positive triple term can coexist with sufficiently negative singletons that the full group remains adverse.

Signs can guide investigation when the comparison is meaningful. Shared depletion may produce a negative pair correction, while compatible actions can remove a bottleneck and produce a positive one. Which explanation applies requires the response model and physical evidence. The transform alone does not identify a cause.

26.6 The EBU sign comes from the potential difference

For nonempty T , the constant $V(x_0)$ cancels in the alternating sum. Therefore

$$m_E(T) = - \sum_{S \subseteq T} (-1)^{|T|-|S|} V(x_S). \quad (26.6)$$

The minus sign is essential. A positive potential interaction represents an extra burden in the composite endpoint landscape; its EBU coefficient has the opposite sign. Retaining this convention throughout will connect the same coefficient to derivatives, log probabilities and, under narrower assumptions, medium entropy.

An additive constant in V changes no EBU and no nonempty interaction. A positive rescaling of V rescales all values and coefficients, while changing references or coordinate weights can change their relative sizes and signs. Interactions are meaningful within their declared denomination and field version.

26.7 Established mathematics, a specific EBU application

Rota's incidence-algebra treatment gives the general inversion language. Cooperative-game theory calls the corresponding coefficients Harsanyi dividends; equivalent representations also occur in pseudo-Boolean functions, capacities and multilinear game extensions [31, 32, 33, 34, 35]. Epistasis studies use related decompositions to describe effects that depend on combinations of factors [36].

EBU's contribution here is a disciplined application: define one physically interpretable endpoint table, retain the sign of potential reduction, and connect that table to the other declared layers. It is not a claim to have invented inclusion–exclusion, cooperative-game dividends or high-order interaction. Nor should a baseline-specific dividend be confused with a distribution-averaged sensitivity index; the averaging measure would add another declaration.

What this law makes possible

The transform gives a lossless decomposition of a complete coalition table. It prevents pair effects from being renamed three-way effects and prevents interaction from being added twice to the group value. Its practical usefulness depends on valid alternatives and adequate measurements.

We have a formula for every order. The next step is to understand why each higher order asks how a lower-order relationship changes in a new context.

When the relationship itself changes

27.1 A third action can change a pair

A water delivery and a refrigeration repair may have a certain combined effect when a clinic already has electricity. If the electrical connection is absent, their relationship may be different. The question is not merely what the third action contributes alone. It is whether the third action changes what the first two contribute together.

This is the intuitive meaning of the third-order coefficient. Higher order continues the same pattern: one more action changes a lower-order interaction. The recursion gives a way to teach, calculate and interpret the hierarchy without treating every order as an unrelated formula.

27.2 A finite difference adds one action

For $i \notin S$, define

$$(\Delta_i E)(S) = E(S \cup \{i\}) - E(S). \quad (27.1)$$

The operator compares a row with and without i while holding its background S fixed. For distinct actions i, j , the difference operators commute wherever all required rows exist. Expanding either order gives the same four entries.

For disjoint T and S , write $\Delta_T = \prod_{i \in T} \Delta_i$. Then

$$(\Delta_T E)(S) = \sum_{A \subseteq T} (-1)^{|T|-|A|} E(S \cup A). \quad (27.2)$$

It is the interaction among the actions in T at background S . At the empty background it is exactly $m_E(T)$.

Theorem 27.1 (Discrete difference and recursion). On a complete Boolean table,

$$m_E(T) = (\Delta_T E)(\emptyset), \quad (27.3)$$

$$(\Delta_{T \cup \{k\}} E)(S) = (\Delta_T E)(S \cup \{k\}) - (\Delta_T E)(S), \quad (27.4)$$

$$(\Delta_T E)(S) = \sum_{R \subseteq S} m_E(T \cup R), \quad (27.5)$$

where $S, T, \{k\}$ are disjoint as needed.

Proof. The first identity is the alternating expansion of the product of differences. The second is the definition of one more difference. For the third, expand each $E(S \cup A)$ by Boolean inversion. A coefficient $m_E(B)$ survives the differences precisely when B contains every element of T and its remaining elements lie in S . Every such coefficient survives once. $\square$

This proof also explains why context matters. A pair difference at a populated background can include the empty-background pair coefficient and higher-order coefficients involving that background. Calling all of it “the pair coefficient” without naming the background hides those contributions.

27.3 The triple written as a comparison of pairs

Define the A–B interaction with C present by

$$I_{AB|C} = E(ABC) - E(AC) - E(BC) + E(C). \quad (27.6)$$

With C absent it is $I_{AB|\emptyset} = E(AB) - E(A) - E(B) + E(\emptyset)$. Their difference is

$$m_E(ABC) = I_{AB|C} - I_{AB|\emptyset}. \quad (27.7)$$

Thus a third-order finding means the pair relationship changed when the third action entered. It does not mean that three standalone benefits were simply added.

27.4 A nonlinear response makes the recursion visible

Consider dimensionless controls $z_i \in \{0, 1\}$ and the constructed response

$$x(z) = z_1 + z_2 + z_3 + z_1 z_2, \quad V(x) = \frac{1}{2}x^2. \quad (27.8)$$

Here $x_0 = 0$. The extra $z_1 z_2$ says that the first two controls jointly change the endpoint beyond their separate increments. It is a teaching response map, not a calibrated clinic model. Its complete table is

Group	x_S	$E(S)$
$\emptyset$	0	0
1, 2, 3 individually	1	−1/2 each
12	3	−9/2
13, 23	2	−2 each
123	4	−8

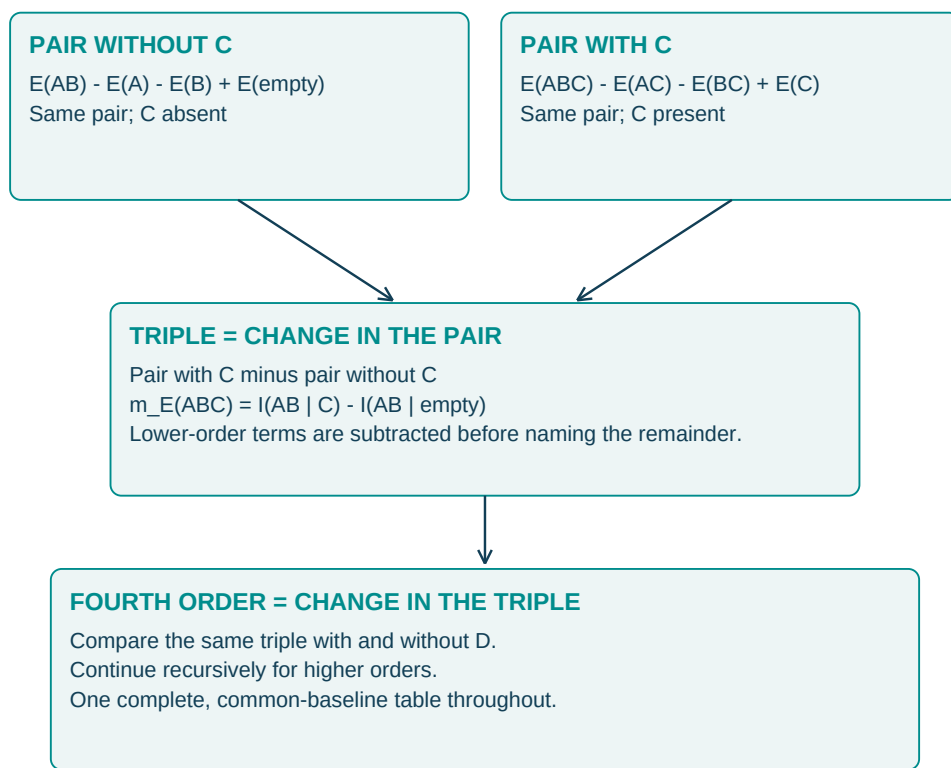


Figure 27.1: Interaction as a changing relationship. The third-order coefficient compares the same pair contrast with and without C. The fourth order asks how this difference changes when D is present. Every box uses one declared coalition table.

The A–B pair contrast without C is $-9/2 + 1/2 + 1/2 = -7/2$. With C it is $-8 + 2 + 2 - 1/2 = -9/2$. Their difference is -1 , the genuine triple coefficient. All endpoint values here are negative because the example moves away from the minimum of V . This is useful: it stops a discussion of structure from quietly turning into a success story. The response illustrates higher-order dependence regardless of whether a particular institution would want the action.

27.5 A quadratic null also has an interpretation

For fixed additive increments and a quadratic potential, the A–B pair coefficient is independent of whether C is included. Therefore the triple difference is zero. The cancellation is a substantive prediction about this declared land-

scape. It is not a statement that C has no standalone effect, that all actors are independent, or that the physical system contains no shared constraints.

If a measured triple differs from zero beyond its uncertainty, one or more assumptions behind that null may fail. The response could be nonlinear, the potential nonquadratic, the baseline inconsistent, or the measurements biased. The result narrows a question; it does not uniquely identify the responsible mechanism. Later chapters make these alternatives explicit.

27.6 Fourth order does not require a new kind of arithmetic

Let $I_{ABC|D} = (\Delta_A \Delta_B \Delta_C E)(\{D\})$. Then

$$m_E(ABCD) = I_{ABC|D} - I_{ABC|\emptyset}. \quad (27.9)$$

The pattern continues at every order. This is an exact recurrence, not an approximation valid only for small actions. It works because the table has discrete absent/present coordinates and all needed rows are supplied.

In an EBU research programme the recursion supports small, interpretable studies. Begin by checking two-action interactions. Add a third action only when the scientific question concerns a change in that relationship. This can reduce the temptation to fit a large opaque model whose high-order coefficients lack a physical interpretation.

27.7 What this could improve in a service system

Some infrastructure failures arise from dependencies among dependencies. A repair can help only when another service works, and that relationship can itself depend on a third condition. A recursive interaction account provides a vocabulary for such situations. It could help identify which combination of interventions deserves joint evaluation and which apparent benefit is already explained by smaller components.

The possible societal gain is better targeting of investigation and investment, not an automatic allocation rule. The table does not decide which community deserves service first. It can provide evidence about how the represented actions depend on one another, leaving need, rights and access as explicit decisions. We now turn this hierarchy into an exact finite analogue of a Taylor expansion.

An exact Taylor language for finite choices

28.1 Why use the word Taylor?

A continuous Taylor expansion builds a function from its value, slopes, curvature and higher derivatives near a point. A coalition table has a related hierarchy: its baseline, single-action changes, pair interactions, and higher interactions. The resemblance becomes an exact mathematical statement when we describe absent/present choices by indicator variables.

There is also a crucial difference. An ordinary truncated continuous Taylor series generally leaves a remainder. The complete multilinear representation of a finite Boolean table leaves none. It reproduces every row exactly because it has exactly the coefficients needed to represent that finite collection of choices.

28.2 From a subset to its indicators

Let z_i be one when action i is included and zero otherwise. Write χ_S for the vector representing S . Define

$$\tilde{E}(z) = \sum_{T \subseteq N} m_E(T) \prod_{i \in T} z_i, \quad z \in \mathbb{R}^n, \quad (28.1)$$

with the empty product equal to one. Each variable appears with exponent at most one, which is what multilinear means here.

Theorem 28.1 (Exact finite discrete Taylor representation). Equation 28.1 is the unique multilinear polynomial satisfying $\tilde{E}(\chi_S) = E(S)$ for all $S \subseteq N$. It has no remainder on the Boolean cube, and

$$\partial_T \tilde{E}(0) = m_E(T) \quad (28.2)$$

for each set of distinct variables T .

Proof. At χ_S , a product is one precisely when $T \subseteq S$ and otherwise zero. Evaluation therefore gives $\sum_{T \subseteq S} m_E(T) = E(S)$. Any multilinear polynomial has one coefficient per such product; evaluating its subsets in increasing size determines these coefficients uniquely by the same inversion. Differentiating

once in each variable of T and evaluating all variables at zero selects exactly its coefficient. $\square$

The theorem allows the whole landscape to be reconstructed from its interactions. It does not require small physical actions, differentiable endpoints or an equilibrium probability law. It is a representation theorem for the finite table.

28.3 A complete quadratic example

For the three equal one-unit couriers of the previous chapter,

$$\tilde{E}(z) = \frac{15}{4}(z_1 + z_2 + z_3) - \frac{1}{2}(z_1 z_2 + z_1 z_3 + z_2 z_3). \quad (28.3)$$

There is no $z_1 z_2 z_3$ term. Setting all indicators to one gives $45/4 - 3/2 = 39/4$, exactly the triple value. Setting only z_1, z_2 to one gives $15/2 - 1/2 = 7$, exactly the pair value.

Now compare this with the physical polynomial for fractional control intensities,

$$E_{\text{phys}}(z) = 4(z_1 + z_2 + z_3) - \frac{1}{4}(z_1 + z_2 + z_3)^2. \quad (28.4)$$

At Boolean vertices $z_i^2 = z_i$, so it reduces to the multilinear formula above. Between vertices the two functions generally differ. For one half-active action, the physical polynomial gives $31/16$, whereas the multilinear extension gives $15/8$. Both agree when the action is entirely absent or present.

This small difference carries a large conceptual lesson: the unique multilinear extension is not automatically the physical law for fractional actions. Binary observations determine one interpolating polynomial in the multilinear class. They do not determine every smooth function through the same vertices.

28.4 A probabilistic interpolation with a declared meaning

There is a useful alternative interpretation. Suppose actions are included independently, with inclusion probabilities $z_i \in [0, 1]$, and then the selected coalition receives its table value. The expected table value is

$$\sum_{S \subseteq N} E(S) \prod_{i \in S} z_i \prod_{j \notin S} (1 - z_j) = \tilde{E}(z). \quad (28.5)$$

The equality follows by taking expectations of the product indicators in Equation 28.1. This is a mathematical randomized selection model with independent inclusions. It does not claim that actual actors choose independently

or that an infeasible group becomes admissible because it has a probability weight.

A fractional physical dose and a probability of a full dose are different interventions. Confusing them can turn an exact interpolation into a false physical prediction. In an EBU system, the record should say whether a slider changes quantity, participation probability, or merely the displayed interpolation.

28.5 Constants, units and reparametrization

The empty coefficient records the baseline table value. In the normalized action table it is zero. With a natural-drift comparator it may not be, and we must retain it or explicitly center the table. Nonempty differences annihilate a constant, but reconstruction of the full value still needs that constant.

The degree also refers to declared action coordinates. Splitting an action into new controls, changing a resolver or choosing nonlinear intensity coordinates can change the apparent polynomial structure of the physical composite. A representation theorem does not make every coordinate choice equally meaningful. Physical interpretation requires the action labels and response map that supplied the table.

28.6 Why this helps an EBU system

A complete finite representation provides a consistency check. If a stored coalition total disagrees with the sum of its stored dividends, either the records refer to different tables or the arithmetic is wrong. It also reveals when a lower-order model is exact: if every coefficient above order two vanishes, singletons and pairs reconstruct every group in that table.

This can reduce storage and measurement effort only when the truncation is justified. Assuming all triples vanish because measuring them is expensive is not a theorem. A system seeking to support shared infrastructure should distinguish a proved structural ceiling, a tested approximation, and an untested convenience. That distinction protects communities from confident recommendations based on omitted dependencies.

28.7 The boundary with ordinary calculus

The word exact belongs to the finite Boolean representation. The next chapter connects a fixed additive physical response to mixed derivatives of the state potential. That connection explains why small interactions have familiar Taylor orders and why quadratic fields give pair-only structure under affine action maps.

The two constructions will meet, but they will not become interchangeable. The discrete polynomial is determined by all vertices. The local continuous derivative is determined near a point. One is a finite table identity; the other is local geometric information with explicit regularity and remainder conditions.

From finite interaction to continuous curvature

29.1 Walk across the table through physical space

The Boolean table records endpoints, but some action families give us more: a continuous path between every combination. Suppose each action has a fixed state increment h_i and the increments add. Turning action i from absent to present then moves continuously along h_i . The pair table forms the corners of a parallelogram in state space; a triple forms the corners of a parallelotope.

Curvature describes how a slope changes across that shape. Repeated integration of mixed derivatives recovers the alternating endpoint difference exactly. This is the bridge between the finite interaction table and the familiar language of continuous calculus.

29.2 Fixed additive increments are a substantive assumption

Let

$$x_S = x_0 + \sum_{i \in S} h_i, \quad G(t) = V \left(x_0 + \sum_{i \in T} t_i h_i \right). \quad (29.1)$$

The h_i do not change when a different subset is chosen. A shared-capacity resolver that reduces one action's quantity when another is present does not satisfy this declaration merely because it returns a vector. It supplies a more general composite map, treated later.

For the physical calculus interpretation, the entire parallelotope must lie in a domain on which the required derivatives are valid. Admissible vertices alone need not supply all admissible intermediate physical states in a nonconvex domain. If an extension of V is used for mathematical interpolation, label that extension and limit the physical claim accordingly.

29.3 Integrating the mixed derivative

Theorem 29.1 (Repeated fundamental theorem of calculus). For $T = \{i_1, \dots, i_k\}$ with fixed additive increments, suppose $V \in C^k$ on a neighbour-

hood of the parallelotope in Equation 29.1. Then

$$m_E(T) = - \int_{[0,1]^k} D^k V \left(x_0 + \sum_{i \in T} t_i h_i \right) [h_{i_1}, \dots, h_{i_k}] dt. \quad (29.2)$$

Here $D^k V(x)[h_{i_1}, \dots, h_{i_k}]$ is the k -linear derivative of V applied to the specified directions.

Proof. For one direction the difference $G(1) - G(0)$ is the integral of its derivative. Apply the same identity in a second coordinate to the difference of those integrals, then continue through all k coordinates. Continuity on the compact cube permits the repeated integrals and their rearrangement. The alternating corner sum is $\Delta_T G(0)$, while the chain rule for the affine map gives the displayed directional derivative. Finally $m_E(T) = -\Delta_T G(0)$. $\square$

For a pair this becomes

$$m_E(ij) = - \int_0^1 \int_0^1 h_i^\top \nabla^2 V(x_0 + sh_i + th_j) h_j ds dt. \quad (29.3)$$

A pair coefficient is therefore an integrated cross-curvature over a finite physical region. It need not equal the Hessian at the initial point times the increments unless the curvature is constant or a controlled approximation is used.

29.4 Why the pair sign is not fixed by convexity

A positive-semidefinite Hessian guarantees $h^\top H h \geq 0$ for the same direction twice. It does not guarantee $h_i^\top H h_j \geq 0$ for different directions. If the directions oppose one another, the cross product can be negative, giving a positive EBU pair correction.

This is the continuous explanation of the cancelling-transfer example. Convexity penalizes a nonzero departure from the reference, but opposite departures can cancel in the joint endpoint. The positive correction repairs the overstatement of separate burdens. It is not evidence of a mysterious extra output.

For EBU, the theorem lets a measured interaction be compared with the field's declared geometry. A mismatch can expose an omitted response dependence or a wrong field. Such a comparison is useful for model diagnosis before any institutional reward is attached to the number.

29.5 Small actions and a controlled local approximation

Set $h_i = \varepsilon d_i$, holding directions d_i fixed while shrinking the common amplitude. Continuity of $D^k V$ at x_0 gives

$$m_E(T) = -\varepsilon^k D^k V(x_0)[d_{i_1}, \dots, d_{i_k}] + o(|\varepsilon|^k). \quad (29.4)$$

The notation $o(|\varepsilon|^k)$ means that the remainder divided by $|\varepsilon|^k$ tends to zero. It does not promise a full extra power of ε under only C^k smoothness.

If $D^k V$ is Lipschitz with constant L on the relevant neighbourhood, the stronger estimate is

$$|R_T(\varepsilon)| \leq \frac{L}{2} |\varepsilon|^{k+1} \left(\sum_{i \in T} \|d_i\| \right) \prod_{i \in T} \|d_i\|. \quad (29.5)$$

Indeed, compare the derivative at $x_0 + \varepsilon \sum t_i d_i$ with that at x_0 , use the Lipschitz bound and multilinearity, and integrate t_i over $[0, 1]$, whose mean is $1/2$.

The assumptions determine the error claim. Replacing little- o by $O(\varepsilon^{k+1})$ without stronger regularity would claim precision the theorem has not supplied. For physical measurements this difference can matter when extrapolating interactions from small interventions to larger ones.

29.6 An exact finite table and a local approximation can both be right

The multilinear expansion in Chapter 28 reconstructs the measured finite table without a remainder. Equation 29.4 approximates a coefficient as action amplitudes shrink. Its remainder concerns the replacement of an integrated derivative by one local derivative, not an incompleteness of Boolean inversion.

A practical comparison is the difference between a complete map of several specified trips and a local estimate of road slope. The map can give exact distances between its listed places while a tangent estimate between nearby points remains approximate. Neither kind of knowledge invalidates the other; their domains differ.

For a system designer, the finite table is appropriate when actions are indivisible or large. A derivative model can help identify how changes scale with size and where a small-action approximation is defensible. The record should say which object was used and carry its error conditions.

29.7 Weaker regularity needs a real theorem

Some physical models are piecewise smooth. Repeated FTC can still hold under suitable absolute-continuity and integrability conditions on the pullback $G(t)$, but “derivatives exist almost everywhere” is not a sufficient replacement by itself. Appendix A states a useful formulation and explains a singular-function counterexample.

The finite Möbius coefficient remains defined from the vertices even when no such derivative theorem applies. This gives a robust order of reasoning: retain the exact finite comparison, then add a continuous interpretation only when its mathematical hypotheses are met. A failed smooth approximation need not invalidate the endpoint account.

29.8 A useful question for the EBU system

If higher-order terms become small as interventions shrink, a local study may efficiently identify the leading geometry. If they remain large, the physical response or field may contain an important nonlinearity. Either finding can guide the scale of a future trial. The theorem does not choose a socially acceptable intervention size; measurement, feasibility and service constraints still govern that choice.

We can now state a particularly useful exact result. When the potential is polynomial and the response affine, sufficiently high finite differences vanish identically, at finite action sizes. The next chapter shows why the Gaussian model is an instructive null case and where its simplicity ends.

What a quadratic can and cannot explain

30.1 A ceiling on interaction order

A quadratic curve has slope and curvature but no third derivative. That familiar fact becomes a statement about action combinations only after the action-to-state map is specified. If the endpoint depends affinely on the action controls, a quadratic potential produces at most pair interactions. If the endpoint depends nonlinearly on the controls, the conclusion can fail.

The distinction is one of the book's most useful experimental lessons. A quadratic potential is a field declaration. Pair-only interaction is a theorem about that field *together with* an affine response. It is not a universal property of everything called Gaussian.

Theorem 30.1 (Polynomial degree ceiling). Let V be a polynomial of total degree at most d on the relevant state coordinates, and let $x(z) = x_0 + \sum_i z_i h_i$. For the complete coalition table $E(S) = V(x_0) - V(x(\chi_S))$,

$$m_E(T) = 0 \quad \text{whenever } |T| > d. \quad (30.1)$$

Proof. The composite $V(x(z))$ is a polynomial in the controls of degree at most d . On Boolean vertices, powers z_i^r with $r \geq 1$ reduce to z_i . This reduction cannot increase total degree. The unique multilinear representation consequently has no monomial involving more than d distinct variables. Its coefficients are the Möbius coefficients, so all higher ones vanish. $\square$

This proof is algebraic. It does not assume infinitesimal increments and carries no Taylor remainder. Restricted physical feasibility still matters: a physical claim about the complete cube requires every row used in the coefficient to be admissible under the stated comparison.

30.2 The quadratic formula in full

Let

$$V(x) = c + g^\top(x - x_0) + \frac{1}{2}(x - x_0)^\top H(x - x_0), \quad H = H^\top. \quad (30.2)$$

Positive definiteness is useful for a homeostatic minimum, but is not needed for the degree argument. Expanding at $x_0 + \sum z_i h_i$ and reducing Boolean squares gives

$$\begin{aligned} m_E(i) &= -g^\top h_i - \frac{1}{2} h_i^\top H h_i, \\ m_E(ij) &= -h_i^\top H h_j, \\ m_E(T) &= 0 \quad (|T| \geq 3). \end{aligned} \tag{30.3}$$

For the Gaussian field, $g = H(x_0 - x^*)$. The pair term is independent of the baseline as long as H and the increments stay fixed. Singleton values change with the baseline because their tangent contributions change.

A diagonal H does not make every pair term zero. Two actions that write the same physical coordinate can have a nonzero weighted overlap. Conversely, with a nondiagonal H , disjoint coordinate writes can interact through the field. The theorem keeps response support and potential coupling distinct.

30.3 Why a null prediction is valuable

Suppose a planned three-action experiment has a quadratic potential, certified fixed additive increments, one common field and baseline, and all eight alternatives. It predicts $m_E(123) = 0$ exactly at the mathematical level. This is a strong null: the prediction was not fitted after looking at a triple coefficient.

A future measurement should then include uncertainty and controls capable of detecting false nonzero values. A nonzero result can reject the conjunction of those assumptions and the observation model. It cannot, by itself, prove that the potential has cubic curvature. The response may have changed with group membership, a source limit may have altered quantities, or the supposed common baseline may have drifted.

The benefit for an EBU system is diagnostic discipline. One can begin with a field simple enough to make a sharp prediction, then investigate deviations rather than treating every interaction as an unexplained new law. This could reduce wasted effort on complicated economic mechanisms built upon misunderstood physical comparisons.

30.4 Cubic curvature supplies a genuine triple

To see an exact higher-order term without a nonlinear response, take the scalar teaching potential

$$V(x) = \frac{\lambda}{6} x^3, \quad x_S = x_0 + \sum_{i \in S} h_i. \tag{30.4}$$

For three increments a, b, c , its third derivative is the constant λ . Repeated FTC gives

$$m_E(123) = -\lambda abc. \quad (30.5)$$

The result is independent of x_0 because the third derivative is constant. It can also be checked by expanding all eight cubes: only the product involving all three distinct increments survives the alternating sum.

With $x_0 = 0$, $a = b = c = 1$ and $\lambda = 1$, singleton values are $-1/6$, pair values $-8/6$, and the triple value is $-27/6$. Therefore $-27/6 + 3(8/6) - 3(1/6) = -1$. These are exact constructed values. The cubic is used on a declared finite domain; it is not claimed to be a globally confining physical equilibrium energy.

30.5 Quartic curvature continues the pattern

For

$$V(x) = \frac{\lambda}{24}x^4, \quad (30.6)$$

the fourth derivative is λ , so four fixed increments give

$$m_E(1234) = -\lambda h_1 h_2 h_3 h_4. \quad (30.7)$$

With unit increments and zero baseline, the alternating coefficient is $[-256 + 4(81) - 6(16) + 4(1)]/24 = -1$ for $\lambda = 1$. All coefficients of order five or higher vanish under the same additive declaration.

The factorials in these examples normalize the relevant derivative to λ . They do not convert stock into EBU by themselves; the units and calibration of λ must match the chosen coordinate. The examples show a structural possibility, not a measured coefficient for society.

30.6 Degree is a property of a representation and its response

A nonlinear coordinate change can turn a simple expression into a complicated one. If it represents the same physical quantity faithfully, the complete endpoint EBU is unchanged, but the split between potential and response derivatives can change. We should therefore resist treating the printed degree of V as a coordinate-free explanation of all observed interactions.

For fixed action coordinates, the relevant object is the composite $G(z) = V(x(z))$. A finite table can even have no high-order coefficients while a chosen smooth extension has nonzero high derivatives between vertices whose integrated effects cancel. Absence of a coefficient is a fact about the declared finite contrast, not necessarily the absence of every local nonlinearity.

The pair-only statement with its full subject

A fixed quadratic potential evaluated on an affine action-to-state map has no Möbius coefficients above order two. Quadratic potential alone is insufficient. Nonlinear response alone does not guarantee a nonzero high-order coefficient.

This precise boundary makes the Gaussian field valuable as a teaching and testing instrument. The next chapter holds the quadratic fixed and changes the response, showing how higher-order interaction can reappear without any cubic term in the state potential.

When actions change one another's physical effects

31.1 The endpoint rule belongs in the mathematics

Two pumps can compete for pressure. One repair can enable a delivery that otherwise stops halfway. A resolver can reduce accepted quantities when a shared source is exhausted. In each case the physical endpoint of a group need not equal the starting state plus a fixed vector for each member.

The appropriate general object is a response map $x(z)$. The controls z describe the actions; the response describes what physical state they produce. The potential then evaluates that state. The composite

$$G(z) = V(x(z)) \quad (31.1)$$

is the landscape seen by the coalition calculation. Keeping both stages visible prevents a nonlinearity in physical response from being misdiagnosed as a new feature of the potential.

31.2 The exact condition for a high-order coefficient

For any declared endpoint map on the Boolean vertices and any nonempty T ,

$$m_E(T) = -\Delta_T G(0). \quad (31.2)$$

Thus a nonzero order- k coefficient occurs exactly when the corresponding k -fold finite difference of the composite is nonzero. This condition requires no smoothness. It is necessary and sufficient for that table and baseline.

It would be too strong to replace it with “nonlinearity creates synergy.” Nonlinearity can create a high-order term, but it can also cancel under composition or vanish on the relevant vertices. A correct result must inspect the actual composite contrast.

31.3 A quadratic potential with a triple interaction

Use a family extending the response of Chapter 27:

$$x(z) = z_1 + z_2 + z_3 + \alpha z_1 z_2, \quad V(x) = \frac{1}{2}x^2, \quad x_0 = 0. \quad (31.3)$$

The square contains $2\alpha z_1 z_2 z_3$ before multiplication by $1/2$. All other terms either involve fewer than three distinct controls or reduce to lower-order Boolean monomials. Therefore

$$m_E(123) = -\alpha. \quad (31.4)$$

At $\alpha = 1$ the table already calculated gives -1 . The field is still quadratic. The third action changes the value of the pair response through the composition.

This is a particularly helpful counterexample for EBU. A finite capacity resolver, a completion rule or an enabling action can change accepted physical effects with membership. A quadratic evaluator downstream of that mechanism does not guarantee a pair-only coalition table. Certifying the evaluator's formula is only half of the structural check.

31.4 Even a linear potential can have high-order response effects

Take a scalar field $V(x) = x$ and the constructed response $x(z) = z_1 z_2 z_3$, again from baseline zero. Every proper subset produces zero, while the full group produces one. Hence $E(123) = -1$ and all proper nonempty values are zero, so $m_E(123) = -1$.

There is no curvature in this potential. The entire three-way dependence is in the response: all three controls are required for the endpoint to change. This example resembles an enabling chain in its logic, but its unit response is a mathematical declaration, not a measured law for any particular machine or institution.

The operational implication is constructive. When a high-order effect is found, examine the response before complicating the field. It may be more faithful to represent the actual enabling mechanism than to force the effect into a fitted potential term.

31.5 A nonlinear response with no interaction at all

Now let

$$x(z) = \sqrt{1 + z_1 + z_2 + z_3}, \quad V(x) = \frac{1}{2}x^2. \quad (31.5)$$

The baseline is $x_0 = 1$. Although $x(z)$ is nonlinear,

$$G(z) = \frac{1}{2}(1 + z_1 + z_2 + z_3), \quad E(S) = -\frac{1}{2}|S|. \quad (31.6)$$

All interactions above singletons vanish. The nonlinear response and the nonlinear field cancel in the composite. This exact example is why a list of nonlinear ingredients is not enough to establish a nonzero interaction.

Nor does a zero coefficient prove that the ingredients are simple. In an application, one may need additional measurements to distinguish additive response from cancellation. The final group values alone identify the finite landscape, not every internal mechanism that could produce it.

31.6 How a resolver changes the scientific question

Suppose A and B each request four units from a source holding six. A declared proportional resolver accepts three each when both are present, while accepting four when either acts alone. The table uses different accepted increments in different rows. That is a well-defined intervention family if the resolver is part of the protocol.

It is not the fixed-increment table of two three-unit actions, nor of two four-unit actions. A theorem about either fixed vector cannot be applied to the resolver table without re-derivation. Even when the resolver's endpoint happens to yield only pair structure in one small case, no general degree ceiling follows from a quadratic field alone.

For a future EBU system, the record should retain request, acceptance and realized endpoint. This helps explain whether a negative interaction reflects a shared physical limit, a field coupling or a decision rule. It could improve transparency around scarce services: people could see which constraint changed the outcome rather than being told only that their group received a lower score.

31.7 What can be learned from a finite table

An exact coalition table determines its Möbius coefficients. It usually does not identify a unique smooth response between the vertices, a unique causal mechanism, or a unique split into potential and response origins. Those stronger claims need additional structure or interventions.

This limitation does not reduce the value of the table. It tells us what the table is good for: reproducible comparisons among declared alternatives, model checks, and focused questions for further measurements. Used carefully, it can inform planning without turning a fitted explanation into an asserted law of society.

To see how the two sources of nonlinearity enter a smooth model, we next differentiate the composite. The resulting chain rule gives a precise explanation, while also revealing why its division into named origins depends on the chosen representation.

Two sources of interaction, one measurable total

32.1 The field bends, and the response can bend too

A curved potential changes how successive increments are valued. A non-linear response changes the increments themselves as controls are combined. Both effects can contribute to the same interaction coefficient. To separate them algebraically, we need a smooth response model between the Boolean endpoints.

For this chapter assume the pullback $G = V \circ x$ is sufficiently smooth on a neighbourhood of the relevant control cube. Write $x_i = \partial_i x$, $x_{ij} = \partial_i \partial_j x$, and similarly for higher derivatives. These symbols denote derivatives of the response, not the endpoint of action i alone. The context and subscripts distinguish them from the earlier physical stock coordinates.

32.2 The pair chain rule

Differentiate once: $\partial_i G = DV(x)[x_i]$. Differentiate again. The derivative can act on DV , because the state changes, or on x_i , because the response direction changes. Thus

$$\partial_{ij} G = D^2 V(x)[x_i, x_j] + DV(x)[x_{ij}]. \quad (32.1)$$

The first term pairs the two first-order responses through field curvature. The second evaluates the mixed response through the field slope. Integrating their sum over the pair face, with a minus sign, gives $m_E(ij)$.

This is useful when diagnosing a measured pair effect. The same total can arise from different combinations of field curvature and response dependence. If quantities are certified fixed and additive, $x_{ij} = 0$ and the second term vanishes. Without that certification, attributing the whole effect to curvature would be unjustified.

32.3 The triple chain rule

Differentiate Equation 32.1 in a third distinct control. The result is

$$\begin{aligned}\partial_{ijk}G &= D^3V[x_i, x_j, x_k] \\ &\quad + D^2V[x_{ij}, x_k] + D^2V[x_{ik}, x_j] + D^2V[x_{jk}, x_i] \\ &\quad + DV[x_{ijk}].\end{aligned}\tag{32.2}$$

All derivatives of V are evaluated at $x(z)$. There is one pure third-derivative term, three terms mixing field curvature with second response derivatives, and one slope acting on the third response derivative.

For a quadratic V , the first term vanishes but the others need not. For an affine response, every response derivative of order two or more vanishes, leaving only the pure derivative of V . The formula therefore unifies the two counterexamples of the preceding chapter without changing the meaning of EBU.

32.4 A mixed example with both contributions

Take $s = z_1 + z_2 + z_3$, response $x = s + \alpha z_1 z_2$, and potential $V = x^3/6$. Boolean expansion gives

$$m_E(123) = -1 - \frac{5}{2}\alpha - \frac{1}{2}\alpha^2.\tag{32.3}$$

At $\alpha = 1$ this is -4 . Both cubic field structure and response dependence contribute.

We can see the split directly from the smooth chain rule. At $\alpha = 1$, $x_1 = 1 + z_2$, $x_2 = 1 + z_1$, $x_3 = 1$, $x_{12} = 1$, and the other relevant higher response derivatives vanish. Since $V''' = 1$ and $V'' = x$,

$$\partial_{123}G = (1 + z_1)(1 + z_2) + x.\tag{32.4}$$

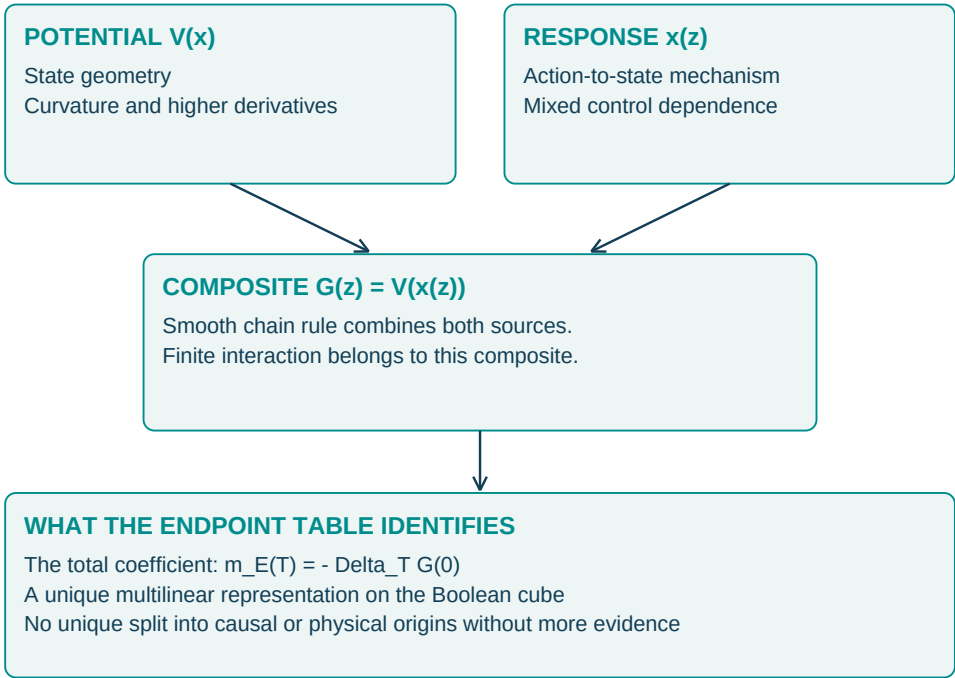
The first term integrates over the unit cube to $9/4$, and the second to $7/4$. Their sum is four, so the EBU coefficient is minus four. This checks the finite table by a different route.

The split is informative within this declared model. It would be misleading to describe $9/4$ and $7/4$ as separately observed physical substances. They are terms in the derivative decomposition of a chosen representation.

32.5 Why the origin split is not unique

Consider two smooth maps on the same pair cube:

$$x^{(a)} = z_1 + z_2, \quad x^{(b)} = z_1 + z_2 + z_1(1 - z_1)z_2,\tag{32.5}$$



Quadratic V + nonlinear response can have higher-order interaction.

Figure 32.1: The coalition coefficient belongs to the composite $G = V \circ x$. Potential curvature and response dependence can both contribute. Their smooth derivative decomposition depends on the chosen coordinates and interpolation; the finite table coefficient does not depend on that interpolation.

with $V = x^2/2$. The extra term vanishes at every Boolean vertex. Both maps therefore produce the same four endpoints and the same pair coefficient $m_E(12) = -1$.

For the first interpolation, the integrated curvature and response terms in Equation 32.1 are 1 and 0. For the second, they are $7/6$ and $-1/6$. They still sum to one. Appendix A gives the elementary integrations.

No observation of the four endpoints alone can choose between these interpolations. Thus the finite interaction is invariant under this change, while the named origin split is not. A nonlinear change of state coordinates can similarly move complexity between V and x . The total composite contrast is the stable object; a causal or physical interpretation of its components needs additional evidence.

32.6 The all-order rule

Let $\Pi(T)$ be the set of partitions of the distinct action indices in T . For a block B , write $x_B = \partial_B x$. The general chain rule is

$$\partial_T(V \circ x) = \sum_{\pi \in \Pi(T)} D^{|\pi|} V(x) [x_B : B \in \pi]. \quad (32.6)$$

Each partition groups together the differentiations that act on one response derivative. The partition into singletons gives the highest field derivative; the one-block partition gives the highest response derivative. All mixed cases lie between them.

For distinct labeled derivatives, each set partition appears once. There is no extra factorial to insert into this form. Appendix B proves the rule by induction and relates it to established multivariable chain-rule work [40].

32.7 What a future diagnosis should report

A scientifically useful report can state the measured finite coefficient, the endpoint model, the smooth extension used for interpretation, and the derivative terms predicted by that extension. It should then say which elements are independently measured and which are inferred. That record is more useful than a categorical claim that an interaction “comes from the potential” when the response is unknown.

For EBU, this offers a disciplined way to locate candidate improvements. A shared pressure limit suggests an engineering response change; a poorly calibrated coupled factor suggests work on the field; unstable baselines suggest better measurement. If those diagnoses are validated, the resulting repairs could improve services and reduce waste. The theorem supplies a way to organize the investigation, not evidence that one intervention will succeed.

We have treated the endpoint response as given. The next part of the argument asks how a declared law of motion can supply it, while keeping the definition of EBU independent of any one generator.

Why a field is not a motion law

33.1 A map does not move the traveler

The potential maps physical states to numbers. Its gradient describes how that number changes near the present state. Nothing in this definition makes a pump run, a household choose a repair, or a government open a water route. Conflating a measurement with a controller would hide the most important design question in EBU: how could the information guide physically possible conduct?

The distinction already appears in mechanics. A height map identifies downhill directions, but a stone's motion depends on inertia, friction and constraints. In thermodynamics, a driving force and a constitutive response are also different objects. The EBU field can describe a directional tendency; a separate account of mobility and choice is required before a trajectory is predicted.

33.2 The Onsager analogy and its boundary

Lars Onsager's 1931 work on reciprocal relations studied near-equilibrium irreversible processes. In that setting, suitably chosen thermodynamic forces and fluxes are connected by linear response coefficients, subject to assumptions including the relevant reversibility conditions [19, 20]. This historical idea is useful here because it shows why naming a "force" does not yet specify a rate: the response coefficients and their domain matter.

As a proposed EBU controller, one might write

$$\dot{x} = -M(x)\nabla V(x), \quad (33.1)$$

where M maps gradients into physically admissible state velocities. If M is symmetric positive semidefinite, the field is fixed and the motion remains in the declared domain, then

$$\frac{dV}{dt} = \nabla V^\top \dot{x} = -\nabla V^\top M \nabla V \leq 0. \quad (33.2)$$

That inequality follows from the proposed dynamics and its assumptions. It does not say that the actual economy follows Equation 33.1, nor that every real

action should move downhill. Essential service, finite reserves and irreversible transformations can create obligations that a one-line gradient flow does not express.

Onsager's point was not that a gradient alone creates a flow. The classical setting identifies physical fluxes, such as heat or matter transport, and forces related to departures from equilibrium. Close enough to the specified equilibrium, the fluxes can be approximated by a matrix acting on those forces. The matrix's entries describe the system's response. Reciprocity is a further statement about paired coefficients under the relevant microscopic assumptions; it is not a universal property of every social decision process. If an EBU implementation calls a numerical gradient a "force," it still owes a measured or designed map from that force to a permissible physical rate.

The analogy to Onsager is therefore structural, not a borrowed experimental validation. The physical force and mobility in a laboratory system are measured for that system. An EBU mobility would have to specify its own units, admissible edges, constraints, response parameters and evidence. Without them, M is a placeholder rather than a natural constant.

33.3 The directed threshold response has its own identity

For the earlier response $J_e = M_e[f_e - \theta_e]_+$ with $M_e > 0$ and $\theta_e \geq 0$, suppose the complete motion is $\dot{x} = u + SJ$. Since $f = -S^T \nabla V$,

$$\dot{V} = \nabla V^T u - \sum_e f_e J_e = \nabla V^T u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (33.3)$$

For an active edge, $f_e = J_e/M_e + \theta_e$; for an inactive edge, its product is zero. This proves the equality for that declared response. With no external drive the potential cannot increase along an admissible continuous solution. Thresholds, disconnected routes and constraints can still prevent arrival at the reference. A later cap or projection changes the vector field and requires checking the actual accepted flux.

For a frozen Euler direction w , a quadratic gives the exact change $\Delta V = h \nabla V^T w + \frac{1}{2} h^2 w^T H w$. When $D = -\nabla V^T w > 0$ and $w^T H w > 0$, nonincrease along this frozen direction holds precisely for $0 \leq h \leq 2D/(w^T H w)$. Physical feasibility remains a separate condition. On a network, $w = SJ$ brings in $J^T S^T H S J$, including shared-node cross terms.

33.4 Conservation restricts motion

Suppose $Ax = b$ expresses conserved homogeneous totals. A permissible internal flow must satisfy $A\dot{x} = 0$. An arbitrary $-\nabla V$ usually will not. One

can construct a mobility whose image lies in the conservation tangent space, or write flows through an incidence matrix, $\dot{x} = Sj$, with $AS = 0$. Either way the conservation law is part of the physical permission layer, not a favourable term that can be traded against EBU.

Additional constraints may bound sources, routes, reserve stocks and simultaneous demand. The menu of feasible actions can therefore be empty even when a mathematical descent direction exists. Conversely, a permitted service action can have negative EBU under the current field. A controller has to say how it treats that case; the potential cannot silently answer it.

33.5 Finite steps require a finite check

Even if a continuous trajectory satisfies $dV/dt \leq 0$, a large discrete step based on its initial slope can overshoot. For the fixed quadratic field, Equation 21.6 shows exactly why: the curvature term grows quadratically with the step. If a policy promises non-increasing V on accepted actions, it must check the finite successor, not merely the sign of the initial gradient.

For example, freeze $g = \nabla V(x)$ and propose one Euler step $\delta = -hMg$ with $h > 0$. The exact quadratic calculation is

$$E(h) = hg^{\top}Mg - \frac{h^2}{2}(Mg)^{\top}H(Mg). \quad (33.4)$$

The first term is the infinitesimal improvement suggested by mobility; the second is the finite curvature correction. When both terms are positive and finite, sufficiently small h improves the field, while an arbitrarily large h need not. This statement still presupposes that the proposed increment is physically feasible. It is not a license to step outside a stock or route constraint.

In one dimension, set $V(x) = x^2/2$ and let mobility be the positive number m . The continuous proposal is $\dot{x} = -mx$, so a solution beginning at x_0 is $x(t) = x_0e^{-mt}$ and its potential is $V(t) = V(0)e^{-2mt}$. The simple finite Euler instruction $x' = x - hmx$ is different. Direct calculation gives

$$E = \frac{1}{2}x^2[1 - (1 - hm)^2] = \frac{1}{2}x^2(2hm - h^2m^2). \quad (33.5)$$

It improves the potential only for $0 < hm < 2$, apart from equality at the endpoints. A larger discrete step can reverse the promised direction even though the continuous law is restoring. This is why a real quote should use the actual finite successor rather than attach an unlimited entitlement to a favourable local slope.

The same lesson appears in the first transfer example of this book. Moving some stock toward the reference improves balance; moving too much crosses

the reference and can worsen it. Exact EBU is therefore a more useful action account than a marginal slogan. Whether a decision-maker chooses the best feasible quantity, a safe bounded quantity, or a quantity required by demand remains a separate rule.

33.6 A usable architecture

A prospective EBU system can keep four questions in a strict order. What physical need or opportunity is present? Which candidate transformations can really occur? What is the exact finite EBU of each accepted physical transition under the declared fixed field? Which actor or institution may select, finance or authorize it? Changing this order can make the field appear to create resources or permit actions that physics forbids.

The architecture is general in ambition: it does not restrict EBU to a laboratory toy. But general applicability would have to be earned through validated state descriptions, measurement, governance and evidence in each consequential domain. A theorem about Equation 33.1 is one component of that programme, not its completion.

The next chapter generalizes this distinction through a generator interface: a declared law supplies endpoints, and the fixed-field definition values them. Later we return to what happens when the yardstick itself changes.

A generator supplies the movement

34.1 From a map of value to a rule of change

The previous chapter of physical dynamics explained why a gradient does not move a pump. We now give a general interface for a declared motion law. Once a law and a duration determine an endpoint for each control setting, that endpoint can enter the same coalition table. The potential definition and Möbius transform need not be reinvented.

A generator therefore sits beside the valuation chain as a source of physical response. It is not the foundation from which EBU must always be derived. Measured finite transitions, fixed increments and jumps can all have endpoint EBU without being solutions of the particular differential equation used here.

34.2 A well-posed flow is needed first

Let a fixed control vector z specify a vector field b_z . Consider

$$\dot{x} = b_z(x), \quad x(0) = x_0, \quad x_z(\tau) = \Phi_z^\tau(x_0). \quad (34.1)$$

For each control used in the coalition table, require a unique solution through the entire declared duration τ , within the admissible state domain. Continuity and local Lipschitz dependence on state are familiar sufficient local conditions; existence up to the chosen horizon and domain invariance still need to be established.

These requirements prevent the endpoint from being ambiguous. The equation $\dot{x} = \sqrt{|x|}$ starting at zero has nonunique solutions: the state can remain zero for an arbitrary waiting time before departing. The equation $\dot{x} = x^2$ starting at one has solution $1/(1 - t)$ and no finite endpoint at or beyond $t = 1$. Writing a generator symbol does not repair either problem.

34.3 The generator-to-coalition theorem

Theorem 34.1 (Flow, value and interaction). Suppose every required control $z = \chi_S$ has a unique admissible endpoint $\Phi_z^\tau(x_0)$, and V is finite there.

Set

$$E_\tau(S) = V(x_0) - V(\Phi_{\chi_S}^\tau(x_0)). \quad (34.2)$$

Then for every nonempty T ,

$$m_{E_\tau}(T) = -\Delta_T[V(\Phi_z^\tau(x_0))]_{z=0}. \quad (34.3)$$

If in addition V and the trajectories satisfy the path theorem,

$$E_\tau(S) = -\int_0^\tau \nabla V(x_S(t))^\top b_{\chi_S}(x_S(t)) dt. \quad (34.4)$$

Proof. The flow supplies a complete endpoint map. Apply the coalition definition and the cancellation of the common constant $V(x_0)$ in every nonempty difference. Along each regular solution, the chain rule gives $dV/dt = \nabla V^\top b_z$; its integral gives the path expression. $\square$

The first statement requires endpoints, not a smooth interpolation across all control values. Control derivatives used later need stronger parameter regularity. Keeping those levels separate makes the theorem usable for a broad range of physically declared response models.

34.4 No control can still mean natural movement

An empty coalition need not leave a dynamical system at rest. If $b_0 \neq 0$, the no-action endpoint $x_\emptyset(\tau)$ can differ from x_0 . Then the raw table has

$$E_\tau(\emptyset) = V(x_0) - V(x_\emptyset(\tau)) \neq 0. \quad (34.5)$$

It measures the whole interval, including natural drift. To compare controls against that same no-action evolution, define the centered table

$$E_\tau^{\text{rel}}(S) = V(x_\emptyset(\tau)) - V(x_S(\tau)). \quad (34.6)$$

Now $E_\tau^{\text{rel}}(\emptyset) = 0$. The raw and relative tables differ by a constant, so all their nonempty Möbius coefficients agree. Their total values and interpretations differ by the baseline drift.

This is not automatically a causal effect estimate. It is a declared dynamical comparator, whose identification requires a valid model or intervention design. A future EBU record should not credit passive natural recovery to an idle actor merely because it occurred during the actor's observation interval.

34.5 Historical force, flux and opposition

Earlier EBU models used a chain from potential to directional force, then mobility or threshold response, then physical flux and state change. In that historical notation J_e denotes an edge flux. An expression involving a scalar Ψ can represent opposition in a particular proposal. Such a scalar is not automatically the vector field b_z in Equation 34.1.

To supply a generator, a model must state how every relevant control, flux, boundary and constraint produces the entire state velocity. The source documents for historical force–flux proposals remain useful context; their response laws are not adopted as universal physical law or as an active economic mechanism by this chapter.

This distinction lets the book preserve a valuable idea without overclaiming it. A transparent force–response architecture can make assumptions testable. Its name or analogy to Onsager does not establish measured transport coefficients, microscopic reversibility, or a successful social controller.

34.6 Ordered actions ask a different question

A flow under simultaneous controls produces one endpoint. Performing A and then B can produce another, and reversing their order can produce a third. These are ordered words, not interchangeable rows of an unordered subset table.

For a concrete smooth example in the plane, let $X = \partial_x$ and $Y = x\partial_y$, starting from $(0, 0)$. Applying X for time τ then Y for time τ reaches (τ, τ^2) . Reversing the order reaches $(\tau, 0)$. Applying $X + Y$ simultaneously for time τ reaches $(\tau, \tau^2/2)$. All statements follow by direct integration of these declared equations.

The order difference is related at small times to the Lie bracket $[X, Y] = DYX - DXY$. For observables, $[L_X, L_Y] = L_{[X, Y]}$, where $L_X f = Df[X]$. Appendix C gives the convention and expansion. The difference between two orders is not itself an unordered Möbius coefficient; one must first define a coalition protocol for every subset.

34.7 Why the interface matters for EBU

The generator interface can connect accounting to realistic duration, capacity and response. It allows the same finite-value law to be used with several explicitly different physical mechanisms while preserving their identities. It also makes a failure informative: an endpoint ambiguity is a response-model problem, while a mismatch in the potential evaluation is a valuation problem.

The social opportunity is more credible comparison of interventions whose physical effects unfold over time. A repair, a delivery and a scheduling change may have different endpoint responses at different durations. That information could support better service planning. It does not determine which planner should have authority, or prove that a particular actor policy will sustain function.

The next chapter shows why even a generator linear in its controls can create higher-order finite-time interactions. A local short-time hierarchy is useful, but it is not an all-time ceiling.

Why time can create higher-order interaction

35.1 A simple input can have a nonlinear accumulated effect

Turning a control on twice as strongly need not double the endpoint after a finite time. The state changes while the law acts on it, and that changed state affects later response. This feedback through time can create a nonlinear control-to-endpoint map even when the vector field is affine in the controls.

The lesson is closely related to finite curvature. An initial local description can be exact at the instant it is written and still fail to describe the whole finite interval by simple multiplication. The generator makes that time dependence explicit.

35.2 The first few time terms

For one fixed z , write at x_0

$$v = b_z(x_0), \quad A = Db_z(x_0), \quad B = D^2b_z(x_0).$$

Under the smoothness and bounded-neighbourhood assumptions detailed in Appendix C,

$$\begin{aligned} E_z(\tau) = & -\tau DV[v] \\ & - \frac{\tau^2}{2} \{D^2V[v, v] + DV[Av]\} \\ & - \frac{\tau^3}{6} \{D^3V[v, v, v] + 3D^2V[v, Av] \\ & \quad + DV[B[v, v] + A^2v]\} + O(\tau^4). \end{aligned} \quad (35.1)$$

The first term evaluates the starting velocity. The next terms account for curvature of V and change in the velocity as the state moves. The expression concerns one declared flow; taking its coalition differences then reveals interaction orders.

The remainder requires more than the existence of the printed derivatives at one point. A convenient sufficient condition is a C^3 vector field and C^4 potential on a neighbourhood containing the short trajectories, with the needed

bounds. This is an asymptotic statement as duration shrinks, not a promise that three terms describe a long experiment accurately.

35.3 Control-affine does give a short-time order rule

Suppose

$$b_z = f_0 + \sum_i z_i f_i, \quad (35.2)$$

with adequate smoothness and common admissible short-time solutions. For $|T| = k$, the leading result is

$$m_{E\tau}(T) = -\frac{\tau^k}{k!} \sum_{\pi \in \text{Perm}(T)} L_{f_{\pi_1}} \cdots L_{f_{\pi_k}} V(x_0) + O(\tau^{k+1}). \quad (35.3)$$

A sufficient regularity choice is C^k fields and $V \in C^{k+1}$ on a suitable common neighbourhood. Appendix C proves the result by expanding iterated generators.

Why can no lower time order contain all k actions? A product of fewer than k control-affine generator factors cannot contain every one of the k distinct control labels. The k -fold Boolean difference removes those lower-degree terms. At order k , every surviving word contains each action field once, in some order. Drift words enter at later orders because including drift would use a place needed for one of those action labels.

This is useful for designing a local diagnostic. It does not say that high-order coefficients stay zero at a finite duration. Nor does it promise a nonzero leading coefficient: Lie-word contributions can cancel.

35.4 An exact counterexample at every order

Take the scalar controlled equation

$$\dot{x} = ux, \quad u = \sum_{i \in S} 1 = |S|, \quad x_0 = 1, \quad \text{quad} V(x) = \frac{1}{2}x^2. \quad (35.4)$$

The generator is affine in the controls and the potential is quadratic. Nevertheless the endpoint is $x_S(\tau) = e^{|S|\tau}$, so

$$E_\tau(S) = \frac{1}{2}(1 - e^{2|S|\tau}). \quad (35.5)$$

For every nonempty T of size k , binomial expansion gives

$$m_{E\tau}(T) = -\frac{1}{2}(e^{2\tau} - 1)^k. \quad (35.6)$$

At every positive duration this is nonzero at all available orders. As $\tau \rightarrow 0$, it is $O(\tau^k)$, exactly consistent with the short-time theorem.

The coefficient does not contradict the quadratic pair-only result: the endpoint map is exponential, not affine in the controls. The example is an exact analytic response illustration, with no claim that uncontrolled exponential growth describes a viable EBU economy.

35.5 What really suffices for pair-only structure at all times

A stronger condition is a common linear state operator with affine control inputs:

$$\dot{x} = A(t)x + b_0(t) + \sum_i z_i b_i(t). \quad (35.7)$$

The operator $A(t)$ must be the same for every coalition. Variation of constants gives

$$x_z(\tau) = \bar{x}(\tau) + \sum_i z_i h_i(\tau), \quad (35.8)$$

where $\bar{x}$ includes the initial state and the common drift, and each h_i is the response to input b_i . Thus the endpoint map is affine in the controls at every admitted horizon. A fixed quadratic potential then gives no coefficients above order two.

This conclusion needs existence and admissibility for the required coalitions. A coalition-dependent matrix A_z , saturation, projection or changing boundary rule can destroy the affine endpoint structure. Describing a system as “linear” without specifying what is linear in which variables is insufficient.

35.6 Nonlinear control dependence can appear immediately

Consider $\dot{x} = z_1 z_2 z_3$, starting at zero, with $V(x) = x$. The full group reaches τ while every proper subset remains at zero. Its triple coefficient is $-\tau$, not $O(\tau^3)$.

The control-affine assumption has failed, so its short-time order theorem does not apply. This example helps interpret a time-scaling experiment: unexpectedly low-order scaling can indicate a direct joint control term, but measurement, baseline and model alternatives still need checking. Scaling is a diagnostic, not a unique mechanism identifier.

35.7 What time-aware interaction could do for a system

A future EBU system may need to distinguish an immediate delivery from an intervention that changes the response over hours or months. The generator

framework can state those differences without changing the endpoint definition. It can also warn against extrapolating a short-duration calibration to a long intervention with accumulated nonlinear effects.

The potential social gain is better planning of coordinated work: identifying when a combination's value depends on timing, or when a response that looks additive locally becomes coupled over time. This requires a validated response law and a relevant service boundary. Neither a small-time theorem nor a closed-form example proves that the resulting institutional choices will be fair or effective.

Before probability enters the argument, we return to structure. A physical network, a dependency graph and a coalition interaction table each describe a different kind of connection. Their relation determines how much of the landscape can be studied locally.

Which connections matter for interaction?

36.1 A pipe drawing is only one map

A network diagram can show where water may flow without showing which measurements a potential reads, which actions share a machine, or which combinations are feasible. These are different structures. Confusing them can make an account appear local while it omits a remote dependency, or make two actions appear independent while they compete for one capacity.

The physical graph lists transformations. The factor graph lists dependencies of the valuation. The feasible-action poset records which subsets are permitted. The value-interaction hypergraph records which sets have nonzero coefficients. A hyperedge can join three or more actions; it is not necessarily a pipe, a causal link or a contractual group.

36.2 Factor support can impose an exact ceiling

Suppose the composite landscape has a declared decomposition

$$G(z) = \sum_{a \in \mathcal{A}} G_a(z_{K_a}), \quad (36.1)$$

where K_a contains every action control on which factor a depends. The dependence must be checked after composing the state response with the potential. A local state factor can acquire broad control dependence through a coupled response.

Theorem 36.1 (Interaction support cancellation). For nonempty T , if no K_a contains T , then $m_E(T) = 0$.

Proof. For each factor there is an index in T on which it does not depend. Taking the difference in that index gives zero. Therefore its full T -difference vanishes. Linearity of differences gives $\Delta_T G = 0$, and the EBU sign relation gives the result. $\square$

This is a sufficient structural condition, not a converse. Even when a factor depends on all of T , its finite difference can cancel. A drawn connection means an interaction is possible under that factor structure; it does not guarantee a nonzero coefficient.

36.3 Shared sources and distant factors

Under the diagonal quadratic, two deliveries from the same source interact through the source coordinate, even if their destinations are far apart. Their pair term contains the product of the two source withdrawals weighted by the source curvature. The relationship comes from shared state support.

A coupled potential can instead link actions with disjoint write supports. For example, a factor representing jointly required water and energy can change when either service changes. A nonlinear response can add a third layer: one action may alter how another affects a remote coordinate. The correct support is obtained by tracing the actual response and factor dependencies, not by measuring distance on the physical drawing.

For an EBU system this matters both computationally and socially. Complete affected-factor evaluation can reduce unnecessary data collection. But declaring a narrow support merely to avoid inconvenient measurements can conceal an external effect. Mathematical locality should be demonstrated from the model and then challenged against the physical claim.

36.4 Small motifs can be studied without claiming universal compression

A motif is a small pattern worth examining repeatedly: two actions sharing a source, a relay, an enabling triple, or a small coupled factor. If two motifs have the same complete coalition table up to a relabeling, their coefficients match under that relabeling. This follows immediately by substituting the bijection into the alternating sum.

The premise is stronger than graph isomorphism. Two identical pipe drawings can have different stocks, capacities, field scales, boundary conditions, histories and timing rules. Any of those differences can change the table. Reusing a cached coefficient requires a certificate that all result-affecting inputs and semantics agree, together with invalidation when a dependency changes.

The earlier EBU motif programme requires its A1–A8 conditions jointly for the stronger runtime and history reuse claims. These include complete boundary and composition information, the necessary subset interactions, cache identity and provenance, dependency invalidation, and history-wide boundary equivalence. This book does not replace that full certificate with a drawing or implement a compressed runtime. The algebraic relabeling result proves only what its premise states.

36.5 Aggregation can hide the relationship one wants to measure

Combining two tanks into a single total preserves mass but loses their internal distribution. A transfer between them then leaves the coarse state unchanged even though the fine potential changes. To preserve the fine value, a coarse representation must retain enough information about internal deviation or include a justified summary of it.

The same issue appears in interaction tables. A region summarized only by its total may hide which local actions share a constrained source. A group may then appear additive at the coarse level because the variables carrying the interaction were discarded. Coarse additivity is not proof of fine independence.

There is a useful exact criterion. If a summary map $r(x)$ and coarse potential $\bar{V}$ satisfy $V(x) = \bar{V}(r(x)) + c$ on all compared states, then every endpoint difference is preserved. If not, a residual must be retained or the claim narrowed. This statement does not require a particular compression algorithm; it gives a test that any proposed one must meet.

36.6 The exponential burden is real

An arbitrary order- k coefficient contains 2^k table entries, each with a nonzero sign. Without extra assumptions, omitting one leaves the coefficient unidentified: changing that missing entry changes the answer while preserving every observed row. This is an information requirement, not merely a slow implementation.

Known low degree, verified factor support, symmetry or a certified reusable motif can reduce the necessary work. An assumed low-order model can also reduce work, but then its error needs testing. A large system cannot be declared scalable simply because the transform has an elegant formula.

A sensible EBU programme begins with small, physically meaningful motifs and a question that justifies the measurement burden. It may then extend only where the extra order resolves a consequential uncertainty. This approach could support more accountable infrastructure decisions while avoiding a costly collection of high-order numbers with no clear service meaning.

36.7 From topology to a physical probability interpretation

Everything so far has concerned states, responses and finite values. None of it has required an equilibrium distribution. That is deliberate: a coalition table can be useful in a deterministic or driven setting without claiming canonical physics.

We can now add a probability layer where independently justified. The next chapter shows exactly what the canonical relation contributes and why it does not, by itself, prove reversibility, a motion law, or a universal economic denomination.

When a potential is also a probability landscape

37.1 Two ways of learning about the same states

A physical model can assign energy to a state. Repeated observations can estimate how often nearby states occur. In a properly specified thermal equilibrium benchmark, these two descriptions are related. This offers a powerful way to test a potential against information that was not obtained merely by evaluating the potential itself.

The relationship is conditional. A useful inventory potential does not become a thermal energy by resemblance to a Gaussian curve. A stationary histogram does not establish detailed balance. We first state the density relation, then distinguish the dynamical and thermodynamic claims that need additional evidence.

37.2 Canonical normalization

Suppose a system has a specified energy $U(x)$, one fixed temperature $T > 0$, an appropriate canonical ensemble, and a declared reference measure ν . Assume its density relative to that measure is

$$p(x) = Z_U^{-1} \exp[-\beta_{\text{thermo}} U(x)], \quad \beta_{\text{thermo}} = \frac{1}{k_B T}, \quad (37.1)$$

with finite nonzero normalizer. Define the dimensionless potential

$$V(x) = \frac{U(x) - U(x^*)}{k_B T}. \quad (37.2)$$

The constant reference energy can be absorbed into the normalizer, giving $p = Z^{-1} e^{-V}$.

Here β_{thermo} has units of inverse energy. It is not the dimensionless coefficient β_{bridge} used to compare an independently defined EBU potential with relative surprisal. Keeping those symbols separate prevents a normalization convention from looking like a fitted universal constant.

Corollary 37.1 (The normalized bridge in its canonical scope). Under the canonical assumptions and Equation 37.2, define $J_{\text{prob}}(x) = -\ln p(x)$ and relative surprisal $J_{\text{rel}}(x) = J_{\text{prob}}(x) - J_{\text{prob}}(x^*)$. Then

$$J_{\text{rel}}(x) = V(x), \quad \beta_{\text{bridge}} = 1. \quad (37.3)$$

Proof. Taking logarithms gives $J_{\text{prob}} = V + \ln Z$. Subtracting its value at x^* removes the constant and uses $V(x^*) = 0$. $\square$

This is known canonical physics expressed in the chosen EBU normalization. It is not evidence that every ecological or economic field has the same relation. A broader cross-field claim remains a hypothesis requiring independent calibration, comparable coordinates and measurements.

37.3 A density is relative to a measure

For a continuous variable, the probability of an exact point is normally zero. A density tells us probability per unit of a declared measure. Saying a state has density twice another state's is meaningful only within that coordinate and measure convention.

If $y = f(x)$ is an invertible smooth change of coordinates and densities are reported relative to the new Lebesgue measure, then

$$p_y(f(x)) = \frac{p_x(x)}{|\det Df(x)|}. \quad (37.4)$$

A nonconstant Jacobian contributes a nonconstant log term. It need not cancel in an endpoint ratio or an interaction contrast. Transforming the reference measure consistently with the state description preserves the original scalar relation; silently changing the measure does not.

A density-of-states factor creates the same kind of issue. If an observed macrostate density is proportional to $g(x)e^{-V(x)}$, its negative log is $V(x) - \ln g(x)$ plus a constant. One must account for g rather than announce a failure or confirmation of $J = V$ from a comparison using incompatible definitions.

37.4 Curvature agrees more generally than covariance does

Where the density relation holds smoothly in the declared coordinates,

$$\nabla^2 J_{\text{prob}}(x) = \nabla^2 V(x) = H(x). \quad (37.5)$$

This follows by differentiation of the log relation. It does not require a Gaussian density, and H may vary with state.

A stronger result applies when $V = \frac{1}{2}(x - x^*)^\top H(x - x^*)$ on all of $\mathbb{R}^d$, with constant symmetric positive-definite H , Lebesgue reference measure and no truncation. Then the normalized density is a global Gaussian with

$$\text{Cov}(x) = \Sigma = H^{-1}. \quad (37.6)$$

Completing the quadratic, or transforming to unit-normal coordinates, proves this covariance statement. It is not a general consequence of matching Hessians at one point.

Truncate the Gaussian to nonnegative stocks, impose a finite-mass boundary, or change the reference measure, and the global covariance need not equal the ambient inverse Hessian. A local quadratic approximation can still be useful, but it must be labeled as an approximation. In particular, the two-cell closed stock world of the opening chapters is not an unconstrained full-support Gaussian distribution.

37.5 Conservation changes which directions can fluctuate

If physical states satisfy $Ax = b$, an interior accessible displacement lies in the null space of A . Choose an orthonormal basis matrix Q for that tangent space and write $x = x_c + Qy$. The restricted potential has Hessian $Q^\top H Q$ in the linear coordinates y .

Only in a full untruncated Gaussian on that affine subspace, with its induced measure and positive-definite restricted Hessian, does

$$\text{Cov}(y) = (Q^\top H Q)^{-1}, \quad \text{Cov}(x) = Q(Q^\top H Q)^{-1}Q^\top \quad (37.7)$$

follow. The ambient covariance is singular along conserved directions. For curved manifolds, coordinate curvature and the chosen intrinsic measure need explicit treatment; one should not casually apply this affine formula.

This is a practical protection against interpreting a conserved total as a failed fluctuation measurement. The inaccessible direction is not supposed to have the variance of an unconstrained system.

37.6 The same density can support different dynamics

A probability density tells us how mass is distributed across states, not how it circulates among them. For a diffusion with constant positive diffusion coefficient D , a drift of the form

$$b = -D\nabla V + c, \quad \nabla \cdot (cp) = 0, \quad (37.8)$$

can preserve $p \propto e^{-V}$ under appropriate boundary conditions while carrying stationary current cp . A nonzero current generally violates the reversible equilibrium interpretation for ordinary even state variables.

Appendix D gives a rotating Gaussian example and a driven jump ring. They show why density agreement, stationarity, detailed balance, local detailed balance and entropy accounting must be treated as separate claims. A histogram alone cannot establish all of them.

37.7 What this bridge offers EBU

A canonical benchmark offers a controlled place to compare two independently obtained descriptions: physical energy divided by $k_B T$, and negative log density in a specified measure. Agreement can validate a calibration and analysis route within that benchmark. It cannot establish economy-wide applicability by analogy.

The prospective benefit is a firmer foundation for measurement. Before asking an institution to rely on an EBU denomination, one can ask whether its physical and statistical readings agree where theory makes a precise prediction. The next chapter carries that same comparison through the complete coalition table.

Interaction as a contrast of probabilities

38.1 The same table in a second language

Under the canonical density relation, a decrease in potential corresponds to an increase in equilibrium density. Each coalition value can therefore be read as a log density ratio. Applying the same alternating signs to the log densities gives the same interaction coefficient.

This is a continuity result: the object being transformed has changed language, while the finite coalition operation has stayed the same. It is not a new source of value to add to the physical account. When its assumptions hold, the two expressions describe one comparison.

Theorem 38.1 (Canonical-density Möbius identity). Assume a complete coalition table, one fixed reference measure and support, $0 < Z < \infty$, and positive finite densities $p(x_S) = Z^{-1}e^{-V(x_S)}$ at all required endpoints. Then

$$E(S) = \ln \frac{p(x_S)}{p(x_0)}, \quad (38.1)$$

$$m_E(T) = \sum_{S \subseteq T} (-1)^{|T|-|S|} \ln p(x_S) \quad (T \neq \emptyset). \quad (38.2)$$

Equivalently, the coefficient is the logarithm of the product of densities raised to the corresponding alternating powers.

Proof. The density relation gives $\ln p(x) = -V(x) - \ln Z$. Subtracting the two endpoint logs gives Equation 38.1. A nonempty Boolean difference annihilates both constant normalization terms. The remaining signed log sum is the EBU coefficient. Exponentiating the sum gives the product form. $\square$

The algebra uses the density relation at the named points. To call it a physical canonical result, the ensemble and calibration conditions of Chapter 37 must be justified independently. Neither stationarity nor reversibility is proved by this finite algebra.

38.2 The pair product

Writing $p_S = p(x_S)$ and $p_0 = p(x_0)$, a normalized two-action table gives

$$m_E(AB) = \ln \frac{p_{AB}p_0}{p_Ap_B}. \quad (38.3)$$

The denominator combines the two standalone density readings. The numerator combines the joint endpoint with the shared baseline. The ratio equals one precisely when this pair contrast is zero.

For the courier table, potential values are 16, 9, 9, 4. Their densities have a common unknown normalization. Substitution gives $\ln[e^{-4}e^{-16}/(e^{-9}e^{-9})] = -2$, with the normalizer canceling. This checks the physical pair correction in probability language. It is an algebraic illustration using the stipulated density relation, not an independent empirical validation of that relation.

38.3 The triple product

The eight-term contrast becomes

$$m_E(ABC) = \ln \frac{p_{ABCP}p_{APBPC}}{p_{ABP}p_{ACP}p_{BCP}p_0}. \quad (38.4)$$

The same recursive interpretation survives: it measures how a pair log contrast changes when the third action is present. The powers and signs are fixed by Boolean inversion. They must not be guessed from a verbal claim that synergy should be positive.

When densities are very small, direct products can be numerically unstable. A future implementation would normally add log densities with their signed coefficients. That is a numerical representation choice, not a change in the physical definition. Zeros require separate treatment: a zero-density endpoint produces an undefined or infinite logarithm, not a finite coefficient to be smoothed away without a model.

38.4 Endpoint densities are not action probabilities

The entries $p(x_S)$ describe density at physical endpoints. They are not generally probabilities of the binary action pattern S . Different coalitions can reach the same endpoint, and a physical density need not sum to one when sampled at a handful of endpoints.

One can define artificial normalized weights $q(S) = p(x_S)/\sum_R p(x_R)$ on the action patterns. Their nonempty log contrasts match the endpoint contrasts because this new normalizer also cancels. But that construction does not

show that actors actually choose according to q , or that the resulting binary variables are causally independent.

If a positive binary joint distribution q is independently specified, its pair contrast on a fixed face is an ordinary log odds ratio. A zero pair contrast in the face with all other variables absent does not establish unconditional independence. For example,

$$\ln q(z) = \text{constant} + az_1z_2z_3 \quad (38.5)$$

has zero pair contrast for actions 1 and 2 when $z_3 = 0$, but pair contrast a when $z_3 = 1$. The context changes the relationship exactly as the recursion predicts.

38.5 Several quantities called information remain different

A pointwise log-density contrast is not an alternating sum of marginal entropies. Interaction information uses marginal entropy combinations; multi-information compares a joint distribution with the product of its marginals. Connected information compares successive maximum-entropy models. Cumulants use a log moment-generating function and a different combinatorial structure.

These constructions can all describe dependence, but they answer different questions and need not give the same number or sign. A claim that EBU interaction is one of them requires a precise derivation under the relevant distribution and coding convention. The common word “information” does not establish the equality.

This is especially important in communication with non-specialists. A positive coalition coefficient should not be translated into “more information” or “more cooperation” without defining that statement. A future system benefits from labels that preserve what was measured rather than borrowing the authority of a familiar term.

38.6 The prior-art connection

Gibbs and log-linear representations, Möbius inversion, cooperative-game dividends and interaction decompositions have extensive antecedents. Jansma explicitly connects higher-order energy interactions, probability contrasts and game-theoretic decompositions in this general mathematical family [38, 39]. The EBU formula here should be read in that context, not presented as a newly invented inversion principle.

The derivation above fixes its own sign from $E = V(x_0) - V(x_S)$ and $\ln p = -V - \ln Z$. This provides a direct check without relying on an isolated intermediate equation in another text. The versioned equation-level source comparison remains available in the project's prior-art appendix [30].

38.7 A two-branch measurement could test the connection

A future experiment could calculate a coalition contrast from independently calibrated physical potential and estimate the corresponding log-density contrast from separate observations. Agreement would be meaningful only if the first branch was not defined by fitting the second to agree, and if endpoints, reference measure and uncertainty were compatible.

Such a test could expose faulty calibration, an omitted density-of-states term, drift or a response mismatch. Its possible benefit to EBU is a more trustworthy physical denomination. It would still be evidence about the tested system and field, not proof that unlike resources share one universal scale.

The remaining thermal step is narrower. Under the accepted conservative single-reservoir benchmark, the same potential difference can be expressed as a medium entropy change. The next chapter keeps every assumption visible while carrying the interaction transform through that equality.

The entropy connection and its boundary

39.1 A thermal statement needs a thermal model

The word entropy is often used loosely for disorder, waste or difficulty. Here it has a precise role in a specified physical benchmark. A conservative overdamped system coupled to one reservoir at fixed temperature can exchange energy as heat while its stochastic system entropy changes. Under the accepted equilibrium conditions, both changes are related to the same endpoint potential difference, with opposite signs.

This is a narrow and useful connection. It does not say that every EBU improvement is universal entropy production or that an economy's total entropy vanishes. The earlier accounting identities remain valid wherever their own assumptions hold, even when this thermodynamic interpretation is unavailable.

39.2 State the complete benchmark before the formula

For each coalition entry, assume the same fixed conservative potential energy U , the same fixed temperature T , a conservative overdamped single-reservoir model, and the stationary canonical density at both endpoint times. Require compatible paths and boundary conventions, with no omitted external work, hidden reservoir, or additional entropy channel. Use $V = (U - U(x^*))/(k_B T)$ in one declared reference measure.

These assumptions must hold for every row of the coalition comparison. If applying a control changes the trap or injects work, the energy balance needs that work term. Merely knowing both endpoint potentials does not establish that all released energy went into the same medium in the way the benchmark assumes.

Theorem 39.1 (Accepted P4 entropy relation). Under the preceding conditions, for each compared process,

$$\begin{aligned}\Delta s_{\text{med}}(S) &= k_B E(S), \\ \Delta s_{\text{sys}}(S) &= -k_B E(S), \\ \Delta s_{\text{tot}}(S) &= 0.\end{aligned}\tag{39.1}$$

Consequently, applying the same Möbius transform to the entropy tables gives

$$\begin{aligned} m_{\Delta s_{\text{med}}}(T) &= k_B m_E(T), \\ m_{\Delta s_{\text{sys}}}(T) &= -k_B m_E(T), \\ m_{\Delta s_{\text{tot}}}(T) &= 0. \end{aligned} \tag{39.2}$$

Proof. With no additional work channel, conservative heat delivered to the medium is $U(x_0) - U(x_S)$. Division by T gives $k_B E(S)$. Stochastic system entropy is $s_{\text{sys}}(x) = -k_B \ln p(x)$; its endpoint change is $-k_B \ln[p(x_S)/p(x_0)] = -k_B E(S)$. The two changes cancel. Möbius inversion is a linear signed sum, so applying it to each entrywise equality gives the coefficient relations. $\square$

The second part is not a new thermodynamic law. It is linearity applied to an already conditional physical identity. Stochastic thermodynamics supplies the heat, work and entropy distinctions; the EBU notation organizes their relationship to the finite account [43, 44].

39.3 Why the signs make sense

Suppose the endpoint energy is lower. Under this benchmark, energy leaves the system as heat to the medium, so medium entropy increases and EBU is positive. The lower-energy endpoint has greater canonical density, so its stochastic surprisal and stochastic system entropy are lower. Their changes have opposite signs.

The cancellation does not mean that nothing happened, nor that every thermal trajectory in every model is reversible. It uses the stationary equilibrium density at both endpoint times and the specified conservative one-reservoir balance. A nonequilibrium initial distribution changes the stochastic system entropy calculation. Driving, multiple reservoirs or unrepresented internal variables can change the total balance.

Individual stochastic entropy changes are also distinct from ensemble averages. If a future study reports an average, it must specify the sampled process and averaging protocol. An endpoint formula should not be turned into an unqualified claim about the entropy of society.

39.4 An interaction changes units, not its logical meaning

If an admissible pair in this benchmark has $m_E(AB) = -2$, its medium-entropy contrast is $-2k_B$. That is a contrast between the entropy changes of four declared processes. It is not necessarily the entropy change of one

newly executed process called “the interaction.” The coefficient retains its counterfactual table meaning.

The same is true for a triple. Multiplying by k_B gives the thermal units under the benchmark; it does not turn an eight-corner comparison into a causal allocation among three actors. The joint outcome must still be distinguished from an attribution or institutional payment.

39.5 A current-carrying system exposes the limit

A stationary density can agree with e^{-V} while probability circulates. A driven ring can have uniform stationary probabilities and constant state potential while its clockwise and counterclockwise rate ratios carry nonzero cycle affinity. Its density-based EBU differences vanish, but its dynamics can produce entropy through the drive.

This is not a defect in the endpoint definition. It shows that a state-potential difference and a general nonequilibrium path functional are different objects. Appendix D gives the ring explicitly. A loop returns the state potential to its start, while dissipative currents need not cancel their thermodynamic cost.

If a broader account includes nonconservative work, it must declare that work and its boundary. One cannot add an unspecified process charge to the fixed physical EBU definition and claim the original theorem proved the new expression. A separately defined net functional can be analyzed, but its additional terms require their own physical justification.

39.6 Macrostate entropy is another declaration

A constrained macrostate model may use a relation such as $S = S_{\text{eq}} - \kappa V$. Its entropy can concern a coarse-grained ensemble with internal multiplicity. That object is not automatically the stochastic endpoint entropy $-k_B \ln p(x)$ used above. The ensemble, state resolution, reference measure and coefficient must be reconciled before equating them.

This distinction is important for EBU’s broader ambition. A physically meaningful account of living systems may need several scales and internal states. The thermal benchmark offers a precise testbed; it does not license erasing those additional structures when moving to ecology or institutions.

39.7 What the conditional connection is good for

Within its scope, the entropy relation links the EBU scale to a known physical quantity and supplies a second consistency check on signs, calibration and

boundary accounting. It can help a future experiment distinguish a correct canonical endpoint analysis from a mistaken assertion about total entropy production.

The possible societal benefit is indirect but important: institutions should rely on physical claims that survive such distinctions. A clear boundary makes it harder to justify a policy by saying that thermodynamics has already proven its desirability. The mathematics can contribute trustworthy information while people retain responsibility for goals, rights and acceptable tradeoffs.

We have now reached the end of the fixed-field continuity chain. Before bringing the links together, we need to address two practical complications: the field itself can change, and the measured coefficients carry uncertainty.

When the yardstick changes

40.1 Learning about the world is not the same as changing it

A community can improve a water reserve by moving water. It can also learn that its previous reserve target was unsuitable. The first changes the physical state; the second changes the field used to interpret that state. Both can be important, but only one is the physical action described by the original receipt.

Let $V(x, \theta)$ be a family of potentials, with θ identifying references, scales and other declared field parameters. A fixed-field action still has $E_\theta = V(x_0, \theta) - V(x_1, \theta)$. Once θ changes, an account spanning both versions needs to show that change explicitly.

40.2 The full differential has two components

Assume V is C^1 along a regular path $(x(t), \theta(t))$. The chain rule gives

$$dV = \nabla_x V \cdot dx + \partial_\theta V \cdot d\theta. \quad (40.1)$$

Consequently,

$$V(x_1, \theta_1) - V(x_0, \theta_0) = \int \nabla_x V \cdot dx + \int \partial_\theta V \cdot d\theta. \quad (40.2)$$

The total is an exact state-function difference on the enlarged state-and-parameter space. The two component integrals are generally path dependent separately. One cannot discard the parameter component and retain a universal endpoint theorem for the physical-action component across changing fields.

For example, with $V(x, \theta) = \theta x$, the physical component is θdx and the parameter component is $x d\theta$. Moving x first and changing θ second allocates different amounts to the two components than doing them in reverse order. Both sums give the same final minus initial product.

40.3 A finite version makes the ordering explicit

For a transition from (x, θ) to (y, θ') , insert the intermediate value $V(y, \theta)$:

$$\begin{aligned} V(x, \theta) - V(y, \theta') &= [V(x, \theta) - V(y, \theta)] \\ &\quad + [V(y, \theta) - V(y, \theta')]. \end{aligned} \quad (40.3)$$

The first bracket values the physical change under the original field. The second records revaluation at the final physical state. An alternative ordering evaluates the physical change under the new field and revalues at the initial state. Both are valid decompositions with different meanings. A protocol must choose and record its ordering.

At (14, 6), changing both Gaussian scales from two to four changes potential from four to one without moving water. That three-unit difference is a field-change entry. An actor who was idle during the revision has not performed a three-unit restoration.

40.4 A closed physical loop can exploit an untracked scale change

Let $W(0) = 0$, $W(1) = 1$, and define fields $V_A = 2W$, $V_B = W$. An action $1 \rightarrow 0$ under A gives +2. A later reverse action $0 \rightarrow 1$ under B gives -1. Returning the field to A completes a loop in both physical state and field version, but the two action entries alone sum to +1.

The missing field-change entries reconcile the loop. No material was created and the fixed-field theorem was not violated: its premise of one field did not hold for the action-only sum. Treating those heterogeneous entries as a persistent spendable claim without a bridge would create an accounting opportunity from changes in the ruler.

This example identifies a concrete institutional risk without deciding a universal remedy. A canonical capacity reference, an explicit conversion rule or separated accounts may each be considered in a future design. Each needs a justified meaning and its own analysis. The book does not adopt one by naming the problem.

40.5 Historical receipts retain their original meaning

A later field can provide a useful retrospective diagnostic, but it should not overwrite what the original account said. Keep the original state, field version and signed value; append the new calculation with its purpose and authority. Otherwise genuine physical improvement and a change in interpretation become impossible to distinguish.

This matters for trust. A community should be able to learn that a reference was wrong without being told that its historical work has silently changed units. An institution should also be able to correct a mistaken measurement without erasing the earlier evidence. Measurement correction, physical reversal and field revision are separate events.

40.6 A common coefficient across fields remains a hypothesis

The canonical benchmark gave $J_{\text{rel}} = V$ under a particular energy normalization. For more general fields, one might investigate

$$J_{\text{rel},\theta}(x) \stackrel{?}{=} \beta_{\text{bridge}} V_{\theta}(x) \quad (40.4)$$

with one coefficient across the specified fields. The question is stronger than agreement within one fitted field. It requires independently defined potentials, compatible density measures and a justified common denomination.

A curvature comparison $K_{\theta} \stackrel{?}{=} \beta_{\text{bridge}} H_{\theta}$ can be part of that programme, with accessible tangent directions treated correctly. Local curvature agreement alone does not establish a global identity or a universal conversion among resources. Nor may the potential branch be defined from the probability branch and then presented as independent evidence of agreement.

40.7 Why a fixed yardstick is not the final social answer

Freezing a field forever would simplify accounting, but knowledge and physical conditions change. Drought can alter feasible reserve levels; technology can change the relationship between energy and service; new evidence can expose an omitted harm. A trustworthy system needs a way to improve its model without presenting every revision as an actor's achievement.

The positive opportunity is an account that remains both relevant and inspectable. Versioned fields can make learning visible, while separated revaluation entries preserve the meaning of physical history. That could support more informed public decisions and prevent windfalls caused solely by hidden scale changes. The rules governing persistent claims remain an institutional research problem.

The next chapter turns from a changing field to imperfect knowledge of a fixed one. Even when the declaration is held constant, measurements and shared calibration can make an interaction uncertain.

How much confidence belongs beside an interaction?

41.1 Exact subtraction is not exact observation

A Möbius coefficient is an exact signed sum of its input table. If the inputs are estimated, the coefficient is estimated too. Alternating signs do not automatically cancel measurement errors, and a shared baseline can create correlations that must be retained.

This matters especially at high order. A genuine signal may shrink as the k th power of a small action amplitude while the contrast still combines 2^k noisy readings. A theorem can be exact and its empirical target difficult to resolve. The appropriate response is a measurement design with uncertainty, not a claim that algebra has removed it.

41.2 A linear contrast has a complete covariance formula

Collect the required estimates into a vector $\hat{e}$ and let $c_S = (-1)^{|T|-|S|}$. Then $\hat{m} = c^\top \hat{e}$. If the estimates have finite covariance matrix C , elementary bilinearity gives

$$\text{Var}(\hat{m}) = c^\top C c. \quad (41.1)$$

This is exact for the specified linear contrast, whether or not the errors are Gaussian. It includes variances on the diagonal and pairwise error covariances off the diagonal.

Proof. Subtract the mean from each estimate and expand the square of $\sum_S c_S(\hat{e}_S - \mathbb{E}\hat{e}_S)$. Taking expectations gives $\sum_{S,R} c_S c_R \text{Cov}(\hat{e}_S, \hat{e}_R)$, which is the matrix expression. $\square$

For independent equal-variance readings of all 2^k potentials, each with variance σ^2 , the corresponding alternating potential contrast has variance $2^k \sigma^2$. Independence is an assumption about errors, not a property supplied by the transform.

41.3 Why the shared baseline does not simply disappear

For a normalized triple table, $E(\emptyset) = 0$ is exact. The seven nonempty values may be calculated as $\hat{E}(S) = \hat{V}_0 - \hat{V}_S$, all using the same estimated baseline potential. If one treats these seven estimates as independent, one ignores their shared error.

The signed coefficients of the seven nonempty rows sum to one. Therefore the baseline estimate enters the triple contrast once. If the eight potential estimates are mutually independent with equal variance σ^2 , the result has variance $8\sigma^2$: one baseline contribution and seven endpoint contributions.

Equivalently, each nonempty $\hat{E}(S)$ has variance $2\sigma^2$, and two different rows have covariance σ^2 through the baseline. Substituting this full covariance into Equation 41.1 again gives $8\sigma^2$. It is not $14\sigma^2$, obtained by incorrectly treating those differences as independent, and not $7\sigma^2$, which would describe seven independently measured E-values of variance σ^2 under a different design.

What cancels, and what does not

A deterministic constant added to every table entry cancels in a nonempty difference. An uncertain baseline reused inside normalized differences can leave a correlated error contribution. These statements concern different data-generating constructions.

41.4 Shared calibration creates further dependence

All endpoints may use the same estimated reference, temperature or Hessian. Errors in those parameters can move several table entries together. A calibration uncertainty is not automatically reduced by taking more endpoint samples; it can be systematic across the entire comparison.

If $m = f(\eta)$ depends smoothly on an estimated parameter vector η , a first-order delta-method approximation gives

$$\text{Var}(\hat{m}) \approx \nabla f(\eta)^\top \text{Cov}(\hat{\eta}) \nabla f(\eta). \quad (41.2)$$

Its approximation status, parameter uncertainty and dependence assumptions should be reported. Close to boundaries, with nonlinear estimation or sparse densities, a more appropriate registered uncertainty analysis may be needed. This book does not choose a universal confidence threshold.

For the probability branch, log transformation can amplify relative error in small densities. Finite-bin probabilities estimate integrals over bins, not exact point densities. Bin size, smoothing and density-estimator choices can alter

the target and its bias. Those choices belong in the prospective analysis design, not in a post hoc effort to make the two branches agree.

41.5 What a nonzero triple can establish

Suppose the null prediction requires quadratic V , fixed additive increments, one field, one baseline, an admissible complete cube and accurate compatible valuation. A statistically resolved nonzero triple is inconsistent with that conjunction. It does not uniquely diagnose which premise failed.

The alternatives are concrete: nonlinear response; a resolver that changed accepted quantities; a nonquadratic field; drift between comparisons; omitted coordinates; calibration error; or an unsuitable measurement model. A good design includes observations or controls that can distinguish some of these explanations. If it cannot, the uncertainty belongs in the conclusion.

A zero triple is also limited. It can be a genuine structural null, cancellation in that motif, or insufficient power. It does not certify absence of all higher-order effects at other baselines or durations. The interpretation must match the tested face of the landscape.

41.6 Choosing a manageable question

A complete arbitrary order- k table requires 2^k corners. Repetitions, controls and calibration add further work. This burden motivates pair studies and carefully chosen triples before large interaction catalogues. Factor support or an independently justified degree ceiling can reduce work, but only within its proven scope.

For EBU, the best early question may be small and decisive: do independently measured fixed increments under a calibrated quadratic reproduce the predicted pair correction and triple null? Such a test can teach more than a broad dashboard of poorly resolved coefficients. The social payoff, if the measurement succeeds, is a more reliable basis for later decisions, not a claim that the small experiment already validates an economy.

41.7 Communicating a result people can challenge

A useful interaction report names the group, baseline, field, physical endpoints, coefficient, uncertainty and source of every row. It distinguishes observed endpoints from modeled counterfactuals and reports missing or infeasible rows explicitly. A positive number without those details is difficult to contest and easy to overinterpret.

This transparency can help communities understand why a shared intervention is judged promising, how confident that judgment is, and what would change it. It also gives an institution a way to acknowledge uncertainty without concealing the physical account. The next chapter gathers the exact identities and their conditional bridges into one readable hierarchy.

What is the glue?

42.1 One comparison carried through several languages

We began with an ordinary concern: how can an account of action say something about the physical conditions that keep life and service possible? The answer developed here is deliberately specific. Declare a state, a fixed potential and a complete physical transition. The before-minus-after difference is its EBU. Every later mathematical layer either represents that same difference or supplies a declared way to construct its endpoints.

The glue is therefore not a metaphor about balance. It is the preservation of one mathematical object across changes of representation. The chain rule connects endpoints to a path integral. Telescoping connects successive transitions. Möbius inversion connects group values to irreducible finite interactions. Smooth response and potential derivatives explain those interactions when their hypotheses hold. A canonical density can express the same values as log ratios, and the accepted thermal benchmark can express them as medium entropy changes.

Each arrow has conditions. Keeping those conditions visible is what makes the continuity rigorous rather than merely persuasive.

42.2 The central continuity statement

Theorem 42.1 (Fixed-field EBU continuity hierarchy). For a complete represented transition $x_0 \rightarrow x_1$ and one finite, fixed potential V , the defining account is $E = V(x_0) - V(x_1)$. Under the additional assumptions stated beside each expression below, this same number has the following representations.

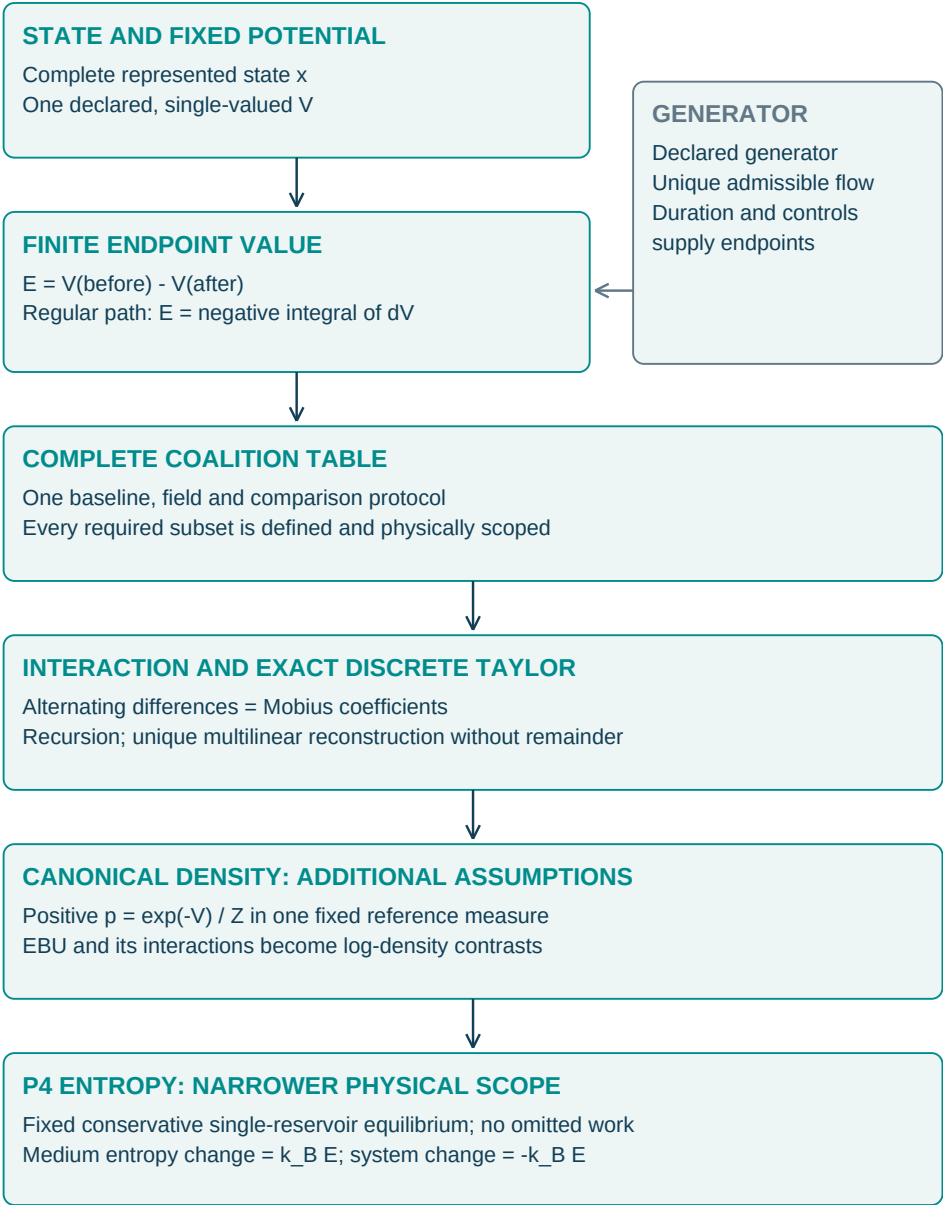


Figure 42.1: The continuity hierarchy. The generator is an optional interface supplying endpoints. The main finite theory needs neither equilibrium nor an actor account. Canonical probability and P4 entropy are additional conditional branches of the same declared comparison.

Central continuity: preserve the assumptions

D: definition. One fixed field and complete endpoints:

$$E = V(x_0) - V(x_1).$$

P: regular path. Theorem 14.1:

$$E = - \int_{\gamma} dV = - \int_{\gamma} \nabla V \cdot dx.$$

Q: canonical density. Fixed measure, positive endpoints, $p = Z^{-1}e^{-V}$:

$$E = \ln[p(x_1)/p(x_0)].$$

T: accepted P4 scope. Fixed conservative overdamped single-reservoir equilibrium, same temperature and no omitted work or entropy channels:

$$E = \Delta s_{\text{med}}/k_B, \quad \Delta s_{\text{sys}} = -k_B E.$$

The proof consists of the already established chain rule, log-density identity and conditional heat balance. It does not remove any premise when the results are placed together. In particular, D does not imply Q or T. A stock account can satisfy D and P without being a thermal canonical system.

42.3 The interaction form of the same continuity

For a nonempty action set T , use one complete coalition table and write $G(z) = V(x(z))$. The exact finite statement is

Central interaction continuity

B: complete Boolean table.

$$m_E(T) = \Delta_T E(\emptyset) = -\Delta_T G(0).$$

A: additive response and $C^{|T|}$ potential.

$$m_E(T) = - \int_{[0,1]^{|T|}} D^{|T|} V(x(t)) [h_i : i \in T], dt.$$

Q: canonical density at every corner.

$$m_E(T) = \sum_{S \subseteq T} (-1)^{|T|-|S|} \ln p(x_S).$$

T: accepted P4 scope for every entry.

$$m_{\Delta s_{\text{med}}}(T) = k_B m_E(T).$$

The derivative formula is one specialization of the general composite difference. With nonlinear smooth response, integrate $\partial_T(V \circ x)$ instead and use its partition chain rule. With no smooth extension, the finite table identity still stands. With a restricted feasible poset, use the applicable incidence inversion rather than assume a missing Boolean cube.

42.4 What the generator contributes

The generator interface declares a law of motion, checks well-posedness and supplies $x_S = \Phi_{x_S}^\tau(x_0)$. The finite definition then values those endpoints. The interface does not sit between potential and definition as a mandatory foundation; it is one way of specifying what an action does.

This placement prevents two errors. First, a valid finite action account does not need to invent a universal microscopic motion law. Second, a proposed motion law does not acquire empirical credibility merely because its endpoints can be valued exactly. The response and the field each need their own evidence.

42.5 What is conserved, and what merely closes

Physical carrier conservation concerns quantities such as mass or energy within a declared boundary. EBU telescoping concerns differences of a state function. Probability normalization concerns total probability mass. An

institutional balance identity concerns a chosen update rule. These are four distinct statements.

A lossless water transfer can conserve litres while lowering a potential. A closed state cycle can have zero total fixed-field EBU while consuming fuel outside a too-small boundary. An exact aggregate attribution can coexist with unfair individual shares. A canonical density can be normalized while its dynamics carry nonzero current. The continuity hierarchy does not collapse these differences.

The benefit is that each failure has somewhere to be located. A missing litre is a physical-account problem; a reused baseline is a comparison problem; a changed scale is a field-version problem; an unjust entitlement is an institutional problem. Keeping those questions distinct makes the whole system more inspectable.

42.6 What this does not prove

The hierarchy does not prove that a chosen V captures every ecological consequence, that all resources share one denomination, or that a potential minimum is an attracting state. It does not prove decentralization, fair ownership, causal entitlement, universal scalability or a successful currency. It does not turn a probability identity into reversibility, or a thermal benchmark into universal entropy accounting.

These are substantive remaining questions. Some need additional theorems, some need physical measurement, and some need explicit public choices. They should not be treated as small implementation details after the equations are finished. A system intended to improve society must confront them directly.

42.7 What the continuity does offer

The sequence provides a disciplined common language for local measurements, joint physical outcomes and their conditional statistical interpretation. Exact cancellation can prevent duplicate claims. Interaction coefficients can expose dependencies that isolated quotes miss. Response models can distinguish a changing mechanism from a changing valuation. Uncertainty can be carried through the same finite contrasts.

If these capabilities are implemented and validated, they could help identify useful maintenance, coordinate complementary infrastructure and reveal when an apparent gain shifts harm outside the boundary. Their social benefit depends on access to those opportunities, protection of essential services and governance of the measurements. The laws supply information and consistency constraints; they do not choose society's aims.

The next chapters therefore return to people and institutions. We will first distinguish the joint physical result from any rule for dividing it, then explain what a trustworthy implementation and a future experiment would have to establish.

Dividing a result does not define an entitlement

43.1 The joint account is already known

We have now defined a complete coalition landscape and its interaction hierarchy. None of that analysis required an actor wallet. A joint transition has one value before anyone decides how to divide it. We can therefore examine an attribution rule without mistaking its closure for a new physical law.

A path attribution can be useful for reporting contributions under a declared convention. Its assumptions, path and child decomposition must be retained. It is not automatically an observed causal contribution, a fair share or a payment.

43.2 One declared path through the joint change

Consider the straight path

$$\gamma(s) = x^b + s\delta_G, \quad 0 \leq s \leq 1. \quad (43.1)$$

The parameter s says what fraction of the joint increment has been included in this mathematical path. If the physical model explicitly stipulates constant-rate simultaneous execution, that path can also be its realized state path. If only endpoints are specified, it is a declared interpolation convention.

For each child, define the simultaneous-path field attribution

$$R_a = - \int_0^1 \nabla V(x^b + s\delta_G)^T \delta_a ds. \quad (43.2)$$

The minus sign follows the book's convention: reducing potential gives positive field value. The integral sums the child's directional contribution along the common path. Every child uses that same path; choosing a different path for each would require a different argument for closure.

This construction has Aumann–Shapley-type common-path mathematics. Naming that relationship is an attribution of mathematical structure, not an adoption of a fairness theorem or a claim that physical matter follows a cooperative-game allocation rule.

43.3 Why the path contributions close

Add equation (43.2) over children. Finite summation and integration commute, and the child increments sum to $\delta_{\mathcal{G}}$:

$$\sum_a R_a = - \int_0^1 \nabla V(x^b + s\delta_{\mathcal{G}})^T \delta_{\mathcal{G}} ds \quad (43.3)$$

$$= - \int_0^1 \frac{d}{ds} V(x^b + s\delta_{\mathcal{G}}) ds \quad (43.4)$$

$$= V(x^b) - V(x^b + \delta_{\mathcal{G}}). \quad (43.5)$$

The last step is the fundamental theorem of calculus, under the differentiability and path-domain assumptions. Our fixed quadratic satisfies the required smoothness.

Common-path closure

For fixed additive child increments and one declared differentiable common path of the stated form, the signed derivative integrals sum to the exact group field reduction. The theorem establishes the sum. It does not establish ownership, causal necessity, fair shares or permission to spend an attribution.

The closure result concerns the complete state and potential that were declared. If a group's physical operation changes additional energy, loss or equipment coordinates, those coordinates belong in its complete increment and in any potential that claims to value them. The field integral does not authorize a second, unspecified charge. Nor does its exact sum determine a unique owner allocation.

43.4 The Gaussian integral is explicit

Because the Gaussian gradient is affine,

$$\nabla V(x^b + s\delta_{\mathcal{G}}) = \mu(x^b) + sH\delta_{\mathcal{G}}.$$

Substitute into equation (43.2) and integrate 1 and s :

$$R_a = -\mu(x^b)^T \delta_a - \frac{1}{2} \delta_a^T H \delta_{\mathcal{G}}. \quad (43.6)$$

No numerical quadrature is needed for this quadratic. The expression includes the way each child overlaps the full group increment. It does not merely divide the total by the number of names on the record.

For the symmetric two-courier example, each attribution is 6 and their sum is 12. The same-baseline singleton value was 7, while the first and second sequential values were 7 and 5. These are three different questions with compatible arithmetic.

43.5 Unequal quantities on the same edge

Let A send one unit and B send three from the same baseline (18, 2). The group still sends four units and has field value 12. The singleton values are

$$E_A^{(0)} = 4 - 1/4 = 3.75, \quad E_B^{(0)} = 12 - 9/4 = 9.75,$$

summing to 13.5. The cross-term correction is -1.5 .

Along the common path, A's increment is one quarter of the total and B's is three quarters, so their field attributions are 3 and 9. This proportionality follows here because the increments are collinear and use the same path. It is not a universal justification for dividing all group values by sent quantity.

If the actions use different destinations, different scales or different physical maps, equal sent quantities need not have equal directional field effects. A proportional policy can still be defined, but it answers a policy question distinct from the common-path calculation.

43.6 A fan-out example with distinct destinations

Use three locations with $x^b = (18, 2, 10)$, references (10, 10, 10) and scales all equal to 2. Let A send two units from location 1 to 2 and B send two from location 1 to 3. The increments are $(-2, 2, 0)$ and $(-2, 0, 2)$.

Initial potential is 16. The joint successor is (14, 4, 12), whose terms are 2, 4.5 and 0.5, totaling 7. Group value is 9. Acting alone, A gives 7 and B gives 3, totaling 10. Their one-unit discrepancy comes from the shared source coordinate.

The common-path attributions are 6.5 to A and 2.5 to B. They sum to 9. Equal sent quantities did not yield equal attributions because one destination began below reference while the other began at reference. The arithmetic describes the declared path; whether these are appropriate economic shares remains a separate question.

43.7 Why an allocation remains an additional choice

Many numerical allocations can sum to 12. Equal shares, proportional shares and some marginal constructions all close in a symmetric example. Closure

alone cannot distinguish them. An institution may also consider responsibility, measured effort, guaranteed access, bargaining power and risk.

Even attractive algebraic properties have precise scope. Relabeling fixed children leaves the common-path formula unchanged. Proportionally splitting a fixed child preserves its summed attribution. But changing requests can change physical acceptance, group composition and affordability. Those effects require analysis of the entire mechanism.

An earlier prospective Gaussian programme proposed using a declared path attribution in an action-capacity rule. That proposal can be studied without pretending it has already solved economic fairness. A negative result about the mechanism would not invalidate the calculus identity; a correct identity would not rescue a poorly performing mechanism.

43.8 Dividends can inform an allocation without choosing it

For a normalized cooperative table, one familiar allocation divides every dividend $m_E(T)$ equally among its members and sums the assigned pieces. This gives the Shapley allocation

$$\phi_i = \sum_{T \ni i} \frac{m_E(T)}{|T|}, \quad \sum_i \phi_i = E(N). \quad (43.7)$$

The sum follows because each dividend is distributed exactly once. This is established cooperative-game mathematics, not a rule adopted for EBU ownership [32, 33]. Other rules can close on the same total under different axioms or priorities.

For the symmetric two-courier table, equal dividend splitting gives six each, as does the common straight path. Coincidence in a symmetric example does not make all allocation methods equivalent. Nonlinear response, asymmetric actions and different baselines can distinguish them.

43.9 What a future system could use this for

An explicit allocation convention can make shared work easier to inspect: readers can reconstruct how a known group total was distributed and challenge the rule separately from the physical calculation. That may support cooperation when people agree on the convention and have fair access to participate.

The possible benefit does not come from declaring the formula morally correct. It comes from making the choice visible. Rights, effort, risk, essential service and bargaining power remain matters for institutional design and evidence. The following chapters carry the account through routes and persistent institutions while preserving that boundary.

Routes, resource accounts and learning

44.1 A local equation in a connected world

The antidote a clinic needs may be made far away. A pump can depend on a spare part, an electrical supply and a skilled repairer. A water transfer can depend on a chain of transformations that began with fuel. Such chains prevent the local equation from becoming an isolated tank puzzle.

Local accounting does not mean that the world consists of disconnected actions. It means that each calculation uses a declared physical boundary and the complete factors affected within it. Links between actions retain sequence, dependence, quantities and provenance. They do not automatically make the whole chain one indivisible action.

This chapter explains what can already be said about composition and what would need a larger model. It preserves the practical questions about routes, infrastructure, cooperation, poverty and institutions while keeping them distinct from the Gaussian identities.

44.2 A route is an ordered physical record

Let a stock travel from A to a hub and then to B. The path has two directed edges. The first action decreases A and increases the hub; the second decreases the hub and increases B. Each action has a finite quantity, time interval and predecessor state.

If the actions occur successively under one fixed potential and no intervening external change, their field differences telescope. The hub may temporarily carry extra stock, but its intermediate potential cancels when the route total is added. This cancellation requires using the actual intermediate state, not evaluating both edges independently against the original one.

Suppose the hub already contains stock. The later outgoing action may use stock that was present earlier rather than a physically distinguishable parcel just received. A quantity ledger can still track flows. A provenance claim about which shipment served which patient needs an explicit tracing convention or evidence. Algebraic endpoint equality does not identify the history of individual molecules.

44.3 A complete two-edge Gaussian route

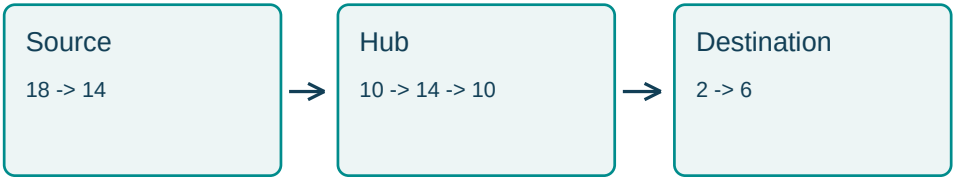
Use three cells with references $(10, 10, 10)X$, scales $(2, 2, 2)X$ and initial state $(18, 10, 2)X$. The order is A, hub, B. Potential is $8 + 0 + 8 = 16$.

Send four units from A to the hub. The state becomes $(14, 14, 2)X$, with potential $2 + 2 + 8 = 12$. The first action's value is 4. Now send four from the hub to B. The state becomes $(14, 10, 6)X$, with potential $2 + 0 + 2 = 4$. The second action's value is 8. Their route total is 12.

A direct lossless transfer of four from A to B has the same endpoint and the same field reduction. Under this ideal field alone, adding an intermediate hub has not created a geographical fee. If the route has additional energy use, time effects or material losses, those are additional physical facts that must be represented and compared.

Stage	State	V	Field value
Before	$(18, 10, 2)$	16	–
A to hub	$(14, 14, 2)$	12	4
Hub to B	$(14, 10, 6)$	4	8

The example provides a calculation, not a recommendation that every delivery should pass through a hub. It omits time, handling and equipment. The scope is stated precisely so that a later extension can identify what changed.



First edge: field value 4

Second edge: value 8

Route total 12; intermediate potential cancels.

Figure 44.1: The relay route has two real transformations and two consecutive baselines. Their signed values add to the endpoint difference.

44.4 Distance enters through physical mechanisms

A longer route may require more energy, more storage, a different transport mode or greater exposure to delay. Distance matters through those physical relationships. The present equation does not contain a universal inverse-square distance law or an automatic per-kilometre burden.

Two routes of the same length can have different grades, loads, efficiencies and reliability. Two routes of different lengths can deliver the same service with different total physical effects. A route comparison must therefore state the transformations and constraints rather than use distance as a substitute for them.

For a time-sensitive medicine, a shorter delay can matter even when transported quantity is equal. If the model ignores expiry or temperature, it cannot certify that the two deliveries are equivalent. A travel-time label in a table is not enough unless the intended service comparison actually uses it.

44.5 A loss ledger is a physical extension

The main Gaussian examples are lossless. To understand the boundary, consider a separate quantity-only route where ten units pass through efficiencies 0.9, 0.8 and 0.95. Delivered amounts are 9, 7.2 and 6.84. The successive missing quantities are 1, 1.8 and 0.36, totaling 3.16.

Those quantities need named carriers, sinks or boundary crossings. The physical ledger can then show $10 = 6.84 + 3.16$ in the relevant stock units. Whether any “loss” remains useful somewhere else depends on the carrier and boundary. Water entering another basin is not annihilated water; usable medicine becoming waste is a transformation of condition or classification.

The product $0.9 \times 0.8 \times 0.95 = 0.684$ is an end-to-end quantity retention ratio. EBU values are not multiplied that way. Each finite value depends on its own before and after field. The lossless Gaussian controller or attribution formulas cannot simply inherit every interpretation from a heterogeneous-efficiency world without re-derivation.

Scope of the loss illustration

The route calculation above closes physical quantities only. It adopts no loss-aware runtime, no valuation of the sink, and no residual dissipation charge. Physical accounting and valuation need separate declarations, especially where an older process-burden rationale assumed a missing loss coordinate.

44.6 Fuel, electricity and water stay linked and distinct

A generator converts fuel energy into electricity and other outputs. A delivery system conveys electricity. A pump uses electricity to move water and generates heat. These are linked transformations, but the fuel quantity, electrical energy and water quantity do not share one stock unit.

An adequate physical description names each input and output, including boundary exchange and losses of useful form. It does not call the entire chain one water action while treating the upstream transformation as invisible. Nor does it subtract an additional burden at every arrow merely because the arrows exist.

The accounting boundary determines which effects enter the physical state and the valuation. Double-counting can arise if the same electrical consumption lowers a represented energy account and is also subtracted unchanged as a generic water-process cost. The finite EBU of the declared complete transition is the potential difference, not that difference followed by a second hidden deduction.

The original book's practical lesson survives: decompose enough to measure physical transformations, and preserve enough links to understand the service. Neither a disconnected list nor an all-purpose mega-action provides both.

44.7 Several resource accounts form a vector

Suppose one project reports +8 in a declared water field and −2 in a declared energy field. The honest record can display

$$\mathbf{E}_a = \begin{pmatrix} +8 \text{ EBU}_{\text{water}} \\ -2 \text{ EBU}_{\text{energy}} \end{pmatrix}. \quad (44.1)$$

Even if both numbers are mathematically dimensionless after standardization, their scales and interpretations can differ. Adding them to obtain 6 chooses a conversion or weighting rule. Dimensionless notation alone does not justify that choice.

A scalar combination may be useful for a specified decision, but its weights need an explicit meaning and sensitivity analysis. If one resource is non-substitutable for an essential function, an easy gain in another account cannot automatically compensate for its loss. Physical feasibility and minimum service conditions may remain separate constraints.

Within each resource account, the relevant endpoint and process terms can close. The vector preserves those separate reconciliations. It allows comparison without pretending that one universal token has already solved the problem of valuing heterogeneous systems.

44.8 Planning is a different use of the records

A planner can compare feasible routes, examine repair strategies or forecast future needs. Such planning may use information that a local action quote does not: deadlines, inventories elsewhere, expected failures and future scenarios. This is acceptable when the layer and its assumptions are declared.

The planner should not be smuggled into the EBU formula. The formula evaluates an action given its state and field. A route optimizer chooses among possible sequences using an objective. A random actor chooses by a different rule. Both can use the same value calculation without becoming the same experiment.

If a planner's prediction later differs from reality, the forecast and actual record should be compared. Replacing the actual physical result with the predicted favourable one would destroy the account's purpose. A forecast can justify a decision under uncertainty while remaining a forecast.

44.9 Receipts as evidence for maintenance

A series of well-specified records may reveal that a pump repeatedly uses more electricity for the same delivered water, that a route often misses a deadline or that emergency parts come from one fragile source. These patterns can motivate investigation.

They do not automatically identify a cause. A higher apparent burden may reflect changing loads, altered references, better measurement or a genuine equipment problem. Comparisons need consistent conditions and versioned definitions. A report should retain these explanatory variables instead of ranking devices by one unqualified average.

Maintenance itself has physical consequences. It can consume parts now and improve later performance. Its present construction or repair record and the later operating records are connected, but the later improvement cannot be issued in advance as an observed current field reduction unless a justified present capacity coordinate already represents it.

44.10 Infrastructure changes the menu of possible actions

Building a bridge, laying a pipe or adding a cold store can change which actions are feasible. This is a deeper change than selecting a different quantity on an existing edge. The action menu, physical map and possibly the represented state must be updated with explicit versions.

The field can describe a consequence only through its declared variables. If a new bridge changes travel time but the model contains neither time nor service

constraints, the benefit is not captured by merely adding an edge label. The limitation should be reported and the representation revised where justified.

Once the new infrastructure operates, future receipts can supply evidence about actual use and consequences. That evidence may differ from the investment forecast. A good archive lets readers study the difference, including unused capacity, maintenance demands and unexpected access effects.

44.11 Cooperation is a hypothesis to examine

Two clinics might share a cold store, transport capacity or specialist repair service. If doing so meets the same needs with lower total represented burden, the group account can show that difference. It still does not determine how to distribute costs, access, responsibilities or gains.

Shared provision can create dependencies and vulnerabilities as well as efficiencies. A failure at a common store can affect both clinics. A powerful participant can control access. Geographic concentration can disadvantage a remote community. These concerns require service and institutional analysis alongside the physical account.

Thus the reasoned hypothesis is conditional: better physical information may help identify beneficial cooperation. Whether it produces stable, fair cooperation depends on mechanisms and behaviour that must be studied. A calculation showing a lower potential is not a proof of voluntary participation or equitable bargaining.

44.12 Poverty can constrain the choice set

A household may use an inefficient device because it cannot obtain a replacement. Another household may have capital, secure housing and access to shared infrastructure. Their physical records can differ because their feasible opportunities differ, not because one values the environment more.

An institution that attaches consequences to EBU records must confront that asymmetry. It could provide access, collective investment, support for essential services or other arrangements, but the equation does not choose among them. A physical burden should remain visible while responsibility and assistance are considered separately.

This is why socioeconomic questions were placed at the beginning of the revised book. They are not a decorative appendix to an otherwise finished currency. An account intended to serve life must be assessed for how people actually encounter its rules, including people with disability, limited resources or unavoidable high-burden needs.

44.13 Observation does not require unlimited surveillance

A record can identify a transformation without publicly exposing every personal detail of its beneficiaries. The information needed to verify a physical event depends on its quantities, timing, boundaries and evidence. Some identity links may be held by authorized auditors; other information may be aggregated or withheld.

The choices need explicit governance: purpose, access, retention, correction and appeal. Mathematical locality offers a way to describe necessary physical inputs. It does not itself design secure storage or justify collecting personal data unrelated to the calculation.

More observations can improve an estimate, but more data can also multiply inconsistent definitions and opportunities for misuse. The quality of a physical account depends on what its measurements mean and how they can be challenged, not simply on how many records exist.

44.14 Money and physical accounts can coexist

A payment can finance a repair while a physical account describes its effects. The two ledgers may inform the same decision without being interchangeable. Money can redistribute claims or fund access; it cannot retroactively change the litres moved or the field used in an observed account.

One possible research sequence begins with observational physical accounts in a narrow domain, examines measurement and interpretability, and only later studies institutional consequences. This is a possible path, not an adopted deployment plan. Broader claims require broader evidence.

An institutional capacity rule is not identical to the physical account. If a future system uses balances to decide which offers are affordable, that rule must state whose balances, which obligations, and how an essential service is protected. Property rights, income support, appeals, labour agreements and democratic authority do not follow from the physical identity; they have to be designed and assessed openly.

44.15 Guided problem: compare complete routes

Two routes move the same amount of water between the same endpoints under one fixed field. The first uses less electrical energy than the second. If energy is represented in the complete state and the potential, their final states differ, so their EBU values may differ too. Without the energy coordinates and their valuation, the difference cannot honestly be assigned a number. If both routes

reach exactly the same complete represented endpoint, their EBU values are equal; a hidden route charge cannot change that identity.

Now suppose the lower-energy route misses a required deadline. It no longer satisfies the same service problem. A report should say that it fails the deadline, not rank it as the better solution by ignoring time. If time-dependent degradation changes the physical endpoint, even equality of the water quantities is insufficient.

44.16 Practice and reflection

Recalculate the two-edge Gaussian route with quantity two. The first state becomes (16, 12, 2) with potential 13; the second becomes (16, 10, 4) with potential 9. The values are 3 and 4, totaling 7, equal to the direct endpoint reduction.

Then choose a familiar service, such as drinking water or a refrigerated meal. Draw three linked physical transformations and name the carrier at each. Identify one quantity ledger, one possible service constraint and one institutional question that cannot be read from the physical ledger.

Finally describe an infrastructure improvement and separate its present construction effects, measured capability, future forecast, authorization and compensation. State which claim can be observed now and which must wait for later evidence.

What to carry forward

Routes compose physical actions under explicit states and times. Resource accounts retain their meanings as a vector unless a conversion is justified. Records can inform learning and cooperation, but adaptation, access and governance require their own evidence and choices.

The route links physical records across space and time. The next chapter asks a further institutional question: when could a signed history alter anyone's future capacity to act, and which parts of that choice remain outside the physical law?

From a signed history to a capacity institution

45.1 A record is not yet a permission to spend

The physical account gives a signed value to a complete transition. A declared attribution can place parts of a joint value beside actors. A further institutional rule might use those records to alter future decision capacity. These are successive choices, not synonyms.

A signed historical account can record harmful and beneficial contributions without requiring its balance to stay nonnegative. A spendable capacity balance can instead impose a nonnegativity rule, a lending rule or another constraint. The potential theorem selects none of them. This distinction is essential if an EBU institution is to support people whose necessary actions sometimes have negative physical value under the current field.

45.2 One conditional balance identity

Suppose a separately declared policy maintains accounts B_i and updates them so that $\sum_i \Delta B_i = E_t$ for the complete actor transition at event t . Let $A_{\text{ext},t}$ accumulate external potential changes, using $D_{\text{ext},t} = V(x_t^b) - V(x_t)$ before the actor's transition $x_t^b \rightarrow x_{t+1}$. Under one fixed field,

$$V(x_t) + \sum_i B_{i,t} - A_{\text{ext},t} = K \quad (45.1)$$

is invariant when the updates and initial value K are defined consistently.

Proof. The change in the left side is $V(x_{t+1}) - V(x_t) + E_t - D_{\text{ext},t}$. Substitute $E_t = V(x_t^b) - V(x_{t+1})$ and the definition of $D_{\text{ext},t}$. Every term cancels. $\square$

This is a conditional accounting theorem, not adoption of a wallet. External audit is not a pool that the next actor may spend. Different ownership and update rules can satisfy the same aggregate identity while giving people very different opportunities.

45.3 Positive group value does not settle owner permission

Imagine an attribution rule gives two owners changes $(+3, -1)$, totaling a positive group value of two. Under a proposed rule requiring each post-event capacity to remain nonnegative, two initially empty accounts would refuse the group because the second owner cannot cover the debit. If one owner held both lines and same-owner netting were permitted, the net $+2$ could pass.

The physical endpoint and group EBU have not changed. Ownership and netting changed the institutional decision. A policy that pools all owners, lends capacity or protects an essential action would produce another decision. Those alternatives deserve explicit comparison rather than being hidden inside a routine called settlement.

The possible social benefit of separating these layers is practical: one can retain the physical cost of a necessary service while designing access rules that do not punish its recipient. Hiding the negative physical line would make the system less informative; treating it as an automatic prohibition would make an unproved moral choice.

45.4 Zero genesis is a narrow observation

At a strict quadratic reference, every nonzero net displacement has negative EBU. If all balances begin at zero and a policy requires every account change to be nonnegative, their sum cannot equal that negative group value. Such an action is therefore unaffordable under that particular policy.

This does not prove attraction toward the reference. After a disturbance, positive records may create capacity, and the same policy may permit uphill actions later. A finite menu may also leave no affordable restoring action. A stationary physical state need not imply stationary account balances under continuing external drive.

These possibilities explain why a capacity mechanism requires a dynamical study beyond its invariant. Closure constrains the process but does not choose its behavior. A correctly implemented mechanism can cycle, ration, refuse or perform badly on service.

45.5 Repeated damage and repair need complete provenance

Under a fixed field, a complete closed physical cycle has zero aggregate EBU. This can block one simple accounting exploit when every signed transition is recorded. It does not by itself identify who caused damage, prevent an actor from concealing the negative event, or guarantee that gains and losses are assigned to the same owner.

A person who damages a shared asset and another who repairs it may have opposite historical records. Whether compensation or responsibility follows needs evidence and governance. The aggregate theorem cannot infer intent from endpoints. Essential maintenance can also recur because of genuine external wear or demand, so repeated positive restoration is not automatically suspicious.

A useful institution would therefore preserve event identities, causal evidence where available, correction procedures and appeals. These are social and engineering requirements supporting the account, not conclusions secretly contained in the equation.

45.6 Atomic commitment protects a joint account

If an institution authorizes one joint group, its physical and account records must refer to that exact group. Crediting one child and losing another child's debit can manufacture a claim. Executing the physical action twice after a receipt failure can duplicate the consequence. An eventual operating protocol needs durable identity and explicit recovery semantics.

Here "atomic" means a coherent all-or-nothing institutional commitment under the declared protocol. It does not mean concealing a chain of physical transformations inside one undifferentiated event. Fuel, electricity, transport and water effects still need traceable boundaries.

No operating protocol is adopted by this discussion. The practical feature is a requirement that an implementer can inspect: a committed entitlement must be linked to the authorized complete event and applied once. If implemented and audited, that feature could reduce both accidental duplication and deliberate overclaiming.

45.7 Access changes whether information can help

A household may be unable to replace an inefficient heater even when the physical benefit is obvious. A remote clinic may depend on a costly route because no alternative exists. Giving those actors better scores does not give them capital, infrastructure or bargaining power.

An EBU institution could use physical evidence to help target public investment, support maintenance or identify shared services. It could also worsen exclusion if adverse records become automatic barriers to care or mobility. The same accurate measurement can enter very different institutions.

The constructive question is therefore how the information changes feasible opportunities. Does it help finance a repair that lowers future burden? Does it

reveal an unrepresented service failure? Can affected people challenge the field or the measurement? Are people with unavoidable needs protected? These are concrete criteria for later institutional studies, not benefits established by the balance identity.

45.8 A place for learning without retrospective rewriting

If a field turns out to omit an important consequence, revise it through a visible process. If an allocation produces unfair outcomes, revise that rule while retaining its historical version. The separation of physical value, attribution and capacity makes such changes more intelligible: one can identify which layer failed and what evidence motivated the revision.

That separation is a strength of the proposed architecture. It gives a future society a way to learn without pretending that either a mathematical theorem or a political rule was infallible. The next chapter translates the same discipline into the responsibilities of an implementation.

Making the account reproducible

46.1 The program must calculate the object the theorem names

An exact equation can be implemented incorrectly by selecting the wrong baseline, omitting a factor, using requested rather than accepted quantities, or giving a selector information it was not meant to see. More decimal places do not correct those errors. The implementation needs records and interfaces that preserve the scientific meaning of every operand.

The older EBU books devoted substantial attention to these boundaries. The central lessons remain relevant, even though their historical engines, random-affordable proposals and execution statuses are not the active authority for a new system. We retain those lessons here without importing old runtime assumptions or claiming that the present book has executed a scientific model.

46.2 Typed records prevent plausible-looking category errors

Physical stocks, reference parameters, route capacity, signed historical contributions and spendable institutional balances should be separate objects. A cached potential is a derived value, not another physical carrier. A numerical field called “budget” is insufficient if it can mean either litres available at a source or EBU an owner may spend.

For one accepted event, the essential record connects the predecessor state, field version, action identity, requested and accepted quantities, complete successor, affected factors and measurement evidence. If a group is involved, retain its children and the declared comparison or attribution semantics. An evaluator should be able to reproduce the value without changing the world.

This design can make errors easier to discover. A quantity with units of stock per hour should not be inserted where a finite stock amount is expected. A later field version should not silently replace the one attached to a receipt. Refusing such substitutions is more reliable than trying to infer the author’s intention from a convenient number.

46.3 Separate permission, evaluation and choice

A physical checker determines whether the complete proposed transformation satisfies the declared constraints. An evaluator computes its value. A selector chooses among eligible alternatives under an explicit policy. An execution mechanism applies exactly the authorized transformation. A recorder preserves what occurred and its evidence.

The boundaries matter because the components answer different questions. A physical checker that silently selects the most favorable quantity is also a controller. An evaluator that changes stock while computing a quote is also an executor. A logger that edits inputs until an invariant closes is changing the scientific event.

A future system can combine these components operationally, but their responsibilities and permitted information must remain inspectable. Local valuation does not establish local selection: a group assembler may still read a global menu. A global auditor can observe outcomes without making those observations available to an actor.

46.4 A small arithmetic audit can expose a large semantic error

The four-unit transfer from (18, 2) to (14, 6) has exact value twelve. Endpoint evaluation, the finite quadratic and the changing-field integral all agree. Returning sixteen suggests that the initial tangent replaced the finite account. Returning fourteen for two two-unit children suggests that independent counterfactual values replaced the group total.

A useful check uses independently derived expected values, not two calls through the same helper. It also includes negative actions, cancellation, unequal scales and non-unit durations. A test that only examines beneficial examples may miss a reversed sign or a hidden rule deleting adverse lines.

For interaction, reconstruct the table from its coefficients and compare with the original entries. Check the quadratic triple null and a constructed nonzero cubic or nonlinear-response case. These are arithmetic checks of mathematical objects. They are not evidence that a deployed system improves service or that an experimental process has been run.

46.5 Randomness requires more than a seed label

If a later registered policy uses random selection, the sample space must be explicit. Uniform choice among edges differs from uniform choice among quantities, groups or owners. Duplicate menu entries change probabilities. A no-

action option changes time occupation even when its immediate field value is zero.

Separate environmental and actor randomness when the design calls for independent or paired exposure. Otherwise extra rejection draws in one arm can shift its later environmental sequence. Even common raw environmental draws do not guarantee common applied disturbances: a proposal can be feasible in one evolving state and infeasible in another. The protocol must record which form of pairing it intends and how state-dependent feasibility is handled.

These are scientific choices, not incidental implementation details. They determine the transition process being studied. A future EBU comparison should preserve them before outcome inspection rather than quietly selecting the version that makes an intervention appear more successful.

46.6 What the implementation must preserve across interruption

A reproducible restart may need physical state, field version, pending event identity, attribution or capacity records, external audit and random-stream coordinates together. Restoring only the stock can leave a different future because the accounts or random position changed.

An invocation, a scientific run and a finalized result are different objects. A program can fail before any state changes, or finish calculations before its evidence packet is durably complete. A study needs an explicit operational contract for these cases. The book does not grant a new attempt by calling it recovery.

Hashes help identify the exact bytes of that contract, implementation and result. They do not establish that a sensor is honest or a model adequate. Their role is to make substitution visible and allow an independent reviewer to know which claim belongs to which artifact.

46.7 Conformance, evidence and adoption

A software test can establish that an implementation follows its specification on the covered cases. A mathematical proof can establish a property for every object satisfying its assumptions. A registered experiment can provide evidence about a physical or simulated regime. Institutional adoption requires further judgments about access, harms and legitimacy.

Historical EBU experiments remain attached to their own source versions and protocols. They should not be relabeled as evidence for the new Möbius-generator hierarchy or a new capacity institution. Useful historical lessons

include the distinction between a soft penalty and a hard permission, preservation and delivered service, performance and efficacy, and valid null results versus invalid comparisons.

For the present edition, the E1a execution pause remains in force in the controlling record. No trajectory, random stream, calibration or campaign is advanced by writing or building this book. Its examples are constructed mathematical objects, and the next chapter describes possible studies without preregistering or authorizing them.

46.8 Why this engineering discipline can matter to society

A physical account intended to influence access or investment must be challengeable. People should be able to ask which measurement changed, which rule applied, why a group was refused, and whether a correction preserved the original evidence. Typed records, independent arithmetic and explicit information boundaries make those questions answerable.

These features could reduce arbitrary decisions and improve trust if paired with effective governance. They are not a substitute for it. A perfectly reproducible system can still implement an unjust rule; a transparent record at least makes the rule and its consequences easier to examine.

Experiments that could test the connections

47.1 A theorem suggests a question, not a result

The hierarchy gives several exact predictions under declared assumptions. Those predictions can guide a future experimental programme, but they do not choose the apparatus, intervention sizes, tolerances or sample counts. Those decisions require a separate prospective design and authorization.

The most useful sequence begins with comparisons whose interpretation is sharp. Establish a pair effect under a controlled quadratic field, test a triple null with certified additive response, and only then seek a positive higher-order benchmark. Here “positive benchmark” means a known nonzero target that the method should detect, not necessarily a positive EBU sign.

47.2 First establish the pair correction

A pair study would compare all four required endpoints under one independently calibrated H and fixed increments h_A, h_B . The prediction is

$$m_E(AB) = -h_A^\top H h_B. \quad (47.1)$$

The design should include conditions under which the correction is nonzero and, where appropriate, a structurally justified zero case. It should verify accepted physical increments rather than assume that identical control settings produce identical changes in every context.

This study could reveal a baseline error, an omitted shared coordinate or a response dependence before the analysis becomes more complicated. Its societal relevance is foundational: a future system must first show it can avoid double-counting a simple shared effect before claiming to value large cooperative projects.

47.3 Then test the additive quadratic triple null

With three fixed increments and a quadratic field, all eight alternatives predict $m_E(ABC) = 0$. A valid physical null requires a complete admissible cube,

common field and baseline, and a covariance model that includes shared calibration and measurements.

A nonzero result would reject the joint null assumptions at the stated uncertainty, not prove one selected explanation. A zero result would support compatibility within the tested resolution and domain, not universal absence of higher-order interaction. Controls should help separate response nonlinearity, field error and measurement drift.

The null is valuable because it can fail. If every nonzero coefficient is removed as a bad example or absorbed into a newly fitted field after observation, the experiment no longer tests the original proposition.

47.4 Introduce a controlled cubic target

A physically justified local cubic contribution can supply the exact algebraic target $m_E(ABC) = -\lambda abc$ under additive increments. A real design must specify its valid domain, stability, boundary and calibration; the unbounded scalar teaching cubic is not itself an apparatus proposal.

The purpose is to show that the measurement and analysis can detect a known higher-order contrast when it should be present. This complements the quadratic null. Passing only a null can mean a method is insensitive; passing only a nonzero case can conceal bias. The two questions belong together.

47.5 Hold the quadratic and change the response

A separate test could retain an independently calibrated quadratic potential while introducing a characterized nonlinear response. The target must come from that response law, not from the assumption that all nonlinear devices produce a triple.

The constructed map $x = z_1 + z_2 + z_3 + \alpha z_1 z_2$ offers an algebraic example with $m_E(123) = -\alpha$. A physical implementation would need its own state, controls and measurements. The experiment should verify the response independently enough to distinguish it from an incorrect potential fit.

This comparison could teach a future EBU system where a dependency resides. Engineering a more additive response and changing a valuation model are different interventions. Knowing which one explains a measured effect can prevent unnecessary changes to an otherwise appropriate field.

47.6 Compare two independent branches at equilibrium

The probability bridge suggests two readings of the same coefficient. Branch A uses physically calibrated potential differences. Branch B estimates the alternating log-density contrast in a declared reference measure. Under the canonical normalization they should agree.

The independence of the branches is essential. If Branch A defines its potential by taking the negative logarithm of Branch B's estimated density, agreement is construction, not validation. Shared measurements or calibration may still create covariance, which must be included in the comparison.

The study also needs compatible endpoint support, positive density where logs are used, density-of-states treatment, and a plan for uncertainty and estimator bias. A thermal entropy interpretation requires the additional P4 conditions; density agreement alone is not a test of reversibility or absence of work.

47.7 Cross-field interaction is a later question

Only after a justified denomination bridge could one compare higher-order contrasts across different fields as quantities on a common scale. Actions must also be matched in physical meaning. Equal labels do not guarantee equal interventions, and mathematically dimensionless potentials need not have interchangeable units of significance.

This leaves an ambitious research direction open without treating it as established. If comparable accounts across domains become defensible, they could help institutions examine tradeoffs that price records alone obscure. If comparability fails, separate vector accounts may remain more informative than an unjustified scalar sum.

47.8 Relation to E1a

The current E1a physical-probabilistic work provides a nearby benchmark and an explicit separation between physical and statistical branches. Its present scientific and execution status remains governed by the frozen repository records. This chapter neither changes its target nor restarts the paused validation track.

The interaction studies described here are proposals beyond the current book integration. No seed, threshold, horizon, instrument or campaign is adopted in these pages. A later design should identify which prerequisites E1a actually establishes and which the new interaction question still needs.

47.9 Beyond one favorable coefficient

An institution aimed at sustaining life support needs evidence about service, reserves, resilience, distribution and access over time. A positive interaction coefficient is one possible piece of information, not a complete success metric. A group can have a favorable interaction while failing to deliver a necessary service or shifting harm outside the boundary.

A long-run study must therefore define its physical domain and comparable alternatives, distinguish natural forcing from actor action, and retain refusal, cycling, adverse and inconclusive outcomes. Boundedness of a continuous potential on a finite closed stock domain is already guaranteed by the domain; it cannot discriminate the success of one actor mechanism.

Similarly, a potential minimum does not prove attraction. A finite action menu can make a compatible reference unreachable. If every allowed transfer changes stocks by an even integer, parity can prevent reaching a reference of the opposite parity. No amount of exact valuation repairs that physical opportunity set.

47.10 A credible route to social benefit

The proposed sequence has a practical purpose: establish that the account measures what it claims, then examine whether people and institutions can use that information well. Small exact benchmarks can reduce uncertainty about the mathematical and physical links. Later studies can ask whether maintenance is better supported, complementary services are coordinated, and essential access is protected.

A failed bridge or adverse institutional result would be useful evidence. It would identify a limitation rather than invalidate every earlier identity. Conversely, a successful benchmark would not authorize broad adoption without further work. The ambition is substantial enough to deserve this sequence of distinct questions.

The final chapters return to the system as a whole: how these pieces might fit into an inspectable institution, and what would count as success beyond a correctly closing account.

From a calculation to an EBU system

48.1 The promise and the missing pieces

Money is useful because it lets people coordinate exchange without re-deriving the full physical history of every purchase. The opening chapters showed why a price alone does not tell us whether the underlying action preserves the conditions of future life. EBU offers a different piece of information: the signed change in a declared measure of physical balance caused by a real transformation. A system built around that information could give people a reason to repair, maintain and regenerate rather than reward only what can be sold now.

The equation is not yet that system. A mathematical result gives a number once state, field and action have been defined. It does not define who may act, whose need has priority, what a person owns, or what happens when a necessary action has negative EBU. Those questions are not embarrassments to be hidden in small print. They are design choices on which the system's usefulness depends.

48.2 Five layers that must remain distinguishable

The first layer is physical description: stocks, flows, carriers, sinks, boundaries and services. Every accepted action has a predecessor and successor state and must pass the relevant conservation and feasibility tests. A useful label is not a substitute for an actual transition.

The second layer is valuation: a declared, versioned potential on the represented state, with its reference values, scales, coupled factors and scientific rationale. Under one fixed field, exact finite EBU is its before/after difference. This layer can be local when affected factors are identifiable; that possibility is important because no society can recalculate a complete planetary state for every small action.

The third layer is attribution. A group of simultaneous actions has one physical successor and one group value. Separate actor records may be useful, but a split of that group value requires its own declared path, rule and meaning. An algebraic decomposition is not automatically a right to spend.

The fourth layer is choice. People and organizations face needs, constraints, histories and incentives. A controller might use an EBU quote to rank physically permissible plans, limit a harmful plan, or make restoration more affordable. It might also protect indispensable service even when the current field gives an adverse sign. The particular policy must be specified before results are observed if it is to be scientifically tested.

The fifth layer is institution: verification, rights, appeal, distribution, public authority and transition from existing money. An accounting unit could co-exist with monetary prices, be used only for disclosure, or become part of a more ambitious capacity system. The physical theorem does not select among these arrangements. Their effects on inequality, coercion and access must be examined rather than presumed beneficial.

48.3 For what would someone receive EBU?

This question needs a concrete answer at the level where the term is used. In the physical account, an action receives a signed EBU record when a verified complete transition lowers or raises the declared fixed potential. It is the event, not the actor's virtue, that is measured. If the event lowers the potential, its EBU is positive; if it raises the potential, negative. A zero record can still be a real action.

In a prospective capacity institution, the further question is whether and how that record changes anyone's future ability to authorize actions. One possibility is that verified restoring work creates a bounded claim, while damaging work uses capacity. But a raw identity cannot settle transfers between owners, initial endowments, essential needs, external disturbances or the possibility of cycling through actions as the field changes. A design that ignores these cases could reward strategic damage or strand people whose urgent needs carry negative field value. That is why a "wallet" must be a separately justified mechanism with adversarial tests, not a renamed physical receipt.

Imagine a town repairing a leaking water line. The first record is not "a repair earns EBU." It describes the line, the water carrier, any energy and material use, the measured pre-repair loss, and the successor condition of the system. A fixed, declared potential then compares the complete states. If the repair genuinely reduces represented deviation, its physical EBU is positive and can be checked by others. The result gives the town information that a repair invoice alone could not convey. A capacity institution could go further and reward the verified contribution, but it must disclose that extra rule and check that the reward cannot be obtained by first creating the leak. This is the distinction between a useful incentive and a circular credit machine.

The same framework does not need to punish a person for drinking water or receiving care. It can recognize service as a reason to act, physical limits as constraints, and the EBU account as information about consequences. Where service and the current field conflict, an institution has to make its priority explicit. An honest negative account can motivate investment in a better supply route or more resilient infrastructure; hiding the negative number would make those improvements harder to see.

48.4 An inspectable event, not an invisible score

A serious record should say which action occurred; which state and field versions were used; which quantities and carriers changed; how conservation closed; which service it was meant to provide; what evidence supports the measurement; and whether the result was a group value or a specifically defined attribution. Corrections should preserve the original event and explain the new evidence rather than overwrite history.

This information is enough to make a calculation challengeable. It is not a demand for public disclosure of personal life. Privacy, data retention and access are institutional design problems. They should be solved with the minimum measurement needed for the claim, independent review and a way to contest errors.

48.5 How an incentive could differ

Under a price-only decision, maintaining a shared water line may be delayed when the payer cannot capture its broad future benefit. With a trustworthy physical account, the same repair can be described by its effect on represented reliability, water availability and energy use. If an institution then rewards demonstrated restoration or uses that information to finance preventive work, the incentive changes. That is a plausible design pathway, not yet an observed economy-wide outcome.

The design must also resist a tempting shortcut: maximizing the EBU number at every moment. A body stays alive through regulated flows, not by maximizing one hormone or emptying every reserve. A society needs service, reserves, redundancy and adaptation as well as balance. The useful question is whether EBU-guided choices, among real feasible alternatives, help preserve these conditions over time better than relevant comparators. We cannot answer it by looking at one favourable equation.

The next chapter states what kind of proof and evidence would turn that promise into a defensible claim.

What would count as success?

49.1 Four statements with different burdens of proof

We can now state the core result without shrinking the ambition. For a declared complete physical transition and a fixed potential, its finite EBU is the potential before minus the potential after. With sufficient differentiability it is also the negative gradient integral along a declared path between those states. Consecutive actions under the same field telescope. These are mathematical statements under assumptions, not promises that a particular society has already been regulated.

The next statement is about an implementation. Software can check source limits, calculate a Gaussian quote exactly for rational inputs, preserve event provenance and reject inconsistent records. Passing those tests establishes conformance to the implemented contract. It does not show that the chosen field is a faithful measure of planetary function or that people will respond well to the incentives.

A third statement is scientific evidence. A registered study may compare EBU-guided action with controls under demand, disturbances, reserves and loss. It should publish every defined outcome, including failure and non-evaluable cases, under a frozen protocol. A demonstration that arithmetic closes is not evidence that a living system remains viable; it is a prerequisite for interpreting such evidence.

The final statement is institutional. To claim that an EBU economy improves long-term life support, one must examine who can participate, which harms cross its boundary, whether essential needs remain available and what happens outside the studied world. Those are not reasons to abandon the project. They identify the work required to make its promise real.

49.2 What a long-run test must ask

A genuine long-run study needs repeated physical opportunities or demands, replenishment or regeneration where those processes exist, disturbances and reserve constraints. It must specify before execution what counts as persistent restoring tendency or sustained function, and distinguish physical possibility

from actor affordability. It should compare policies on the same external conditions without quietly giving one policy a better menu, a different arrival stream or a hidden optimizer.

At least three falsifiers are easy to name. The model may be physically incomplete: a positive EBU action might merely move harm outside the boundary. The policy may be ineffective: even with a faithful field and feasible choices, actors may not preserve service and reserves. The institution may be unfair: it might deny necessary action to those without historical capacity. A useful study is designed to expose these failures, not to make them impossible by definition.

49.3 The special role of exactness

The Gaussian accounting identities can be checked exactly under rational inputs. That is valuable because a failed equality then points to a model, implementation or record problem rather than an adjustable numerical tolerance. Yet exact arithmetic is not exact knowledge of the world. Measurements have error, reference values can be contested, and a formally complete state can still omit a relevant carrier. The right attitude is to be strict about the mathematics and candid about the empirical boundary.

49.4 A future the account is meant to protect

The planet will continue receiving sunlight and changing through cycles whether or not an accounting system exists. What is not guaranteed is that human action will preserve the living and built conditions on which future generations depend. A sale can be successful today while drawing down water, fertility, resilience or access. A repair can matter enormously even when no immediate sale rewards it.

EBU begins with the idea that physical balance should be visible in the decisions that shape those conditions. The field and finite action law make a precise form of that idea possible. They do not require the world to stand still at a perfect point. Homeostasis means sustaining function through change, including the ability to recover and adapt. That is why the ultimate measure of the project is not how many positive receipts it can issue, but whether a properly governed system helps people meet needs without consuming the basis of tomorrow's life.

This book has carried the reader from monetary incentives through physical laws, homeostasis, Gaussian geometry, exact finite accounting and the design of an inspectable institution. The route is continuous, but its destinations remain distinct: a theorem is proved, a mechanism is specified, an experiment

is run, and a society decides whether the result is worth trusting. EBU's ambition deserves all four.

Finite differences, regularity and exact checks

This appendix supplies details that would interrupt the main explanatory route. Every calculation concerns a declared finite table or analytic expression. None is a simulated trajectory or a new experimental result.

A.1 The alternating-sum identity by induction

Let $T = \{1, \dots, k\}$ and let F be defined on the corresponding Boolean cube. For $k = 1$, $\Delta_1 F(0) = F(1) - F(0)$. Suppose the k -fold formula is known. Taking one more difference subtracts the same k -fold sum with the new indicator absent from the sum with it present. The first sum gives all subsets containing the new index with sign $(-1)^{k+1-|S|}$, and the second gives all subsets lacking it with that same rule. Thus

$$\Delta_T F(0) = \sum_{S \subseteq T} (-1)^{|T|-|S|} F(\chi_S) \quad (\text{A.1})$$

at every finite order. No assumption about the values other than their existence is used.

This proof also establishes commutation of distinct difference operators: the resulting signed sum depends only on the set of indices, not the order of expansion. Ordered physical flows can still fail to commute because their endpoint protocol is a different object.

A.2 Repeated FTC with a useful weaker formulation

The main theorem assumes $G \in C^k$ on a neighbourhood of the cube. One may weaken this if the repeated one-dimensional FTC steps remain valid. A sufficient formulation is that, in a chosen coordinate order, the successive mixed partials have absolutely continuous representatives on the relevant coordinate sections, their endpoint traces agree with the lower-order derivatives used in the iteration, and the final mixed derivative is integrable with the hypotheses needed for Fubini's theorem. The conditions must cover the boundary sections appearing in the repeated differences, not merely an unspecified almost-everywhere interior derivative.

Under these conditions, apply the one-dimensional absolutely continuous FTC successively. Each step gives an endpoint difference equal to an integral, and the integrability conditions justify the repeated integration. This is a valid weaker route, but it is not a license to replace the hypotheses with “smooth enough” without checking them in a piecewise or singular model.

The Cantor function illustrates the risk. It is continuous, increases from zero to one, and has derivative zero almost everywhere, yet the integral of that derivative is zero. It is not absolutely continuous. A statement based only on almost-everywhere differentiability would miss its finite change. For a nonsmooth EBU model, the finite endpoint definition remains available even when a proposed differential representation needs repair.

A.3 The remainder estimate in detail

For $h_i = \varepsilon d_i$, factor ε^k out of the multilinear derivative in the repeated-FTC formula. Subtract the derivative at x_0 . If $D^k V$ is continuous, uniform continuity on shrinking compact neighbourhoods gives a remainder $o(|\varepsilon|^k)$.

If $D^k V$ is Lipschitz in operator norm with constant L , the integrand difference is bounded by

$$L|\varepsilon| \left\| \sum_i t_i d_i \right\| \prod_i \|d_i\| \leq L|\varepsilon| \sum_i t_i \|d_i\| \prod_i \|d_i\|. \quad (\text{A.2})$$

Multiplying by $|\varepsilon|^k$ and integrating each t_i gives the $L/2$ bound in Equation 29.5. This argument does not assume the leading derivative is nonzero. If it vanishes, the bound still controls the remainder but does not identify the next nonzero order without more information.

A.4 Checking the mixed cubic example from vertices

For $x = z_1 + z_2 + z_3 + \alpha z_1 z_2$ and $V = x^3/6$, the singleton states are one, the 12 state is $2 + \alpha$, the other pair states are two, and the triple state is $3 + \alpha$. Baseline is zero. Therefore

$$\begin{aligned} m_E(123) &= \frac{-(3 + \alpha)^3 + (2 + \alpha)^3 + 2(2^3) - 3}{6} \\ &= -1 - \frac{5}{2}\alpha - \frac{1}{2}\alpha^2. \end{aligned} \quad (\text{A.3})$$

At $\alpha = 0$ the additive cubic value is -1 ; at $\alpha = 1$ it is -4 . The continuous split in Chapter 32 must add to this vertex result, providing a useful independent sign check.

A.5 Checking interpolation dependence explicitly

For the second pair interpolation in Chapter 32, write the actual map

$$x(s, t) = s + t + s(1 - s)t = s + (1 + s - s^2)t. \quad (\text{A.4})$$

With $V = x^2/2$, the curvature term is $x_s x_t$ and the response term is $x x_{st}$. Here

$$x_s = 1 + (1 - 2s)t, \quad x_t = 1 + s - s^2, \quad x_{st} = 1 - 2s.$$

Integrate in t first. The curvature contribution is

$$\int_0^1 (1 + s - s^2)(3/2 - s) ds = \frac{7}{6}. \quad (\text{A.5})$$

The response contribution is

$$\int_0^1 (1 - 2s) \left\{ s + \frac{1}{2}(1 + s - s^2) \right\} ds = -\frac{1}{6}. \quad (\text{A.6})$$

Their sum is one. For $x = s + t$, the corresponding pair is $(1, 0)$. Both maps agree at all four vertices, so both EBU coefficients equal minus one. The contrast belongs to the vertices; the named derivative split belongs to the smooth interpolation as well.

A.6 Affine constraints and quadratic geometry

For a constant symmetric positive-definite H , minimizing $\frac{1}{2}(x - x^*)^\top H(x - x^*)$ on an affine plane $Ax = b$ is a constrained problem. A stationary interior candidate satisfies

$$H(x - x^*) = A^\top \lambda. \quad (\text{A.7})$$

The sign convention for λ is arbitrary and fixed here by that equation. Solving the constraint gives the appropriate projected point when $AH^{-1}A^\top$ is invertible. Nonnegativity or other active boundaries require their own constrained conditions.

For one conserved total and diagonal $H_{ii} = \sigma_i^{-2}$, the candidate is

$$x_i = x_i^* + \lambda \sigma_i^2, \quad \lambda = \frac{M - \sum_i x_i^*}{\sum_i \sigma_i^2}. \quad (\text{A.8})$$

It solves the nonnegative problem only when every coordinate is nonnegative. Equal nonzero marginals can therefore characterize a constrained stationary point even when the ambient gradient is not zero. This explains why a zero edge field is not universally the same as reaching the declared reference.

A.7 What these checks establish

The calculations verify definitions and constructed examples under explicit assumptions. They supply independent routes for reviewing signs and coefficients. They do not establish a calibrated potential, a measured response law or a successful institution. Their value is to make later empirical disagreements easier to interpret.

The all-order composite chain rule

B.1 Partitions organize every derivative placement

Let T be a finite set of distinct control labels. A set partition π divides T into nonempty disjoint blocks whose union is T . For example, the partitions of $\{1, 2, 3\}$ are

$$1|2|3, \quad 12|3, \quad 13|2, \quad 23|1, \quad 123.$$

The notation records which differentiations act together on one derivative of the response. It is not a partition of physical actors into institutions or a grouping decision for execution.

Assume x and V are C^k on suitable neighbourhoods, with $k = |T|$, and define $x_B = \partial_B x$. The chain rule is

$$\partial_T(V \circ x) = \sum_{\pi \in \Pi(T)} D^{|\pi|} V(x) [x_B : B \in \pi]. \quad (\text{B.1})$$

Since mixed derivatives commute under these hypotheses, the order in which the blocks are listed does not change the term.

B.2 An induction proof

For one label, the only partition has one block and the formula is the ordinary chain rule $\partial_i(V \circ x) = DV[x_i]$.

Assume the formula holds for T and differentiate in a new label j . In any term, the derivative can act on $D^{|\pi|} V(x)$ through x . This creates a new argument x_j and corresponds to adding the singleton block $\{j\}$ to π .

Alternatively, it can act on one argument x_B , replacing it by $x_{B \cup \{j\}}$. This corresponds to inserting j into that block. Every partition of $T \cup \{j\}$ has exactly one of these forms: either j is a singleton, or it belongs to one unique larger block. Removing j recovers a unique predecessor term. Thus every new partition appears once and only once, proving the formula.

This labeled form explains why no factorial coefficient accompanies a partition. Multiplicities arise in alternative versions when several differentiation labels are identified or terms with equal form are collected. Those versions must be translated carefully before numerical coefficients are borrowed.

B.3 The fourth order displayed

For labels 1, 2, 3, 4, the fifteen partitions fall into five patterns:

$$\begin{aligned}
 \partial_{1234}(V \circ x) &= D^4V[x_1, x_2, x_3, x_4] \\
 &\quad + \sum_{\{i,j\} \subset \{1,2,3,4\}} D^3V[x_{ij}, x_k, x_l] \\
 &\quad + D^2V[x_{12}, x_{34}] + D^2V[x_{13}, x_{24}] + D^2V[x_{14}, x_{23}] \\
 &\quad + D^2V[x_{123}, x_4] + D^2V[x_{124}, x_3] \\
 &\quad + D^2V[x_{134}, x_2] + D^2V[x_{234}, x_1] \\
 &\quad + DV[x_{1234}].
 \end{aligned} \tag{B.2}$$

In the six-term sum, $\{k, l\}$ is the complementary pair of labels. The five patterns have counts 1, 6, 3, 4, 1, adding to fifteen.

A quadratic potential leaves the D^2V and DV terms. An affine response leaves only the D^4V term. If both conditions hold, the fourth derivative vanishes, in agreement with the finite degree ceiling. If neither holds, several algebraic origins can contribute to one total.

B.4 From the derivative back to the coefficient

Suppose the complete control cube and the regularity needed for repeated FTC are available. Then

$$m_E(T) = - \int_{[0,1]^{|T|}} \partial_T(V \circ x)(z_T, 0_{N \setminus T}) dz_T. \tag{B.3}$$

Substituting the partition formula gives a finite sum of integrated derivative terms. This can support a model-based diagnostic, but it still depends on the chosen smooth response across the cube. The vertices alone determine the sum, not its decomposition into those terms.

A nonzero local mixed derivative also need not produce a nonzero finite coefficient: positive and negative contributions can cancel in the integral. Conversely, a nonzero coefficient proves that the integrated mixed derivative is nonzero when the representation applies, not that its sign is uniform throughout the cube.

B.5 How to use the formula without overclaiming it

The formula can identify which additional response measurements a proposed explanation needs. A claim that a triple is entirely due to D^3V should justify why the mixed response derivatives vanish. A claim of direct three-way

response should identify x_{123} rather than infer it solely from the total coefficient.

The theorem is established multivariable calculus; Hardy's treatment explains the combinatorics of such partial derivatives [40]. In EBU it serves as an interpretation tool inside a declared model. Its practical benefit is a more precise diagnosis of dependencies, while its limitation protects the book from presenting a coordinate-dependent split as a unique physical discovery.

Time expansions, Lie words and ordered response

C.1 Differentiating one solution

For an autonomous vector field b with solution $x(t)$, differentiate the equation at $t = 0$. With $v = b(x_0)$, $A = Db(x_0)$ and $B = D^2b(x_0)$,

$$x'(0) = v, \quad (\text{C.1})$$

$$x''(0) = Av, \quad (\text{C.2})$$

$$x'''(0) = B[v, v] + A^2v. \quad (\text{C.3})$$

For $F(t) = V(x(t))$, the ordinary chain rule then gives

$$F'(0) = DV[v], \quad (\text{C.4})$$

$$F''(0) = D^2V[v, v] + DV[Av], \quad (\text{C.5})$$

$$F'''(0) = D^3V[v, v, v] + 3D^2V[v, Av] + DV[B[v, v] + A^2v]. \quad (\text{C.6})$$

Subtracting the Taylor expansion of F from $F(0)$ proves Equation 35.1. A C^3 field and C^4 potential on a common neighbourhood containing the short trajectories give a bounded fourth time derivative and hence the stated $O(\tau^4)$ remainder. With less regularity, the remainder must be weakened rather than retained by notation alone.

The calculation does not solve a trajectory numerically. It derives local terms for a declared differential equation. Its empirical usefulness depends on whether that equation describes the physical response over the tested interval.

C.2 The observable generator

For a smooth observable f , define $L_b f = Df[b]$. Along the flow, $df(x(t))/dt = L_b f(x(t))$. Iteration gives the formal short-time expansion

$$V(\Phi_z^\tau(x_0)) = \sum_{r=0}^k \frac{\tau^r}{r!} L_{b_z}^r V(x_0) + O(\tau^{k+1}) \quad (\text{C.7})$$

under the corresponding smoothness, existence and boundedness assumptions. This is a finite expansion with a controlled remainder, not an unconditional claim that an infinite series converges.

If $b_z = f_0 + \sum_i z_i f_i$, then $L_{b_z} = L_{f_0} + \sum_i z_i L_{f_i}$. The operator product of r such factors expands into ordered words. Each word carries a control monomial of degree at most r . The operators need not commute, so their order must be retained.

C.3 Proof of the control-affine order result

Take a nonempty set T of k labels. The T -difference annihilates every term of degree less than k . Therefore all time orders $r < k$ vanish from the coefficient.

At order $r = k$, a surviving monomial must contain each label in T at least once. There are exactly k factors, so it contains each once, no label outside T , and no drift factor. The surviving operator words are all permutations of those k action fields. The endpoint EBU sign gives

$$m_{E_T}(T) = -\frac{\tau^k}{k!} \sum_{\pi \in \text{Perm}(T)} L_{f_{\pi_1}} \cdots L_{f_{\pi_k}} V(x_0) + O(\tau^{k+1}). \quad (\text{C.8})$$

A finite number of Boolean differences preserves the remainder order. The coefficient can vanish through cancellation; then this expression gives an upper-order bound rather than identifying the first nonzero term.

The generator-series and word language has established antecedents in nonlinear systems and control theory, including Fliess and Sontag [41, 42]. The EBU use here is to apply the finite coalition contrast to the endpoint observable, with the comparison assumptions kept explicit.

C.4 The all-time affine endpoint proof

For $\dot{x} = A(t)x + b_0(t) + \sum_i z_i b_i(t)$, let $\Phi(t, s)$ be the fundamental matrix of the common homogeneous equation. Variation of constants gives

$$\begin{aligned} x_z(\tau) &= \Phi(\tau, 0)x_0 + \int_0^\tau \Phi(\tau, s)b_0(s) ds \\ &\quad + \sum_i z_i \int_0^\tau \Phi(\tau, s)b_i(s) ds. \end{aligned} \quad (\text{C.9})$$

The first line is $\bar{x}(\tau)$ and the integral multiplying each z_i is $h_i(\tau)$. The endpoint is affine in controls. A quadratic potential therefore yields a Boolean polynomial of degree at most two for every horizon for which the equations and physical domain assumptions hold.

If A depends on z , the fundamental matrix does too. It cannot be pulled into a common affine expression in this way. The bilinear example $\dot{x} = (\sum_i z_i)x$ makes that failure explicit and produces all interaction orders at finite time.

C.5 Ordered noncommutativity and the sign convention

Let X and Y be smooth vector fields. Applying X for time τ and then Y gives

$$\Phi_Y^\tau \Phi_X^\tau(x) = x + \tau(X + Y) + \tau^2\{\frac{1}{2}DX X + DY X + \frac{1}{2}DY Y\} + O(\tau^3). \quad (C.10)$$

Reversing the order replaces the cross term $DY X$ by $DX Y$. Hence

$$\Phi_Y^\tau \Phi_X^\tau(x) - \Phi_X^\tau \Phi_Y^\tau(x) = \tau^2[X, Y](x) + O(\tau^3), \quad (C.11)$$

with $[X, Y] = DY X - DX Y$. For observables, $L_X L_Y f - L_Y L_X f = Df[DY X - DX Y]$, so the convention is consistent.

The planar example $X = \partial_x$, $Y = x\partial_y$ has $[X, Y] = \partial_y$, agreeing with the positive τ^2 vertical difference when Y follows X . Keeping the convention explicit avoids an otherwise easy sign reversal.

This is a statement about ordered execution. An unordered coalition table must state whether its group row means simultaneous flow, a designated order, an average over orders or another protocol. The table transform cannot choose that semantics afterward.

C.6 A stochastic generator values expected observables

For a Markov process X_t^z with a well-defined transition semigroup P_t^z , an expected-value table can be declared as

$$\bar{E}_t(S) = V(x_0) - \mathbb{E}_{x_0}[V(X_t^{XS})] = V(x_0) - P_t^{XS}V(x_0). \quad (C.12)$$

Require the expectation to be finite. Its nonempty coefficients are the corresponding finite differences of $-P_t^z V(x_0)$. This is an expected observable, not a particular realized endpoint receipt.

Writing a time expansion in a Markov generator additionally requires that V belong to the relevant generator domains and that the iterated-generator and remainder conditions hold. Differential symbols alone do not establish those functional-analytic facts. Boundary conditions and admissibility can matter as much as the interior formula.

This extension shows how the same table logic can organize stochastic response. It does not establish a stationary law, detailed balance or the thermodynamic interpretation of every Markov process. Those questions belong to the next appendix's physical distinctions.

Density, current and thermodynamic scope

D.1 Stationarity and reversibility are separate properties

For a diffusion with constant scalar diffusion coefficient $D > 0$ and density p , the probability current has the form $j = bp - D\nabla p$. A stationary density satisfies $\nabla \cdot j = 0$, with appropriate boundary conditions. Zero divergence permits circulating current; it does not require $j = 0$.

If $p = Z^{-1}e^{-V}$, then $\nabla p = -p\nabla V$. A drift $b = -D\nabla V + c$ therefore gives $j = cp$. Any sufficiently regular c satisfying $\nabla \cdot (cp) = 0$ and the boundary requirements preserves the same density. Reversible gradient diffusion is the special case with no such current in the usual setting of even state variables.

This mathematical distinction is one reason nonreversible sampling dynamics can share a target density with reversible ones [45]. An observed canonical-shaped density cannot determine which of those mechanisms produced it.

D.2 A rotating Gaussian with the same density

On $\mathbb{R}^2$, take $V(x) = \frac{1}{2}x^\top x$, $D = 1$, and a nonzero constant skew-symmetric matrix

$$Q = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}. \quad (\text{D.1})$$

Consider drift $b(x) = -x + Qx$. The standard Gaussian density satisfies $\nabla p = -xp$. Its current is $j = Qxp$.

To check stationarity, use $\text{tr } Q = 0$ and $x^\top Qx = 0$:

$$\nabla \cdot (Qxp) = \text{tr}(Q)p + (Qx)^\top \nabla p = 0. \quad (\text{D.2})$$

The Gaussian decay supplies the usual integrability and boundary behavior. Yet the current is nonzero away from the origin when $\omega \neq 0$. Density agreement with e^{-V} therefore coexists with circulation.

An EBU endpoint log-density identity still holds for that declared density and potential. The accepted P4 interpretation cannot be inferred from the density alone. In a physical model, the nonconservative rotational drive and its work or entropy accounting need their own treatment.

D.3 Detailed balance for a jump process

For a jump process with rates k_{xy} and a positive stationary mass function p , detailed balance means

$$p(x)k_{xy} = p(y)k_{yx} \quad (\text{D.3})$$

for the paired transitions. When $p \propto e^{-V}$, it gives

$$\ln \frac{k_{xy}}{k_{yx}} = \ln \frac{p(y)}{p(x)} = V(x) - V(y). \quad (\text{D.4})$$

The rate-ratio conclusion requires detailed balance, not merely stationarity. A physical local-detailed-balance relation, in turn, connects rate ratios to specified reservoir entropy changes under a physical model. It should not be used as a synonym for global detailed balance in a driven network [44, 43].

D.4 A driven ring with constant state potential

Take a three-state ring, clockwise rate two and counterclockwise rate one on every edge. Uniform probability $p_i = 1/3$ is stationary: each state has total inflow and outflow three times the same mass. Pairwise detailed balance fails because $p_i k_{i,i+1} = 2/3$ and $p_{i+1} k_{i+1,i} = 1/3$.

The uniform density corresponds to a constant V up to normalization, so every endpoint EBU is zero. Nevertheless each clockwise edge has log rate ratio $\ln 2$, and the clockwise cycle affinity is $3 \ln 2$. If a compatible physical local-detailed-balance model identifies those ratios with reservoir entropy flow, the drive has a nonzero thermodynamic cost. The state potential does not account for it as an endpoint difference.

The example isolates the logical boundary without inventing an EBU charge. A broader physical account must explicitly represent the drive, work and reservoirs. It cannot recover their path dependence by claiming that a constant state potential already contained them.

D.5 Stationary curvature and global Gaussian covariance

If a positive density satisfies $J_{\text{prob}} = V + \ln Z$ in fixed coordinates, Hessian equality follows wherever the derivatives exist. No statement about global covariance follows without additional structure.

For a full-support Gaussian with positive-definite constant H , diagonalize $H = R^T \Lambda R$ with orthogonal R . The variable $u = \Lambda^{1/2} R(x - x^*)$ has density proportional to $e^{-u^T u/2}$, so $\text{Cov}(u) = I$. Transforming back gives $\Sigma = H^{-1}$.

If a boundary truncates the density, the transformed support is not all of $\mathbb{R}^d$. Its moments need not be those of the untruncated standard normal. A locally fitted Hessian can describe local log curvature while the tails and boundaries determine different global moments. This distinction is particularly relevant to nonnegative conserved stocks.

D.6 Different statistical interactions use different operations

A log-linear coefficient on a binary cube is an anchored alternating log-probability contrast. Interaction information instead applies inclusion–exclusion to marginal entropies. Marginalization changes the objects before the alternating sum. One should therefore not expect the two operations to commute or produce equal coefficients.

As a simple structural example, a joint log density with only $az_1z_2z_3$ has no anchored pair term at the all-zero background, but conditioning on the third variable being one produces a pair log odds ratio a . Marginalizing over the third variable produces yet another pair expression involving a sum of exponentials. Background, conditioning and marginalization are distinct procedures.

Connected-information and maximum-entropy decompositions offer another perspective on dependence [46]. They can complement an EBU study when the relevant joint distribution is defined, but their terms should retain their own names and estimands. Pointwise endpoint densities alone do not supply the full statistical objects those methods require.

D.7 The boundary is scientifically productive

These examples do not weaken the canonical-density theorem. They identify which additional observations are needed to advance from density to dynamics and entropy. Currents, rate ratios, work and reservoir conditions can distinguish processes that endpoint histograms cannot.

For EBU, that creates a disciplined route to deeper physical interpretation. Begin with the exact finite account; test its density relation where justified; then test the dynamical and thermodynamic premises separately. Each successful step can support a stronger claim, while a failure remains interpretable rather than hidden behind a universal analogy.

Following one claim all the way through

E.1 A declaration small enough to inspect

Use three dimensionless binary controls and the scalar response

$$x(z) = z_1 + z_2 + z_3 + z_1 z_2, \quad V(x) = \frac{1}{2}x^2, \quad x_0 = 0. \quad (\text{E.1})$$

Assume all eight endpoints are admissible for this constructed example. The declaration specifies a response and a potential, not a real apparatus, a selection policy or an economic institution.

For each subset S , define $E(S) = -V(x_S)$. The table, including every interaction coefficient, is

S	x_S	$E(S)$	$m_E(S)$
$\emptyset$	0	0	0
1	1	$-1/2$	$-1/2$
2	1	$-1/2$	$-1/2$
3	1	$-1/2$	$-1/2$
12	3	$-9/2$	$-7/2$
13	2	-2	-1
23	2	-2	-1
123	4	-8	-1

The coefficients sum on the full group to $-3/2 - 7/2 - 1 - 1 - 1 = -8$. The three-way value is not the whole difference between the group and its singletons. The pair terms must be removed first.

E.2 The exact finite polynomial

The unique multilinear extension is

$$\tilde{E}(z) = -\frac{1}{2}(z_1 + z_2 + z_3) - \frac{7}{2}z_1 z_2 - z_1 z_3 - z_2 z_3 - z_1 z_2 z_3. \quad (\text{E.2})$$

Evaluating this polynomial at any Boolean vertex recovers the table. It has no remainder there. Its values at nonbinary controls need not match the physical smooth composite $-x(z)^2/2$, because the multilinear reduction replaced repeated powers using a Boolean identity.

E.3 The smooth derivative check

For the physical smooth response, the only nonzero mixed second derivative is $x_{12} = 1$, and $x_3 = 1$. Since $V'' = 1$ and $V''' = 0$, the triple chain rule gives $\partial_{123}(V \circ x) = 1$. Its integral over the unit cube is one, so the EBU triple is minus one.

This matches the table and demonstrates response-generated interaction under a quadratic field. It does not prove a unique physical cause in another system whose response has not been measured. The response is known here because it was explicitly declared.

E.4 The probability check is conditional

If, separately, these scalar states are evaluated under the normalized Gaussian density $p(x) = (2\pi)^{-1/2}e^{-x^2/2}$ on $\mathbb{R}$, the density relation holds. The triple log product is

$$\ln \frac{p(4)p(1)^3}{p(3)p(2)^2p(0)} = -1. \quad (\text{E.3})$$

The normalizers cancel. The density readings are at response endpoints; they are not probabilities that each coalition will be selected.

This calculation is conditional algebra because the density was stipulated. An experiment would have to obtain its physical potential and density estimates independently enough to make agreement informative. The table itself contains no evidence of stationarity or detailed balance.

E.5 The entropy check needs still more

If every entry were realized by a process satisfying the full fixed conservative overdamped single-reservoir P4 conditions, the medium-entropy coefficient would be $-k_B$. That conclusion does not follow merely because a Gaussian density was written down.

The response law above does not specify heat, work, a reservoir, or how controls are applied. Therefore the example as declared does not claim that any actual entropy was produced or measured. It illustrates exactly where the thermal premise would enter and what additional evidence would be needed.

E.6 A joint outcome still leaves attribution open

The full group has value minus eight. Many triples of signed numbers sum to minus eight. None is selected by the Möbius decomposition as a unique owner

payment. The dividends describe subset interactions; an allocation rule may use them, but must state how and why.

A common-path decomposition would need a declared path and a decomposition of its velocity into child contributions. A Shapley-type allocation would average conditional increments across orders under its own axioms. A service institution might use another rule entirely. All can be discussed without changing the already determined group value.

E.7 An uncertainty statement needs the measurement design

If the eight potential values were independently estimated with common variance σ^2 , the triple contrast would have variance $8\sigma^2$. If the table instead came from shared fitted parameters, its covariance would differ. The exact table cannot tell us which observation model was used.

Thus a complete empirical report would append the calibration and covariance model, not simply say that the coefficient is exactly minus one. Exactness describes the constructed mathematics here. Measurements would need their own uncertainty and possible bias assessment.

E.8 The claim ledger

Claim	What supports it
Complete endpoint table	The declared response and admissible vertices.
Exact coefficients	Boolean finite differences and reconstruction.
Triple under quadratic V	Nonlinear response and the composite derivative.
Log-density equality	An additionally stipulated canonical density and measure.
Thermal entropy interpretation	Not established by this response declaration; full P4 premises would be needed.
Owner entitlement	Not defined by the table.
Social improvement	Not established by constructed arithmetic.

This sequence is the book’s method in miniature. Start from a clear physical comparison, derive what follows, and name every additional premise when

moving to a stronger interpretation. The result is a more useful foundation for future work because it tells readers both what they can rely on and what they still need to learn.

Plain-language dictionary and symbol guide

Action. A declared physical transformation with a specified boundary, quantity or other input, predecessor state and successor. It can be a transfer or another represented change. Naming a useful activity does not by itself define its state effect.

Actor. The person, organization or modeled decision-maker associated with an action. Its identity is distinct from the action's field value. The field equation does not assign moral worth to the actor or recipient.

Affected factors. Terms in a potential that depend on at least one changed coordinate. A complete local calculation includes every such term. In the separable Gaussian model, these are the changed endpoint terms.

Baseline, x^b or x_0 . The common state from which an action and its no-action alternative are compared. In the prospective forcing-first order it is the post-forcing, pre-action state. A baseline must identify both data and model version.

Capacity account, B_i . A proposed persistent account controlling later modeled affordability. It is not the physical stock, storage capacity, household income or an automatically legitimate entitlement. Its update and use require a separately declared rule.

Cell or node. An indexed location or subsystem holding a represented state. A cell is a modeling boundary rather than necessarily a household, person or geographic square.

Common path, $\gamma(s)$. A single declared state path used to calculate the group integral and its child contributions. The path parameter is not automatically clock time. A physical-path reading needs a physical model that realizes it.

Conservation. A complete physical quantity balance within a named boundary. Lossless internal transfers preserve the sum of homogeneous stocks. Transformations, sinks and boundary crossings must be named when they are present.

Curvature. The way a slope changes as the state changes. It is represented locally by the Hessian $H(x)$, constant for a quadratic. Curvature explains why an initial slope times a finite quantity differs from the exact finite value.

EBU, E . The signed difference $V_{\text{fixed}}(x_{\text{pre}}) - V_{\text{fixed}}(x_{\text{post}})$ across a declared complete physical transition. In the Gaussian example it is dimensionless

standardized-deviation reduction. The name does not make it physical energy or money.

Edge. A declared possible directed physical process. For a lossless transfer, its incidence column has -1 at the source and $+1$ at the destination. A line drawn between places is not sufficient evidence that a real process is available.

Equilibrium. A property of a specified dynamical system, typically a state unchanged by its evolution rule. It is distinct from a declared reference or a minimum of a chosen potential unless the required relationship is established.

External audit, A_{ext} . The proposed cumulative signed potential change from external forcing. It explains externally introduced or removed deviation. It is not a spendable “nature wallet.”

Factor. One term in a decomposed potential, with a declared set of input coordinates. A factor can be local without depending on only one coordinate. Its dependencies determine the required evaluation neighbourhood.

Feasibility. Satisfaction of the declared physical conditions on an action, such as nonnegative stock, permitted edges and shared capacity. A favourable EBU does not establish feasibility.

Field. The declared state, reference geometry and relevant action relationships used for valuation. Its gradient supplies local directional information. The word does not assert a new fundamental physical force.

Field attribution, R_a . A child’s signed derivative integral along a declared group path. It sums to the group field reduction under the stated conditions. It is not automatically a uniquely observed causal contribution or an ownership share.

Forcing. A declared external physical change preceding the action baseline in the prospective study. Raw random proposals and physically applied forcing can differ if feasibility rules reject some proposals.

Functional balance. A description of conditions under which relevant functions can continue. A single numerical reference is one representation of part of that idea, not a complete definition for every living system.

Gaussian geometry. The quadratic relative negative-log shape used here to measure standardized deviation. It does not establish an observed normal distribution of actual states, nor an autonomous restoring process.

Gradient, $\mu = \nabla V$. The vector of derivatives of the potential. Each entry describes local change in potential per small change of its coordinate. The gradient is not an actor, a physical stock or a selection algorithm.

Group, $\mathcal{G}$ or S . Accepted actions evaluated through a common physical transition because of their declared dependencies. The child actions remain identifiable. A group total does not by itself determine every child allocation.

Hessian, H . The matrix of second derivatives. For the separable Gaussian model, $H_{ii} = 1/\sigma_i^2$ and off-diagonal entries vanish. Shared action coordinates can still produce group cross terms.

Homeostasis. Maintaining relevant function through continuing change and regulation. The term does not mean that every component is static, that resources are unlimited or that any proposed accounting rule has already achieved regulation.

Incidence column, S_e . A signed template turning an edge quantity into its state increment. Multiplying S_e by a finite quantity gives the source and destination changes in stock units.

Locality. A statement about necessary support or information. Mathematical locality, distributed computation and decentralized governance are distinct properties. None should be inferred from the others without a specific argument.

Marginal field, f_e . For a lossless directed transfer, $\mu_i - \mu_j$. It gives the initial rate of field reduction per added action quantity. Exact finite value also includes curvature and all represented physical effects.

No action. A zero control can still permit natural drift. A zero physical increment has zero fixed-field EBU; a drifting no-control comparator must be retained or explicitly centered as in Chapter 34. Missing evidence is not verified inactivity.

Potential, V . The declared scalar function of state. In this model, $V = \frac{1}{2} \sum_i ((x_i - x_i^*)/\sigma_i)^2$. It measures represented standardized discrepancy, not total welfare or conserved matter.

Physical carrier or sink. A represented place where energy, matter or useful capacity goes during an action. It must be accounted for in the state or boundary when the corresponding physical effect is in scope; it is not a second subtraction from the finite EBU identity.

Quantity, q . The finite source-side amount in this edition. A rate j has stock-per-time units and becomes quantity after integration over time. For a constant rate, $q = \Delta t j$.

Receipt. A provenance-linked record of a physical event, its accepted comparison, evidence and signed calculation. A record can exist before an institution has decided whether its number changes a transferable balance.

Reference, x_i^* . A declared local comparison value. It may be informed by functional evidence, but it is not automatically an observed mean, ethical optimum or dynamic equilibrium.

Scale, $\sigma_i > 0$. A fixed positive stock-unit width determining how much a given deviation contributes to the potential. It is separate from measurement uncertainty unless an explicit model establishes a relationship.

Service. A separately specified outcome such as delivery of a required quantity before a deadline. A positive field difference can coexist with unmet service; a negative one can coexist with necessary service.

State, x . The ordered collection of represented physical coordinates. Complete state identity includes units, meanings and ordering. Equal potential does not imply equal state.

Support, W . Coordinates directly changed by the action. With coupled factors, the set of coordinates required to evaluate the effect can be larger than this write support.

Telescoping. Cancellation of intermediate values in a sum of consecutive differences. It requires consistent definitions and correctly linked boundaries; it does not reconstruct missing physical history.

Vector account. Separate resource-specific results retained together without an undeclared conversion. Even dimensionless scores can have different calibrations and meanings.

Coalition. A subset of declared actions whose endpoint is specified under one comparison protocol. An unordered subset does not automatically specify an execution order.

Möbius coefficient, $m_E(T)$. The alternating finite contrast that removes all proper-subset contributions before naming the interaction of T .

Composite, $G(z) = V(x(z))$. The potential evaluated on the action-to-state response. Its finite differences determine interaction, including when the response is nonlinear.

Discrete Taylor representation. The unique complete multilinear polynomial reproducing a Boolean table. It has no remainder at the vertices; it need not be the physical response at fractional controls.

Generator, b_z or L_z . A declared state-velocity law or its action on observables. It supplies response, not the definition of EBU. Historical J_e denotes a flux; a scalar opposition Ψ is not automatically a vector generator.

Canonical density. A density of the form $p = Z^{-1}e^{-V}$ relative to a specified measure, with finite normalizer. A physical canonical interpretation needs the appropriate ensemble and calibration.

Relative surprisal, J_{rel} . $-\ln[p(x)/p(x^*)]$ in a fixed density convention. Absolute $J_{\text{prob}} = -\ln p$ differs by a reference constant.

Stationarity and reversibility. Stationarity preserves a probability law under a declared process. Reversibility is a stronger symmetry of the process; stationary currents can distinguish them.

Thermal and bridge coefficients. $\beta_{\text{thermo}} = 1/(k_B T)$ has inverse-energy units. β_{bridge} is the dimensionless comparison coefficient between a declared EBU potential and relative surprisal. The canonical normalized benchmark gives the latter as one.

Entropy tables. Medium, stochastic-system and total entropy changes have separate meanings. Their EBU scaling holds only under the stated P4 physical conditions for every compared process.

The compact symbol card

Symbol	Object	Units or interpretation
x_i, x_i^b	Stock at a location	Stock, X
x_i^*	Declared reference	Stock, X
σ_i	Positive reference scale	Stock, X
q	Finite action quantity	Stock, X
j	Action rate	Stock per time
δ	State increment	Stock in each coordinate
V	Declared state potential	Dimensionless account value
μ_i	Marginal potential	Inverse stock
H	Curvature matrix	Inverse squared stock
E	Signed finite value	Compatible potential units
R_a	Common-path attribution	Compatible potential units
B_i	Proposed action balance	Declared capacity-account units
A_{ext}	External potential audit	Compatible potential units
z_i	Action control or indicator	Dimensionless; distinct from stock baseline
$G(z)$	Composite potential	Potential units
$m_E(T)$	Interaction coefficient	Same units as E
p, Z	Density and normalizer	Relative to declared measure
J_{prob}	Negative log density	Declared density convention
β_{bridge}	Bridge coefficient	Dimensionless
k_B	Boltzmann constant	Entropy units

Notation alone establishes neither physical calibration nor spending rights.

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How to follow a mathematical claim

The social and environmental references support the bounded claims attached to them in the opening chapters, not a claim that EBU succeeds. The project sources identify definitions, conditional mathematics and historical records. Source versions, PDF hashes and the full revision disposition accompany the editable manuscript.

The local-to-global argument in this edition is the cancellation of unchanged factors. Its Gaussian examples use a newly declared quadratic; they do not reinterpret historical threshold fixtures. The exact finite expansion is derived directly from that quadratic. The path sum uses linearity, the chain rule and the fundamental theorem of calculus. The sequence and closed-cycle identities use consecutive endpoint cancellation.

The field definition, the path convention and any proposed capacity rule have different roles. A proof about the sum of path contributions cannot supply a missing ownership map. Likewise, a physical conservation equation cannot

by itself choose which consequences belong in a potential. These boundaries are part of the claim, including when a source file is committed in Git.

The continuity chapters rewrite the cleared R and S-MG derivations into a teaching sequence [28, 29, 30]. The proofs in the text are explicit; the references locate antecedents and physical scope. No theorem is presented as newly discovered on the basis of this integration.

Edition and scientific-status note

This expanded continuous volume starts from the 160-page *What an Economy Must Keep Alive* manuscript prepared in the author’s “Book writer” chat. Its exact source and PDF identities are recorded in the repository integration report. The earlier separate volumes, historical facsimiles and original worktree remain preserved outside this book. The present edition extends the explanatory argument rather than binding those facsimiles into it or adding studio chapters to reach a page target.

The central integration uses the repaired R-stage source at commit `6c1d082` and the S-MG theorem and prior-art package at commit `a7655f3`, with the independent-clearance disposition supplied in the author’s integration request. Their original reports are unchanged. Clearance of those inputs does not constitute independent audit of this newly edited book.

The governing definition remains $E = V(x_{\text{pre}}) - V(x_{\text{post}})$ for a complete declared transition under one fixed field. The frozen physical foundation and working theory baseline retain precedence over exploratory or historical material. The manuscript changes no scientific authority, runtime, preregistration, seed, execution seal or result artifact.

The finite and interaction theorems are conditional mathematics. The canonical density and accepted P4 entropy relations carry their additional physical assumptions. Cross-field denomination, general nonequilibrium interpretation, actor mechanisms, economic effectiveness and institutional fairness remain separate research questions. No completed experiment or successful economy is inferred from this book’s rendering.

All new numerical examples are constructed teaching calculations. No scientific random stream, model trajectory, synthetic campaign or optical-trap experiment was executed for this integration. The current E1a execution pause is unchanged. A future study requires its own scientific and operational authorization.

The eight-part editorial map remains a map of teaching responsibilities and future work. This one-volume Book 1 consolidates the author’s latest introductory synthesis and develops the central continuity hierarchy. It does not renumber the later series or claim to replace their deeper measurement, dynamics, topology and institutional programmes.

Book 1 theoretical integration complete — independent book audit required.